Information Design and Pricing in Lead-Selling Platforms with Supply Competition

Yanwei Sun* Niloofar Zamani-Foroushani* Zhe Liu Jiahua Wu

Imperial Business School, Imperial College London

{yanwei, n.zamani-foroushani20, zhe.liu, j.wu}@imperial.ac.uk

We analyze a type of platform common in the service sector, where customers can freely post tasks, and service providers pay a fee to purchase job leads and unlock the ability to communicate with these customers. We adopt a Bayesian persuasion framework to explore how the platform can leverage its informational advantage about the provider base and pricing to strategically influence the decisions of service providers, aiming to maximize its revenue. We begin the analysis by assuming that prices are exogenously given and characterize the structure of the optimal signaling mechanism. Specifically, the platform recommends full participation when the realized number of potentially interested providers is below a certain threshold. When this number exceeds the threshold, the platform's recommendation randomizes between two adjacent participation levels, and the mixing is independent of the realized market size. This behavior is driven by a unique feature of our model, where the beliefs of the platform and service providers are non-identical: a provider's knowledge of their own participation inflates their belief about the total number of service providers. Next, we endogenize the platform's pricing decision and consider the policy space where the signaling mechanism and (possibly randomized) state-dependent pricing are jointly optimized. Notably, within this space, we find that the optimal signaling mechanism with fixed pricing is optimal. This result implies the superiority of the optimal signaling mechanism over the commonly adopted contingent pricing policy. We show that the performance discrepancy between these two policies is particularly pronounced when supply competition is intense.

Key words: lead-selling platform, Bayesian persuasion, pricing, supply competition, heterogeneous beliefs

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1. Introduction

This paper focuses on a type of online platforms prevalent in service sectors, as exemplified by platforms like Angi, Thumbtack, and Bark from the US, MyBuilder, MyJobQuote and RatedPeople from the UK, and GetNinjas from Brazil. These platforms have become an important intermediation channel in local services markets. In the United States, Angi reports substantial revenue from selling

*Equal contribution.

leads and related advertising to service professionals: its 2024 domestic "Ads and Leads" revenue was $962.6 million, with a total company revenue of $1.185 billion.[1] Another large U.S. marketplace, Thumbtack, reports a nationwide network of roughly 300,000 local service professionals and claims that projects initiated on its platform exceed 90 million.[2]

This type of platforms typically operate as follows: A customer in need of completing a short-term task, such as house renovation or maintenance, posts the job on the platform. Service providers on the other side can view job postings but cannot directly contact the customers. Instead, they must pay a fee to the platform to unlock the ability to communicate with the customers. The purchase of leads to these jobs is non-exclusive, allowing multiple service providers to buy access to the same job lead. Once a lead is purchased, the negotiation phase begins, during which service providers can learn more about the job, discuss pricing, and, ultimately, the customer may select a service provider to complete the task.

These platforms choose to operate in this manner for several reasons. First, in terms of business models, alternatives such as charging commissions based on actual transaction values are difficult to implement in these service sectors due to the risk of platform disintermediation. Once service providers are matched with customers, there is nothing to prevent them from bypassing the platform and completing the transaction offline. Second, rather than matching a customer directly with a single provider using proprietary and often opaque algorithms, the platform shows providers all the job leads that match their trades, skills, and work areas. This level of transparency helps the platform address recent controversies and backlash against algorithmic decision-making, such as concerns over privacy breaches, bias in recommendations, and lack of transparency, allowing both sides of the market greater flexibility in how they connect with each other.

A major operational challenge for these platforms involves the effective management of service providers. From the platform's viewpoint, having a large number of service providers buy leads is desirable, as the primary revenue stream for the platform originates from the sales of job leads. Furthermore, competition among these providers not only facilitates the delivery of high-quality services but also contributes to lowering prices. Such dynamics, known as the network effect, are beneficial in attracting new customers to the platform. However, service providers may hesitate to purchase leads if they anticipate intense competition. Using data from a platform employing this mechanism, Besbes et al. (2023) empirically estimate a structural model and document that providers respond strongly to supply competition, with these responses affecting platform-level outcomes.

In this paper, we develop a theoretical framework to study how a platform can jointly leverage information and/or pricing levers to manage provider competition, with implications both for platform

[1] https://www.sec.gov/Archives/edgar/data/1705110/000170511025000018/angi-20241231.htm

[2] https://www.thumbtack.com/about

revenue and for welfare. The platform holds an informational advantage, possessing a comprehensive understanding of the provider base, including those with specific interests or specializations relevant to the posted jobs—details that remain hidden from the providers. Our base model focuses on the informational aspect, with the assumption that prices are fixed and exogenously determined. Thus, the challenge of maximizing revenue essentially comes down to maximizing the number of service providers that purchase each lead. Adopting a Bayesian persuasion framework, we first identify that for any given signaling mechanism, there exists an obedient and symmetric mechanism that yields the same expected supply. This insight allows us to formulate the problem as a linear programming problem. We demonstrate that under the optimal mechanism, the platform recommends full participation when the realized number of potentially interested providers is below a certain threshold. When this number exceeds the threshold, the platform's recommendation randomizes between two adjacent participation levels, and the mixing is independent of the realized market size. This behavior is driven by a unique feature of our model, where the beliefs of the platform and providers are non-identical: a provider's knowledge of their own participation inflates their belief about the total number of interested service providers.

Additionally, we compare the optimal signaling mechanism against public mechanisms, where the same signal is sent to all providers. Because the optimal public mechanism is generally *not* directly solvable via a linear program, under a linear reward function, we develop a novel belief-based *reduction* that transforms the multi-dimensional concavification problem into a tractable *continuous knapsack* problem. We fully characterize the optimal public mechanism, which admits a structure of *probabilistic lower-censorship* (PLC). This PLC mechanism pools sufficiently low states, and may randomize at a cutoff state, into a single public message, while truthfully revealing high states. Through extensive numerical comparisons, we demonstrate that the optimal private mechanism can substantially outperform the optimal public mechanism generally.

Next, we endogenize the platform's pricing decision and consider the policy space where the signaling mechanism and (possibly) randomized prices, which can depend on the number of interested providers, are jointly optimized. Notably, we find that the optimal signaling mechanism with fixed pricing is optimal within this policy space. This result implies the superiority of the optimal signaling mechanism with fixed pricing over the commonly adopted contingent pricing policy, where the price charged by the platform depends on the number of interested providers. We show that both policies can extract all expected surpluses from participating providers. However, the dominance of the optimal signaling mechanism is mainly driven by its ability to alleviate coordination failures that arise from providers' self-selected participation decisions under the contingent pricing policy. The ratio between the expected revenues from the optimal signaling mechanism with fixed pricing and the optimal contingent pricing policy, which we refer to as the *coordination gap*, is more pronounced

when supply competition is intense. We are able to characterize the upper bound of the coordination gap when the providers' reward function is linear. On the practical side, we show that the optimal signaling mechanism can be easily implemented by setting a limit on the maximum number of providers who can purchase the lead.

Lastly, we explore several extensions of the base model. First, we study a two-stage dynamic extension of the base model. In Stage 2, we show that for any (not necessarily optimal) stage-1 mechanism, the platform's optimal stage-2 policy *fully reveals* realized stage-1 participation and sets a stage-2 price that depends on stage-1 participation but is *independent* of the stage-2 market size. This independence yields a signaling structure analogous to that in the base model. In Stage 1, when the resulting objective, which depends on both the stage-1 and stage-2 reward functions, is unimodal, information design with a single price remains optimal within the general policy space in which information and state-dependent pricing are jointly optimized. Second, we generalize the base model to a competitive market in which a pool of potential providers competes for two leads. The resulting problem shares the same core structure as the base model but introduces two additional *cross-lead incentive-compatibility* constraints, which ensure that providers recommended to purchase one lead do not deviate to purchase the other. These constraints are absent in the base model and substantially complicate the analysis. We show that the main insight from the base model, i.e., the optimality of information design with fixed pricing, continues to hold when either the two leads share the same reward function or the leads are sufficiently popular. We also consider a setting where the objective is to maximize providers' welfare instead of the platform's revenue. We demonstrate the stark contrast in the optimal signaling mechanism when the objective changes. Finally, we consider heterogeneous providers who differ in their utility functions. We identify the conditions under which the optimal signaling mechanism with fixed pricing for each type of providers remains optimal within the policy space where the signaling mechanism and state- and type-dependent pricing are jointly optimized. In the interest of space, we relegate the latter two extensions to the appendix.

Related Literature. Our work contributes to the rapidly growing literature on information design (also known as Bayesian Persuasion), particularly its applications in operations and online platforms. Since the seminal work of Kamenica and Gentzkow (2011), the framework of information design has been adopted to study how a player with an informational advantage can use it to influence the decisions of others by committing to a mechanism a priori. This approach has been applied in a wide range of operational contexts, including but not limited to: retailing (Lingenbrink and Iyer 2018, Drakopoulos et al. 2021), queuing (Lingenbrink and Iyer 2019, Anunrojwong et al. 2023), resource allocation (Yang et al. 2019, Candogan and Wu 2023, Ashlagi et al. 2025), pandemics control (Alizamir et al. 2020, De Véricourt et al. 2021), product review (Papanastasiou et al. 2018, Boyaci

et al. 2024), sequential search (Xiao et al. 2023), and supply chain management (Candogan and Gurkan 2023, Song et al. 2024). For a more comprehensive review, see Candogan (2020).

The studies closely related to our work are those examining how platforms use informational levers to improve operational efficiency. Romanyuk and Smolin (2019) study a dynamic matching platform in which short-lived buyers arrive over time and are randomly routed to long-lived sellers, who decide whether to accept a request subject to capacity constraints. The platform commits to an information policy that determines what information about each buyer's type is disclosed to sellers before they accept. They show that full disclosure can lead to excessive rejections ("cream skimming") and hence to inefficiency, whereas suitably coarse disclosure can restore efficiency. Küçükgül et al. (2022) focus on platforms running sales campaigns over fixed periods and study how revealing information about sales history can influence future customers to maximize platform revenue. Bimpikis et al. (2024) explore information design in two-sided platforms with quality-differentiated sellers and customers with different valuations for service quality. They find that the revenue-optimal information provision policy often involves delaying the disclosure of information regarding the quality of available providers. Papanastasiou and Sekar (2024) investigate the effect of ranking in two-sided platforms where both sellers and consumers are strategic. They assume that the platform holds private information about the match probabilities between sellers and consumers and can strategically prioritize certain products in the rankings to receive more consumer attention. While the above-mentioned studies consider that platform's users are expected-utility maximizer, Anunrojwong et al. (2024) extend the Bayesian persuasion methodology to settings where users are risk-conscious. Lian et al. (2025) develop a game-theoretic model of platform–worker interactions to study how earnings transparency policies shape equilibrium behavior and pay outcomes in the gig economy.

Besides, Bimpikis and Mantegazza (2023) consider a two-sided platform that can disclose information on the demand side to influence the entry and pricing decisions of its potential providers. Although we also focus on the supply side of the platform, similar to Bimpikis and Mantegazza (2023), our context and modeling framework differ significantly. One interesting feature of lead-selling platforms is that the demand uncertainty is lower than the uncertainty on the supply side. With a job post on the platform, demand is already known to some extent, and the major driver influencing providers' decisions is the competition on the supply side. To model this supply-side uncertainty, we consider a model with a random number of providers. Games with an unknown number of players are less explored in the literature (McAfee and McMillan 1987, Myerson 2000, Lingenbrink and Iyer 2018). These games have a unique property: players' own participation influences their beliefs, causing a deviation from the true distribution. Our analysis shows that this shifted belief plays an important role in shaping the structure of the optimal signaling mechanism.

Our work also contributes to the relatively underexplored literature that compares information design with other operational levers, especially pricing. Drakopoulos et al. (2021) consider a two-period model where a seller can charge different prices in each period and share information about inventory quantities to influence customers' purchasing decisions. They find that, in certain cases, the optimal private signaling mechanism with publicly posted prices achieves the same expected revenue as a mechanism where prices are personalized in the first period and fixed for all customers in the second period. Similarly, Lingenbrink and Iyer (2019) show that in the $M/M/1$ queuing setting, optimal information design with a fixed price generates the same expected revenue as the optimal state-dependent pricing mechanism. This complements Banerjee et al. (2015), who show that static pricing remains optimal even when the platform can adjust prices dynamically, using a queueing-theoretic model of a two-sided ride-sharing platform. In contrast, in our setting, we demonstrate that the revenue under the optimal signaling mechanism with fixed pricing is higher, and can be significantly higher, than that under the optimal state-dependent pricing.

In another study, Gur et al. (2023) model a dynamic interaction between a seller and a platform that maximizes consumer surplus. Buyers arrive over time and are either *patient* or *impatient*; impatient buyers consider only the promoted product. The platform commits to (i) an information policy governing what is revealed to the seller about demand conditions (in particular, the value of being promoted) and (ii) a (possibly history-dependent) promotion policy. They introduce a class of *confounding* promotion policies that make the seller's sales observations uninformative about underlying demand parameters, thereby limiting seller learning and enabling the platform to improve long-run average consumer surplus. Gur et al. (2023) primarily study how a platform can combine information disclosure with a *promotion policy* to maximize *consumer surplus*. Our focus is on how a platform combines information design with a *commission fee* to maximize *revenue*. Thus, while both papers pair information design with an additional policy instrument, they analyze different instruments (promotion versus commissions), objectives (consumer surplus versus revenue), and primitives governing the strategic interaction.

There is also a small but growing body of literature that studies online platforms selling job leads. Besbes et al. (2023) empirically estimate a structural model using data from a large lead-selling platform and document that providers respond strongly to supply competition. Building on this behavioral evidence, we develop a theoretical framework to characterize the platform's optimal policy when it can jointly design information disclosure and pricing, and we allow for a richer information environment than the (arguably) binary specification considered in Besbes et al. (2023). By contrast, Shi (2024) takes a more aggregate perspective, focusing on how lead pricing balances supply and demand on a lead-selling platform. Our approach is micro-level: we focus on a single job lead and the platform's ability to manage competition for that lead. This micro-level perspective allows us to study

a broader platform decision space in which the platform can use both (possibly supply-contingent) pricing and information about supply conditions to shape providers' participation decisions. As a result, we obtain an easily implementable policy that optimally manages supply competition.

2. Model Setup

For an intuitive grasp of our model, think about a service platform as described in Section 1. There is a finite set $\mathcal{U} = \{1, 2, \ldots, u\}$ of providers on the platform. Consider one representative customer who posted a job on the platform. This job can be of interest to a set $\mathcal{I} \subseteq \mathcal{U}$ of service providers. The likelihood of a given provider being interested in the job post may depend on factors such as their specialization, geographical location and other characteristics, which are assumed to be exogenously given. We denote the probability of the set of interested providers being $\mathcal{I} \subseteq \mathcal{U}$ as $\beta(\mathcal{I})$. $\beta(\cdot)$ is common knowledge for the platform and all providers.

The realized set of interested providers is obscured from the providers; however, it is known to the platform, who has comprehensive knowledge about the registered provider base.[3] Due to the informational advantage, the platform can reveal some information about the pool of interested providers to incentivize participation. We consider a Bayesian persuasion framework where the platform is able to commit to a signaling mechanism a priori, the details of which are formalized in the rest of this section.

Actions and Utility Function. An interested provider may opt to buy the lead from the platform for a commission fee $p_0 \geq 0$. Acquiring a lead enables providers to initiate contact with the customer, gain further insights into the job requirements, and potentially negotiate pricing. Ultimately, the customer selects one provider to complete the task. We relax the assumption of fixed commission fee and discuss contingent pricing, where the commission fee depends on the number of interested providers, in Section 5.

We assume providers are homogeneous; we later discuss extensions to heterogeneous providers in Appendix F.2. Given this assumption, a sufficient statistic for the decision-making of the platform and the providers is the total number of interested providers $|\mathcal{I}|$, i.e., the cardinality of $\mathcal{I}$. We use N to denote this random number of interested providers for the focal job post, which is referred to as the *state* or *thickness* of the market. Let $\Omega = \{\underline{n}, \ldots, \bar{n}\} \subseteq \mathbb{N}_+$ (induced by all possible subsets $\mathcal{I} \subseteq \mathcal{U}$) denote the finite set of underlying states containing all possible *total numbers of interested providers*. We assume $|\Omega| \geq 2$ to avoid trivial problems.

[3] On lead-selling platforms, the workflow typically proceeds as follows. When a provider registers, they specify their service categories (i.e., the types of jobs they are willing and able to perform—for example, plumbing repairs, electrical work, painting and decorating, appliance installation, carpentry, or bathroom/kitchen renovation) as well as their service area (e.g., postcode districts, a radius around a home base, or a set of neighborhoods). When a customer posts a job, the platform matches the job's category and location to these provider profiles and makes the lead visible to the subset of providers whose stated specialties and geographic coverage are consistent with the job. As a result, the platform can form a reasonably accurate view of which providers are *potentially* interested in the lead.

For provider $i \in \mathcal{I}$, let $a_i \in \{0,1\} =: \mathcal{A}$ be provider i's action representing whether to participate ($a_i = 1$) or not ($a_i = 0$). "Participate" means purchasing the lead in our context, and we use them interchangeably throughout the rest of the paper. Without loss of generality, we normalize the utility of not participating (namely the outside option) to zero. When there are $N = n$ interested providers, let $a^n_{-i} \in \mathcal{A}^{n-1}$ be the actions of the remaining $n-1$ interested providers. Given the action profile (a_i, a^n_{-i}), the utility function of provider $i \in \mathcal{I}$ is

$$u_i(a_i, a^n_{-i}) = \begin{cases} R(\sum_{j\in\mathcal{I}} a_j) - p_0, & \text{if } a_i = 1, \\ 0, & \text{if } a_i = 0, \end{cases} \tag{1}$$

where $R : \mathbb{R}_+ \to \mathbb{R}$ is the reward function depending on the total number of participants and $p_0 \geq 0$ is the commission fee charged by the platform. For notational convenience, let $h(x, p_0) := R(x) - p_0$.

Assumption 1. *We assume the following for the reward function:*

(i) $R(1) > 0$ *and* $R(x)$ *strictly decreases in* $x \in \mathbb{R}_+$*;*

(ii) $xR(x)$ *is concave in* $x \in \mathbb{R}_+$*.*

Assumption 1 (i) is innocuous: $R(1) > 0$ eliminates the trivial situation where no provider would participate even at a price (i.e., commission fee) of zero, and the decreasing property of $R(\cdot)$ indicates that each provider receives a lower reward as competition intensifies. The concavity of $xR(x)$ serves primarily for technical convenience. Nevertheless, it encompasses many plausible forms of $R(\cdot)$ relevant to our particular context. Below, we provide one micro-foundation for the provider utility that satisfies Assumption 1; our results are by no means limited to this single case.

Example 1. Suppose the utility of a provider from securing and completing the job is given by r. Assume there are x providers who opt to purchase the lead from the platform. The providers are homogeneous in the sense that each possesses an equal probability[4] of $1/x$ to successfully win the job from the customer. Associated with the purchase of the lead is an upfront cost α, possibly encompassing the cost incurred in communicating/negotiating with the customer. Therefore, given x competing providers, the expected reward for each is given by $R(x) = r/x - \alpha$. It is easy to verify that Assumption 1 is satisfied as long as $r > \alpha$.

Appendix A provides an alternative micro-foundation for the reward function $R(\cdot)$ by modeling the providers who purchase the lead as subsequently engaging in price competition for the job under multinomial logit demand. In this setting, $R(\cdot)$ coincides with each provider's equilibrium profit. Under this model, $R(x)$ is decreasing in the number of participating providers x, and $xR(x)$, i.e., the participating providers' total profit, is unimodal in x. Note that, except for Section 3, most of our results require only unimodality of $xR(x)$, rather than full concavity. Additional numerical

[4] Shi (2024) demonstrates that under market-clearing prices, the platform should match customers *randomly* to available service providers, even with heterogeneous providers.

experiments based on this micro-foundation also suggest that $xR(x)$ is concave in x under general conditions, providing further support for Assumption 1(ii).

Prior Beliefs. One unique and interesting aspect of our model is that the platform and the providers share different prior beliefs of the underlying state $N \in \Omega$. To see this, denote the platform's prior belief (probability mass function (PMF)) about the underlying state N as $\mu_0 := (\mu_0(n))_{n\in\Omega} \in \Delta(\Omega)$. It is easy to verify that the platform's belief of there being n interested providers is given by $\mu_0(n) = \sum_{\mathcal{I}\subseteq\mathcal{U},|\mathcal{I}|=n} \beta(\mathcal{I})$. On the other hand, denote $\mu_{\mathsf{p}}^i \in \Delta(\Omega)$ as provider $i \in \mathcal{I}$'s belief of the number of interested providers. Knowing that they are in the market would alter their belief, making it different from the platform's belief μ_0. Below is a toy example to illustrate the forces at play.

Example 2. Consider a platform with two providers, $\mathcal{U} = \{1,2\}$. For each provider $i \in \{1,2\}$, let $I_i \in \{0,1\}$ indicate whether provider i is interested in the job. The pair (I_1, I_2) follows a bivariate Bernoulli distribution:

$$\mathbb{P}(I_1 = 1, I_2 = 0) = \mathbb{P}(I_1 = 0, I_2 = 1) = 0.25, \qquad \mathbb{P}(I_1 = 1, I_2 = 1) = 0.5, \qquad \mathbb{P}(I_1 = 0, I_2 = 0) = 0,$$

which implies a correlation coefficient of $-1/3$. Then, the probability that the set of interested providers is $\{1\}$, $\{2\}$, or $\{1,2\}$ is given by $\beta(\{1\}) = \beta(\{2\}) = 0.25$ and $\beta(\{1,2\}) = 0.5$. The platform's prior belief is given by $\mu_0(1) = \mu_0(2) = 0.5$. Now consider provider 1, and suppose that they are interested in the job. The fact that they are in the pool of interested providers rules out the event of $\{2\}$. Based on Bayes' rule, their belief of the number of interested providers being one and two is given by $\mu_{\mathsf{p}}^1(1) = 1/3$ and $\mu_{\mathsf{p}}^1(2) = 2/3$, respectively. The same applies to provider 2.

For the rest of the analysis, we assume that prior beliefs of interested providers are *symmetric*, meaning that $\mu_{\mathsf{p}}^i(n) = \mu_{\mathsf{p}}^j(n), \forall\, i,j \in \mathcal{I}$ and $\forall\, n \in \Omega$. This is a reasonable assumption when providers are homogeneous, and the probability of a provider being interested in the job is independently and identically distributed. Note that this symmetric prior-belief assumption does *not* imply independence of providers' interest in the lead; in particular, as illustrated in Example 2, the events that different providers are interested in the lead may be *inter*dependent, either positively or negatively correlated. We relax this symmetric assumption and discuss heterogeneous providers with asymmetric priors in Appendix F.2. With a bit of abuse of notation, we drop the superscript and denote providers' prior belief as μ_{p}. μ_{p} is updated by the providers (according to Bayes' rule) based on the fact that they themselves belong to the pool of interested providers; however, we still refer to it as providers' *prior* belief, as it is the belief of providers prior to receiving any signal from the platform. To distinguish from the underlying state following the platform's prior belief, $N \sim \mu_0$, denote its counterpart following the providers' prior belief by $\hat{N} \sim \mu_{\mathsf{p}}$.

Lemma 1 **(Service Providers' Prior Belief).** *Under the symmetric prior belief assumption,*

(i) service providers' prior belief μ_{p} *is*

$$\mu_{\mathsf{p}}(n) = \frac{n\mu_0(n)}{\mathbb{E}_{N\sim\mu_0}[N]},\ \forall n \in \Omega;$$

(ii) μ_{p} *first-order stochastically dominates* μ_0.

Lemma 1 (i) is a direct consequence of Lemma 1 in McAfee and McMillan (1987). Lemma 1 (ii) shows that the providers' prior belief has a first-order stochastic dominance over the platform's prior belief; we thus have $\mathbb{E}[\hat{N}] = \frac{\mathbb{E}[N^2]}{\mathbb{E}[N]} \geq \mathbb{E}[N]$. That is, the expected number of interested providers from the providers' perspective is greater than that from the platform's perspective. The underlying driver of this result is clearly reflected in Example 2. A provider's own inclusion in the pool of interested providers rules out all events where this focal provider is absent. These events are more likely to involve fewer providers (due to the absence of the focal provider), without which, the focal provider has an inflated belief about the total number of interested providers. This is commonly observed in games with a random number of players (McAfee and McMillan 1987, Lingenbrink and Iyer 2018).[5][6] Note the resemblance of inflated belief to the inspection paradox by nature.

Signaling Mechanism. The platform can commit to a signaling mechanism that reveals some information about the underlying states to the providers. A signaling mechanism Σ consists of a finite signal space $\mathbb{S}$ and a mapping $\pi : \Omega \to \Delta(\mathbb{S}^n)$, where $\mathbb{S}^n$ represents the nth Cartesian power of $\mathbb{S}$ given that the number of interested providers is $n \in \Omega$. Specifically, $\pi(s^n \mid n)$ is the probability of choosing the signal profile $s^n \in \mathbb{S}^n$ given there are n interested providers. Due to the assumption of homogeneous providers and symmetric prior beliefs μ_{p}, it is without loss of generality to *randomly* reindex the set of n interested providers and denote it as $[n] := \{1, \ldots, n\}$. Let $[n]^+ := [n] \cup \{0\}$. Denote the ith element of the signal profile s^n (received by provider $i \in [n]$) by s_i^n, or simply s_i when the context is clear.

Service Providers' Strategies and Equilibrium. We focus on symmetric equilibria and assume all (homogeneous) providers adopt a common mixed strategy $f : \mathbb{S} \to [0, 1]$, a mapping from the received signal to the probability of participation ($a_i = 1$). Upon receiving signal s_i, given all other providers following strategy f, provider i's expected utility is given by

$$\mathbb{E}^{\Sigma,f}[u_i(a_i, a_{-i}^N) \mid s_i] = \underbrace{(1 - f(s_i))}_{\text{not participate}} \cdot 0 + \underbrace{f(s_i)}_{\text{participate}} \cdot \mathbb{E}^{\Sigma,f}[u_i(a_i = 1, a_{-i}^N) \mid s_i] = f(s_i) \cdot \mathbb{E}^{\Sigma,f}[u_i(a_i = 1, a_{-i}^N) \mid s_i].$$

[5] Note that, Lemma 1 holds under the assumption that Ω is finite. It is shown in Myerson (1998, 2000) that, if $\Omega = \{0, 1, \ldots, \infty\}$ contains infinite number of elements and μ_0 follows a Poisson distribution, then the provider's belief remains unchanged, i.e., $\mu_{\mathsf{p}} = \mu_0$.

[6] The economic insight in McAfee and McMillan (1987), that agents may overestimate competition when the number of competitors is uncertain, is supported across several settings. For example, field experiments on job-search platforms show that providing information about actual competitor counts materially changes behavior, increasing application rates among previously hesitant candidates (especially women) (Gee 2019). Laboratory evidence from auctions similarly finds that concealing the number of bidders induces more aggressive bidding, consistent with bidders behaving as if the field were larger (Dyer et al. 1989). Related experimental work in contest-like environments also reports systematic overestimation of rival counts (Otten 2022).

We use the superscripts to highlight that provider i's expected utility depends on both the signaling mechanism Σ and the participation strategy f. The conditional expectation $\mathbb{E}^{\Sigma,f}[u_i(a_i=1,a_{-i}^N)\mid s_i]$ is taken over provider i's posterior belief (updated by Bayes' rule upon receiving signal s_i) about the number of interested providers N, the signals received by other providers s_{-i}^N, and their corresponding strategy $a_j\sim f(s_j),\forall j\in[N]\setminus\{i\}$.

Because provider i's utility $u_i(a_i=1,a_{-i}^N)$ only depends on the total number of lead-purchasing providers, we can simplify the expected utility by using (1) as

$$\mathbb{E}^{\Sigma,f}[u_i(a_i,a_{-i}^N)\mid s_i]=f(s_i)\cdot\mathbb{E}^{\Sigma,f}[u_i(a_i=1,a_{-i}^N)\mid s_i]=f(s_i)\cdot\mathbb{E}^{\Sigma,f}[h(1+1^\top a_{-i}^N,p_0)\mid s_i],\tag{2}$$

where $1^\top a_{-i}^N$ counts the number of providers other than provider i who purchase the lead.

Provider i purchases the lead if their expected utility is greater than their outside option, i.e., $\mathbb{E}^{\Sigma,f}[h(1+1^\top a_{-i}^N,p_0)\mid s_i]>0$, and does not participate if $\mathbb{E}^{\Sigma,f}[h(1+1^\top a_{-i}^N,p_0)\mid s_i]<0$. When they are indifferent between participating or opting out, provider i adopts a mixed strategy, participating with a probability of $f(s_i)$. We formally summarize the providers' equilibrium below.

Definition 1 (Providers' Equilibrium). Given signaling mechanism $\Sigma=(\pi,\mathbb{S})$ and price p_0, a strategy $f:\mathbb{S}\to[0,1]$ forms an equilibrium if and only if for any interested provider i and signal $s_i\in\mathbb{S}$,

$$f(s_i)=\begin{cases}1, & \text{if } \mathbb{E}^{\Sigma,f}[h(1+1^\top a_{-i}^N,p_0)\mid s_i]>0,\\ 0, & \text{if } \mathbb{E}^{\Sigma,f}[h(1+1^\top a_{-i}^N,p_0)\mid s_i]<0,\end{cases}$$

otherwise, $f(s_i)\in[0,1]$.

Platform's Problem. The platform's revenue originates from providers purchasing the lead. The platform has two levers to influence providers' participation decisions: information via the signaling mechanism Σ and pricing via the commission fee p_0. The optimal information and pricing policies can be derived from solving the following optimization problem:

$$\max_{p_0\geq 0}\ \max_{\Sigma=(\pi,\mathbb{S})}\quad p_0\cdot\sum_{n\in\Omega}\mu_0(n)\sum_{s^n\in\mathbb{S}^n}\pi(s^n\mid n)\sum_{i\in[n]}f(s_i)\tag{$\mathcal{P}$}$$
$$\text{s.t.}\quad f \text{ forms an equilibrium by Definition 1 under } (p_0,\Sigma).$$

The objective function is the platform's expected revenue, subject to the constraint that f is an equilibrium strategy for providers as defined in Definition 1. In the objective function, the probability that there are n interested providers is given by $\mu_0(n)$. In such case, the platform sends a signal $s^n\in\mathbb{S}^n$ with probability $\pi(s^n\mid n)$, and provider $i\in[n]$ purchases the lead with probability $f(s_i)$ upon receiving signal $s_i\in\mathbb{S}$. The platform aims to maximize its revenue by finding an optimal signaling mechanism Σ^* and an optimal price p_0^*. We show later that it is without loss of optimality to restrict attention to a single price with signaling mechanisms, even when the platform can choose a (possibly randomized) state-dependent pricing policy.

Timeline. The timeline of events is summarized as follows. The platform first commits to a signaling mechanism Σ and a price p_0. Then, the set of interested providers $\mathcal{I}$ is realized, and thus the underlying state (market thickness) n. Based on the committed signaling mechanism, each provider $i \in [n]$ receives a private signal $s_i \in \mathbb{S}$ from the platform. Providers update their beliefs in a Bayesian fashion and act according to an equilibrium strategy f as described in Definition 1. The payoffs of the platform and all providers are realized.

Discussions. A few comments are in order regarding the modeling assumptions. First, we consider a single-period static model where the platform reveals information upfront, rather than a dynamic model where signals are released and/or updated over time. This choice is primarily for analytical tractability. A dynamic information design problem is challenging because signals can depend not only on the number of providers who have already purchased the lead, but also on the potential number of future purchases. Our static model is a good reflection of reality, especially when the customers need to complete their jobs quickly and the time window is short. That said, a two-stage dynamic extension of the base model is studied in Section 6.1.

Second, our model considers one single job from a representative customer. Our main reason for focusing on a single lead is that, on lead-selling platforms, *within-lead* competition among providers is often a first-order concern, whereas *across-lead* choice is frequently secondary, for two reasons. First, providers typically pay on the order of \$10–\$100 per lead (depending on job type and local competitiveness) on major platforms, such as Angi and HomeAdvisor,[7] while the revenue from completing the job can be substantial conditional on winning it.[8] A central concern for providers is how likely a purchased lead is to convert into paying work, typically referred to as the *close rate*. Estimated close rates on most of these platforms are commonly reported in the range of 10–30%.[9] Thus, while the per-lead fee may appear modest, the effective acquisition cost per realized job (roughly, fee divided by the close rate) can be substantial. This makes the platform's joint design of information disclosure and lead fees a critical problem. A second reason it is sensible to focus on a single lead is that jobs typically arrive sequentially and timing is critical. The sooner a provider contacts the customer, the higher the likelihood of converting the lead into paid work. As a result, providers often make effectively myopic decisions on a lead-by-lead basis, treating each arriving job as an independent participation decision rather than solving a rich across-lead portfolio problem. That said, we generalize the base model to a competitive market in which a pool of potential providers competes for two leads in Section 6.2.

[7] https://www.getjobber.com/academy/homeadvisor-vs-angi/

[8] Although precise statistics on average revenue per job are difficult to obtain, industry benchmarks provide useful proxies. For example, the average cost of a roof repair job in the U.S. in 2025 is about \$1,150, interior house painting typically ranges from \$966 to \$3,088, and kitchen remodeling averages \$26,969 (https://www.homeadvisor.com/cost/). These figures suggest that per-lead fees are often modest relative to the underlying job value.

[9] https://7ten.marketing/how-much-does-thumbtack-charge-for-leads/, https://www.mbi-deepdives.com/content/files/2022/07/ANGI-Deep-Dive.pdf

3. Optimal Signaling with an Exogenous Price

We start our analysis by assuming that the price p_0 is exogenously given, and focus on determining the optimal signaling mechanism. Later, we utilize the insights obtained from this section to derive policies that jointly optimize signaling mechanism and pricing in Section 4. In this section with exogenously given price p_0, we assume $R(1) - p_0 > 0$ to avoid the trivial case that no provider would participate under p_0. Given price p_0, problem ($\mathcal{P}$) reduces to finding the optimal signaling mechanism such that the expected number of lead-purchasing providers is maximized. The expected number of service providers that purchase the lead is referred to as *expected supply* in the rest of this paper.

3.1. Structure of Optimal Mechanism

In general, problem ($\mathcal{P}$) is hard to solve due to the prohibitively large space of possible mechanisms. In what follows, we first introduce two special types of mechanisms, namely obedient mechanisms and symmetric mechanisms. Then, we show that without loss of optimality, we can restrict our attention to these two types of mechanisms.

Definition 2 (Obedient Mechanism). A signaling mechanism $\Sigma = (\pi, \mathbb{S})$ is obedient if and only if the signal space is the action space $\mathcal{A} = \{0, 1\}$, i.e., $\mathbb{S} = \mathcal{A}$, and providers find it incentive-compatible to follow the recommended action.

Definition 3 (Symmetric Mechanism). A signaling mechanism $\Sigma = (\pi, \mathbb{S})$ is symmetric if and only if $\pi(s^n \mid n) = \pi(\hat{s}^n \mid n)$ whenever $\hat{s}^n$ is a permutation of s^n.

The term *obedient mechanism* is defined in accordance with the terminology used in Bergemann and Morris (2016, 2019). On the other hand, a *symmetric mechanism* is characterized by its independence from labeling the providers. Our next result shows that without loss of optimality, we can restrict our focus on obedient and symmetric mechanisms. The proof of the lemma uses a revelation-principle-style argument (Myerson 1981, Bergemann and Morris 2019).

Lemma 2 **(Symmetry and Obedience)**. *Given the price p_0, for any signaling mechanism, there exists an obedient and symmetric mechanism that achieves the same expected supply.*

Lemma 2 substantially narrows down the signal space to be considered in search of the optimal mechanism. The principle of obedience implies that, in the quest for the optimal mechanism, it suffices to determine the number of providers $k \in [n]^+$ that the platform should ask to participate for each $n \in \Omega$. Following this, the property of symmetry enables the platform to choose k providers through a (uniformly) random selection, recommending the chosen providers to purchase the lead while recommending the rest to abstain from participation.

Consequently, we can reformulate problem ($\mathcal{P}$) as follows. Given that there are n interested providers in total, let $\sigma(\cdot \mid n) \in \Delta([n]^+)$ be the distribution (PMF) of the number of providers being recommended to participate. Specifically, $\sigma(k \mid n)$ represents the probability of recommending $k \in$

$[n]^+$, out of the total n interested providers, to participate. Conditional on $\sigma(k \mid n)$, the probabilities that each provider is recommended to participate and not to participate are $\frac{k}{n}$ and $\frac{n-k}{n}$, respectively.

LEMMA 3 **(LP Formulation)**. *Given the price p_0, the optimal signaling mechanism can be found by solving the following problem ($\mathcal{P}_{\mathsf{LP}}$):*

$$\max_{\sigma} \quad \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot k \tag{$\mathcal{P}_{\mathsf{LP}}$}$$

$$\text{s.t.} \quad \sum_{n\in\Omega} \mu_0(n) \sum_{k=1}^{n} \sigma(k \mid n) \cdot k \cdot h(k, p_0) \geq 0 \tag{join}$$

$$\sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n-1} \sigma(k \mid n) \cdot (n-k) \cdot h(k+1, p_0) \leq 0 \tag{not-join}$$

$$\sigma(\cdot \mid n) \in \Delta\left([n]^+\right), \ \forall n \in \Omega. \tag{distribution}$$

The platform seeks to maximize the expected number of providers that purchase the lead, subject to the constraints that it is rational for all providers to adhere to the recommendations. The objective function reflects the expected supply based on the platform's belief. We have two constraints that ensure providers' incentive compatibility in following these recommendations. Constraint (join) stipulates that the expected utility for a provider who follows the recommendation to participate is no less than their outside option. One noteworthy point is that the expected utility is calculated based on the provider's posterior belief, which is updated upon receiving the signal. In particular, upon receiving a signal to join ($s_i = 1$), the expected utility for provider i to join, $\mathbb{E}^{\Sigma,f}[h(1 + 1^\top a^N_{-i}, p_0) \mid s_i = 1]$, is proportional to $\sum_{n\in\Omega} \mu_{\mathsf{p}}(n) \sum_{k=1}^{n} \sigma(k \mid n) \cdot \frac{k}{n} \cdot h(k, p_0)$, where $\sigma(k \mid n) \cdot \frac{k}{n}$ is the probability that the provider receives a to-join signal when there are n interested providers and the platform recommends k of them to participate. Then, using the relation $\mu_{\mathsf{p}}(n) = \frac{n\mu_0(n)}{\mathbb{E}_{N\sim\mu_0}[N]}$ established in Lemma 1, we get constraint (join). A detailed derivation can be found in the proof of Lemma 3. Similarly, constraint (not-join) guarantees that the expected utility for a provider who deviates from the recommendation of not purchasing the lead and participates instead, will not exceed their outside option. Finally, constraint (distribution) specifies that the randomized signaling mechanism constitutes a properly defined probability distribution.

Throughout the rest of this section, we assume that each provider's ex ante expected utility is non-positive for a given price p_0, i.e.,

$$\mathbb{E}[h(\hat{N}, p_0)] = \frac{\mathbb{E}[Nh(N, p_0)]}{\mathbb{E}[N]} \leq 0, \tag{4}$$

where the equality follows from Lemma 1. When the equality holds or when (4) is violated, the *no-information* mechanism arises as an optimal signaling mechanism because it would be rational for all providers to participate even if the platform shares no information about the number of interested providers. Note that this assumption is only relevant for this section when the price p_0 is assumed

to be exogenously given. When the platform jointly optimizes the price p_0 as well, as we study in Section 4, it is never optimal for the platform to charge a price so low that $\mathbb{E}[h(\hat{N}, p_0)] > 0$. In this case, the platform can increase the price without changing the behavior of providers, thereby achieving a higher revenue. With this, we are able to characterize the optimal signaling mechanism.

THEOREM 1 **(Optimal Signaling Mechanism)**. *Given a price p_0 such that (4) holds, there exist a threshold τ, a critical number k^* and a probability γ such that for any $n \in \Omega$ with $n < \tau$,*

$$\sigma^*(k \mid n) = \begin{cases} 1, & k = n, \\ 0, & \textit{otherwise}, \end{cases}$$

and for any $n \in \Omega$ with $n \geq \tau$,

$$\sigma^*(k \mid n) = \begin{cases} \gamma, & k = \lfloor k^* \rfloor, \\ 1-\gamma, & k = \lfloor k^* \rfloor + 1, \\ 0, & \textit{otherwise}. \end{cases}$$

In general, we do not have a closed-form solution for τ, k^*, and γ. However, we provide an efficient iterative algorithm capable of finding the parameters in at most $|\{n \in \Omega : n \geq R^{-1}(p_0)\}|$ iterations. The details are provided in Appendix B.2.

Theorem 1 shows that the optimal signaling mechanism follows a threshold structure. That is, there exists a threshold τ such that if the number of interested providers is less than τ, the platform would recommend all interested providers to participate. Note that it can be shown that the optimal threshold τ is no less than $R^{-1}(p_0)$, which we term "break-even participation" since $h(R^{-1}(p_0), p_0) = 0$. This is because providers' expected utility from participating would be strictly positive at any state $n < R^{-1}(p_0)$. Thus, at optimality, the platform would recommend all providers to participate for any state $n < R^{-1}(p_0)$, and the positive surplus will be redistributed across the unfavorable states when competition is intense, ensuring it is rational for as many providers as possible to purchase the lead upon receiving a signal to participate. On the other hand, in situations of intense competition where the number of interested providers is more than or equal to τ, the platform's recommendation randomizes between two consecutive integers, namely $\lfloor k^* \rfloor$ and $\lfloor k^* \rfloor + 1$. Note that the recommendation could be deterministic, i.e., $\gamma = 1$, if k^* happens to be an integer. In general, with non-integer k^*, randomization occurs to ensure that constraint (join) is binding. We prove Theorem 1 by verifying the optimality conditions of ($\mathcal{P}_{\mathsf{LP}}$). One interesting byproduct is the following observation, which plays a key role when we tackle the joint optimization problem in Section 4.

OBSERVATION 1. *Given a price p_0 such that (4) holds (with strict inequality), constraint (not-join) is non-binding at optimality. As a result, without loss of optimality, we can discard constraint (not-join) in problem ($\mathcal{P}_{\mathsf{LP}}$).*[10]

[10]For a price p_0 such that $\mathbb{E}[Nh(N, p_0)] = 0$, it can be shown that constraint (not-join) can still be discarded without loss of optimality.

We emphasize that, for $n \geq \tau$, the fact that σ^* randomizes between $\lfloor k^* \rfloor$ and $\lfloor k^* \rfloor + 1$, with these randomization points independent of n, is driven by providers' *inflated prior beliefs* (see Lemma 1). We discuss the intuition behind this in greater detail in Appendix B.3 and provide the corresponding comparative statics in Appendix B.4. On another note, even though providers are assumed to be homogeneous, the optimal signaling mechanism is usually a private mechanism in the sense that different providers may privately receive different signals. In Section 3.2, we study optimal public mechanisms where the platform sends the same signal to all providers, and show that public mechanisms are strictly suboptimal in general.

3.2. Public Mechanisms

In this section, we focus on a specific type of signaling mechanism, namely *public mechanisms*, where the platform sends the same signal to all providers publicly. Public mechanisms have been extensively studied in the literature (Lingenbrink and Iyer 2018, Candogan and Wu 2023), for good reasons. They are easier to implement since the same signal is sent to all providers. They also alleviate the fairness concerns where some providers are recommended to participate while others are dissuaded, as described in Theorem 1. This is particularly pertinent when providers are assumed to be identical ex ante. Note that public mechanisms encompass two important cases commonly adopted in practice: the *no-information* scenario, where providers receive no signal from the platform and base their beliefs about the provider pool size on their prior distribution μ_{p}; and the *full information* scenario, in which the platform reveals the exact number of interested providers.

There are two main challenges in finding the optimal public mechanism. First, characterizing providers' joining equilibrium is difficult under general reward functions. For any given public signal, each provider's best response depends on the expected number of participating providers. In equilibrium, this induces a binomial distribution for the number of participants, and solving the resulting fixed point is challenging for general reward functions (see Section 5.1 for a detailed example even under full information). To ensure analytical tractability, we therefore focus on linear reward functions, which deliver a closed-form joining strategy. Second, the optimal public mechanism can no longer be obtained by solving a linear program (Bergemann and Morris 2016). We therefore adopt the belief-based concavification approach of Kamenica and Gentzkow (2011). This is typically intractable in high-dimensional state spaces, but we address this challenge by proposing a novel reduction under which the optimal public mechanism can be derived by solving a *continuous knapsack* problem.

We begin by defining a class of mechanisms as follows:

Definition 4 (Probabilistic Lower Censorship). A *probabilistic lower-censorship* signaling mechanism, denoted $\mathsf{PLC}(\omega, \alpha)$ with $\omega \in \Omega$ and $\alpha \in [0, 1]$, is defined as follows:

- If $n < \omega$, send the signal L.

- If $n = \omega$, reveal the state $n = \omega$ with probability α, and send L with probability $1 - \alpha$.
- If $n > \omega$, fully reveal the state.

The mechanism $\mathsf{PLC}(\omega, \alpha)$ reduces to the no-information benchmark when $\omega = \bar{n}$ and $\alpha = 0$. Likewise, it reduces to full information when $\omega = \underline{n}$ and $\alpha = 1$. We show below that an optimal public signaling mechanism can be chosen from the class PLC.

THEOREM 2 **(Optimal Public Signaling Mechanism)**. *Fix a price p_0. Under a linear reward function, an optimal public signaling mechanism has the following structure:*

(i) If $\bar{n} \le R^{-1}(p_0)$, then all public mechanisms yield the same expected supply $\mathbb{E}[N]$.

(ii) If $R^{-1}(p_0) \le \underline{n}$, then full information *is optimal, yielding expected supply $\sum_{n\in\Omega} \mu_0(n)\, n\, \frac{R^{-1}(p_0)-1}{n-1}$.*

(iii) If $\underline{n} < R^{-1}(p_0) < \bar{n}$, then:

(a) If $\mathbb{E}[\hat{N}] < R^{-1}(p_0)$, the no-information *mechanism is optimal, with expected supply $\mathbb{E}[N]$.*

(b) If $\mathbb{E}[\hat{N}] \ge R^{-1}(p_0)$, the probabilistic lower-censorship *mechanism $\mathsf{PLC}(\omega^*, \alpha^*)$ is optimal, where (ω^*, α^*) can be computed efficiently via Algorithm 2 in Appendix B.5. The expected supply under this mechanism is*

$$\mathbb{E}[N] - A_1^* + (R^{-1}(p_0) - 1) \sum_{n > R^{-1}(p_0),\, n\in\Omega} \frac{\mu_0(n)\, n\, \pi_H^*(n)}{n-1},$$

where π_H^* is also given by Algorithm 2 and $A_1^* := \sum_{n > R^{-1}(p_0),\, n\in\Omega} \mu_0(n)\, n\, \pi_H^*(n)$.

Theorem 2 characterizes optimal *public* signaling mechanisms under linear reward functions, partitioning the parameter space into four mutually exclusive and collectively exhaustive regions:

$$\text{(i) } \bar{n} \le R^{-1}(p_0); \qquad \text{(ii) } R^{-1}(p_0) \le \underline{n};$$

$$\text{(iii-a) } \underline{n} < R^{-1}(p_0) < \bar{n} \text{ and } \mathbb{E}[\hat{N}] < R^{-1}(p_0); \qquad \text{(iii-b) } \underline{n} < R^{-1}(p_0) < \bar{n} \text{ and } \mathbb{E}[\hat{N}] \ge R^{-1}(p_0).$$

Note that under linear rewards, $\mathbb{E}[\hat{N}] < R^{-1}(p_0)$ is equivalent to $\mathbb{E}[h(\hat{N}, p_0)] > 0$, and $\bar{n} \le R^{-1}(p_0)$ is a sufficient condition for $\mathbb{E}[h(\hat{N}, p_0)] > 0$. Hence, $\mathbb{E}[h(\hat{N}, p_0)] > 0$ holds in both cases (i) and (iii-a). Recall that we assume $\mathbb{E}[h(\hat{N}, p_0)] \le 0$ for Theorem 1 since otherwise the optimal signaling mechanism reduces to the *no-information* case, where it is rational for all providers to participate when the platform shares no information about the number of interested providers. This result is verified in Theorem 2(iii-a). Furthermore, in more favorable situations where $\bar{n} \le R^{-1}(p_0)$, meaning that supply competition is mild and any interested provider would be better off participating under all possible state realizations, all public mechanisms are equivalent. It is straightforward to verify that this result holds when the assumption of linear reward functions is relaxed.

The more interesting cases are (ii) and (iii-b), in which $\mathbb{E}[h(\hat{N}, p_0)] \le 0$. In these cases, providers' inflated beliefs render them sufficiently pessimistic that full participation is *not* an equilibrium absent

informative signaling. This is where the optimal public mechanism and the optimal (private) mechanism diverge. The two forces that shape the structure of the optimal (private) signaling mechanism—information obfuscation and inflated prior belief from providers—also influence the design of the optimal public mechanisms, albeit in a slightly different manner. Theorem 2(iii-b) shows that when there exist sufficiently favorable states in which a provider obtains strictly positive surplus from participating even if all other interested providers participate, information obfuscation can be valuable. In that case, the optimal public mechanism is a PLC mechanism (as defined in Definition 4): it pools all sufficiently low states—and may randomize at a cutoff state—into a single message that induces full participation, while fully revealing high states. The optimal cutoff ω^* and mixing probability α^* can be computed via Algorithm 2 in Appendix B.5; they amount to the solution of a *continuous knapsack* problem, see (ContinuousKnapsack) in the proof for details. By contrast, in case (ii), where such favorable states are absent, it is optimal for the platform to eliminate uncertainty and fully disclose information, as shown in Theorem 2(ii).

Remark 1. Belief-based approaches to optimal public mechanisms with *more than two states* typically impose that the first moment of the posterior belief is a sufficient statistic (Dworczak and Martini 2019, Kolotilin et al. 2022, Candogan and Wu 2023). Our Theorem 2 is, to the best of our knowledge, the first result that characterizes the optimal public mechanism over general state spaces when the first moment is *not* sufficient. In particular, Kolotilin et al. (2022) show that when (i) the first moment is sufficient and (ii) the interim value function is first concave and then convex, *deterministic* lower censorship is optimal. In contrast, in our setting the first moment is *not* sufficient; nevertheless, via a novel reduction we obtain an interim supply function (in a transformed variable) that is linear up to a threshold and convex thereafter, under which a *probabilistic* lower-censorship policy is optimal. We expect the reduction technique developed here to apply more broadly beyond our specific model. □

Additional numerical comparisons between the optimal private mechanism, the optimal public mechanism, and the no- and full-information benchmarks are reported in Appendix B.7. Consistent with Theorem 2, in cases Theorem 2(i) and (iii-a) the optimal public and private mechanisms coincide, whereas in cases (ii) and (iii-b) the optimal private mechanism can substantially outperform the optimal public mechanism.

4. Jointly Optimal Signaling and Fixed Pricing

In this section, we endogenize the platform's pricing decision and characterize the policy that jointly optimizes the signaling mechanism and a fixed commission fee. Following the same argument in Section 3 that leads to the ($\mathcal{P}_{\mathsf{LP}}$) formulation in Lemma 3, problem ($\mathcal{P}$) can be reformulated as

$$\mathsf{REV}^* := \max_{p_0 \geq 0, \sigma} \quad p_0 \cdot \sum_{n \in \Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot k \tag{$\mathcal{P}_{\mathsf{rev}}$}$$

$$\text{s.t.} \quad (\sigma, p_0) \text{ satisfy (join), (not-join) and (distribution).}$$

Observe that problem ($\mathcal{P}_{\mathsf{rev}}$) is a bilinear program, which is NP-hard in general. In what follows, we leverage the structure of the problem to provide an analytical solution. Note first that the optimal price p_0 must satisfy $\mathbb{E}[Nh(N, p_0)] \le 0$; otherwise the platform can increase p_0 until $\mathbb{E}[Nh(N, p_0)] = 0$ without changing the providers' participation incentive, thereby achieving higher revenue. Then, according to Observation 1, constraint (not-join) can be discarded without loss of optimality. Furthermore, by Lemma B.2 in Appendix B.1, constraint (join) can be equivalently set as an equality, i.e.,

$$p_0 \cdot \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot k = \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot kR(k). \tag{5}$$

Note that the objective function in problem ($\mathcal{P}_{\mathsf{rev}}$) is the same as the left-hand side of equation (5), which represents the platform's expected revenue under signaling mechanism σ and price p_0. Thus, problem ($\mathcal{P}_{\mathsf{rev}}$) can be simplified to

$$\max_{\sigma \text{ satisfies (distribution)}} \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot kR(k). \tag{$\hat{\mathcal{P}}_{\mathsf{rev}}$}$$

The platform's revenue maximization problem boils down to maximizing the expected total reward for participating providers. Since there are no constraints linking different states in problem ($\hat{\mathcal{P}}_{\mathsf{rev}}$), it is equivalent for the platform to maximize the expected reward for participating providers in *each* state $n \in \Omega$:

$$\max_{\sigma(\cdot|n)\in\Delta([n]^+)} \sum_{k=0}^{n} \sigma(k \mid n) \cdot kR(k). \tag{6}$$

The objective function is a convex combination of $kR(k)$ with weights $\sigma(k \mid n)$, for $k = 0, \dots, n$. Consequently, given a realized state $n \in \Omega$, the optimal solution is given by $\sigma^*(k^*(n) \mid n) = 1$, where $k^*(n) = \arg\max_{k\in[n]^+}\{kR(k)\}$. Having identified the optimal mechanism σ^*, the optimal price p_0^* can be solved from (5) immediately. Accordingly, denote by $\mathsf{REV}^*(n) := p_0^* \cdot k^*(n)$ the platform's revenue in state n under the optimal signaling mechanism with fixed pricing.

Define $\tau^* := \arg\max_{x\ge 0}\{xR(x)\}$ as the unique maximizer of the function $xR(x)$. The following theorem substantiates $k^*(n)$ and p_0^* discussed above and formally summarizes the optimal signaling mechanism with a fixed commission fee.

THEOREM 3 **(Optimal Signaling Mechanism with Fixed Pricing)**. *The optimal solution to problem ($\mathcal{P}_{\mathsf{rev}}$) is as follows: For any $n \in \Omega$,*

(i) if $n \le \tau^*$, then $k^*(n) = n$, i.e., $\sigma^*(n \mid n) = 1$;

(ii) if $n > \tau^*$, then $k^*(n) = k^* \in \arg\max_{x\in\{\lfloor\tau^*\rfloor, \lceil\tau^*\rceil\}}\{xR(x)\}$, i.e., $\sigma^*(k^* \mid n) = 1$.

The optimal price is given by

$$p_0^* = \frac{\sum_{n\in\Omega, n\le\tau^*} \mu_0(n) \cdot nR(n) + \sum_{n\in\Omega, n>\tau^*} \mu_0(n) \cdot k^* R(k^*)}{\sum_{n\in\Omega, n\le\tau^*} \mu_0(n) \cdot n + \sum_{n\in\Omega, n>\tau^*} \mu_0(n) \cdot k^*}. \tag{7}$$

The optimal expected revenue is

$$\mathsf{REV}^* = \mathbb{E}[\mathsf{REV}^*(N)] = \sum_{n\in\Omega, n\le\tau^*} \mu_0(n)\cdot nR(n) + \sum_{n\in\Omega, n>\tau^*} \mu_0(n)\cdot k^* R(k^*). \tag{8}$$

When the signaling mechanism and price are jointly optimized, the structure of the optimal signaling mechanism is similar to that described in Theorem 1 where price is assumed to be exogenous. There exists a threshold τ^* for the number of interested providers, below which the platform recommends all to participate, and above which a fixed number of providers, k^*, are recommended to purchase the lead.

Despite the similarities, there are two key differences. First, when the price is endogenized, the threshold is independent of the price and is chosen to maximize the expected total reward for all participating providers. This is achieved by maximizing the expected reward for participating providers at each realized state $n \in \Omega$. Since all surplus from participating providers is extracted by the platform at optimality, the platform's revenue is maximized as a result. Second, the recommendation is no longer *randomized* when the realized state exceeds the threshold. In Theorem 1, randomization ensures the binding of constraint (join). However, when the price is endogenized, this approach is strictly suboptimal. Instead of randomizing, say between k^* and k^*+1, the platform can allocate all the weight to the integer with a higher expected total reward for participating providers. Since the platform can extract all surplus from participating providers, it leads to higher expected revenue for the platform as well.

It is worth noting that, although at optimality, the platform's expected revenue equals the expected reward for all participating providers, as derived from problem ($\hat{\mathcal{P}}_{\mathsf{rev}}$), they are not equivalent on a per-state basis. Specifically, at optimality, the platform's expected revenue in state n is not necessarily equal to the expected total reward for participating providers in state n, as derived from equation (6). This discrepancy arises because a provider's expected reward from participation is the highest in state $\underline{n}$ due to the smallest pool of interested providers, and gradually decreases as n increases. The expected reward then remains constant as long as the state n is greater than τ^*. As the platform charges a fixed price, the optimal price is chosen to leave some surplus for the participating providers when n is small. However, this leftover surplus is completely extracted by the platform in states with large n, while still ensuring it is rational for the provider to participate.

On another note, one premise of information design is that the platform must commit to a signaling mechanism a priori and publicly announce it. This is typically a strong assumption. However, the optimal signaling mechanism described in Theorem 3 mitigates this concern. Specifically, the optimal mechanism recommends up to k^* providers to participate. This can be easily implemented in practice by setting a limit on the maximum number of providers who can purchase the lead. Providers typically arrive sequentially, and any provider who arrives before the limit is reached will be recommended to

purchase the lead. In fact, some platforms, such as RatedPeople[11], already set a limit on the maximum number of providers who can purchase the lead, albeit without the knowledge that it is optimal. Our results provide an analytical solution for k^*, which depends on the providers' reward function but is independent of the underlying state. This implies that k^* can be calculated without knowing the provider base, allowing for a straightforward implementation. We also provide comparative statics on how optimal revenue and the optimal mechanisms vary with the reward function in Appendix C.1.

5. Optimality and Comparisons

We now consider a general policy space by jointly optimizing the signaling mechanism and state-dependent pricing. Specifically, we consider the most general setting of this policy space, where both the signal and the state-dependent price can be randomized. Formally, the platform decides $\kappa : \Omega \to \Delta(\mathbb{R} \times \mathbb{S}^n)$, a mapping from each state to a *joint* distribution over prices, charged to all interested providers, and *private* signals, each provider received. Let $\mathcal{M}$ denote the set of all feasible mechanisms κ.

In general, finding the jointly optimal policy is challenging because the state-dependent price itself could reveal information about the underlying state. Nevertheless, we show that the optimal signaling mechanism with fixed pricing analyzed in Theorem 3 is optimal in this expanded policy space.

THEOREM 4 **(Optimality of Signaling with Fixed Pricing)**. *Within the policy space $\mathcal{M}$ that jointly considers signaling and randomized state-dependent pricing, the optimal signaling mechanism with fixed pricing remains optimal, i.e., the mechanism in Theorem 3 is second-best.*

The proof involves three key elements. First, similar to Lemma 2, we show that, given a realized price, it is without loss of optimality to work with obedient mechanisms. In this case, an obedient mechanism maps the underlying state *and the realized price* to a distribution over the number of participating providers, and each provider receives a signal to join (or not) uniformly at random. Second, we formulate the platform's problem in terms of randomized pricing and obedient mechanism, enforcing incentive compatibility for all possible prices and signals. Last, we construct an upper bound on the optimal revenue of this problem and show that the revenue REV^* under the optimal signaling mechanism with fixed pricing achieves this upper bound.

5.1. Optimal Contingent Pricing

We now analyze the practically relevant state-dependent pricing under full information, where the platform can incentivize the participation of interested providers by charging a commission fee that depends on the number of interested providers, referred to as *contingent pricing*. This practice has been successful in improving firms' bottom lines across a wide range of industries and applications

[11] Link: How RatedPeople works for tradespeople?

(Talluri and Van Ryzin 2006). In this section, we study the optimal contingent pricing policy and examine whether pricing could be a more effective lever than information.

Specifically, when there are $n \in \Omega$ interested providers, the platform charges a state-dependent commission fee $p(n)$, and we assume all interested providers have *full information* about the system state n under this contingent pricing setting. Similar to Definition 1 of the providers' participation equilibrium, all interested providers adopt a common mixed strategy $f(n) \in [0,1]$ of participation probability in response to the price $p(n)$ in state n. An adaptation of (2) provides the expected utility for a provider to participate as $\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)] - p(n)$. Noticing that the first term, i.e., the expected reward, is strictly decreasing in $f(n)$ due to the monotonicity of $R(\cdot)$ by Assumption 1, we have $\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)] \in [R(n), R(1)]$. Therefore, an analogue of Definition 1 under contingent pricing gives the equilibrium strategy in state n:

$$f(n) = \begin{cases} 1, & \text{if } p(n) < R(n), \\ \text{unique root of } \mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)] = p(n), & \text{if } p(n) \in [R(n), R(1)], \\ 0, & \text{if } p(n) > R(1). \end{cases} \tag{9}$$

The platform's optimal contingent pricing problem is then formulated as

$$\begin{aligned} \mathsf{REV}^*_{\mathsf{cp}} := \max_{p(\cdot)} \quad & \sum_{n\in\Omega} \mu_0(n) \cdot p(n) \cdot n \cdot f(n) \\ \text{s.t.} \quad & f(n) \text{ satisfies (9)}, \quad \forall n \in \Omega. \end{aligned} \tag{$\mathcal{P}_{\mathsf{cp}}$}$$

With the constraints decoupled over states, the optimal solution can be obtained by maximizing the platform's expected revenue in each state. To this end, let $\mathsf{REV}^*_{\mathsf{cp}}(n) := \max_{p(n)}\{p(n)\cdot n \cdot f(n) : (9)\}$ be the platform's pricing problem in state $n \in \Omega$. Note that this objective implies $p^*(n) \in [R(n), R(1)]$ since $p(n) < R(n)$ and $p(n) > R(1)$ are apparently suboptimal by observing constraint (9). In other words, only the middle case of (9) is relevant. As a result, problem ($\mathcal{P}_{\mathsf{cp}}$) can be simplified to solving, for each state $n \in \Omega$,

$$\mathsf{REV}^*_{\mathsf{cp}}(n) = \max_{f(n)\in[0,1]} \mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)] \cdot n \cdot f(n). \tag{10}$$

The optimal contingent pricing policy is formally summarized in Proposition 1.

PROPOSITION 1 **(Optimal Contingent Pricing)**. *In state $n \in \Omega$, the optimal contingent price under full information is $p^*(n) = \mathbb{E}_{k\sim\mathsf{Bin}(n-1,f^*(n))}[R(1+k)]$, where the equilibrium participation probability is given by*

$$f^*(n) \in \arg\max_{f(n)\in[0,1]} \mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)] \cdot f(n).$$

The corresponding equilibrium expected number of participating providers is $x^*(n) = n \cdot f^*(n)$ and the optimal expected revenue is $\mathsf{REV}^*_{\mathsf{cp}}(n) = p^*(n) \cdot x^*(n)$.

In general, closed-form solutions for optimal contingent prices are unavailable due to the absence of an analytical expression for $\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)]$. In the case of linear reward functions $R(\cdot)$, the solution admits an analytical characterization; we provide the details in Appendix C.2.

5.2. Comparisons and Coordination Gap

As a direct consequence of Theorem 4, we can establish the superiority of the optimal signaling mechanism with fixed pricing over the optimal contingent pricing policy, summarized in Corollary 1.

COROLLARY 1 **(Coordination Gap).** $\mathsf{REV}^* \geq \mathsf{REV}^*_{\mathsf{cp}}$. *Besides, for linear reward functions, we have*

$$\frac{\mathsf{REV}^*}{\mathsf{REV}^*_{\mathsf{cp}}} \leq \frac{1}{\left(1-\frac{1}{R^{-1}(0)}\right)^2},$$

where the upper bound is tight when all states in Ω approach infinity (i.e., $\underline{n}\to\infty$) and τ^* is an integer.

When the reward function $R(\cdot)$ is linear, Corollary 1 provides an upper bound on the expected revenue of the optimal signaling mechanism with fixed pricing relative to that of the optimal contingent pricing. Note that the bound is solely a function of $R^{-1}(0)$, which is the number of participants required to reduce a provider's reward to zero. To some extent, $R^{-1}(0)$ measures the effect of supply competition on a provider's potential payoff. In the context of Example 1, providers need to spend time and effort communicating and negotiating with the customer before possibly securing the job. When this cost relative to the potential payoff is low, i.e. $R^{-1}(0)$ is large, the provider's expected reward remains positive despite intense supply competition. In this case, the difference between the optimal signaling mechanism with fixed pricing and the optimal contingent pricing is relatively small. Conversely, if a few competitors can easily reduce their expected payoff to zero, i.e., $R^{-1}(0)$ is small, Corollary 1 shows that the potential value of the optimal signaling mechanism with fixed pricing is high. The potential value of information design is realized as supply competition intensifies.

5.2.1. State-Wise Comparison. We now compare, on a *per-state basis*, the optimal signaling mechanism with fixed pricing and the optimal contingent pricing, together with a first-best outcome. The comparison sheds light on the underlying drivers for the dominance of the optimal signaling mechanism with fixed pricing. To derive clear insights, we focus on linear reward functions but note that this assumption is not required for our other results unless specified otherwise.

First, consider the *optimal contingent pricing* with full information analyzed in Section 5.1. By (10) and the reward function $R(\cdot)$ being linear, the expected revenue in state n under the optimal contingent pricing is given by

$$\mathsf{REV}^*_{\mathsf{cp}}(n) = \max_{f(n)\in[0,1]} \mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[R(1+k)]\cdot n\cdot f(n)$$

$$= \max_{f(n)\in[0,1]} n \cdot f(n) \cdot R\left(1+(n-1)\cdot f(n)\right) = \max_{x(n)\in[0,n]} x(n) \cdot R\left(x(n)+1-\frac{x(n)}{n}\right), \quad (11)$$

where $x(n) = n \cdot f(n)$ denotes the expected number of participating providers in state n, which can be a non-integer under mixed strategy $f(n)$.

Next, consider the *optimal signaling mechanism with fixed pricing*, which, by Theorem 4, attains the highest possible expected revenue (REV^*) within the policy space of joint signaling and state-dependent pricing. The expected revenue in state n under the optimal signaling mechanism with fixed pricing, $\mathsf{REV}^*(n) = p_0^* \cdot k^*(n)$, has a rather complicate expression with p_0^* given by (7) and $k^*(n) \in \arg\max_{k\in[n]^+} kR(k)$. Therefore, for the purpose of our state-wise comparison, we use the maximum expected total reward for participating providers in state n (given by (6)) as a proxy, and name it the *second-best* revenue in state n:

$$\mathsf{OPT}(n) := \max_{\sigma(\cdot|n)\in\Delta([n]^+)} \sum_{k=0}^{n} \sigma(k \mid n) \cdot kR(k) = \max_{k\in[n]^+} kR(k),$$

where the last equality follows by assigning the whole probability mass 1 to the maximizer of function $kR(k)$ within $[n]^+$. Note that $\mathsf{REV}^*(n) \neq \mathsf{OPT}(n)$ in general. However, by the analysis preceding Theorem 3, in particular equation (5), we know $\mathsf{REV}^* = \mathbb{E}[\mathsf{OPT}(N)]$, i.e., the platform's optimal total expected revenue is equal to the corresponding total expected reward for participating providers. This is because the platform can extract all surplus from participating providers at optimality.

Finally, define the *first-best* revenue[12] in state n as:

$$\mathsf{FB}(n) := \max_{x(n)\in[0,n]} x(n) \cdot R(x(n)).$$

With the same objective but allowing for fractional numbers of participating providers, it is straightforward that $\mathsf{FB}(n) \geq \mathsf{OPT}(n)$. Comparing the objective of $\mathsf{REV}^*_{\mathsf{cp}}(n)$ in (11) with that of $\mathsf{FB}(n)$ then implies $\mathsf{REV}^*_{\mathsf{cp}}(n) \leq \mathsf{FB}(n)$, because the former shifts the linear and decreasing reward function $R(\cdot)$ to the left by $1 - \frac{x(n)}{n} \geq 0$ for $x(n) \in [0, n]$.

We extend the above discussion and summarize the state-wise comparison in Proposition 2.

PROPOSITION 2 **(State-Wise Comparison)**. *Under linear reward functions,* $\tau^* = \arg\max_{x\geq 0}\{xR(x)\} = \frac{R^{-1}(0)}{2}$. *For any* $n \in \Omega$*:*

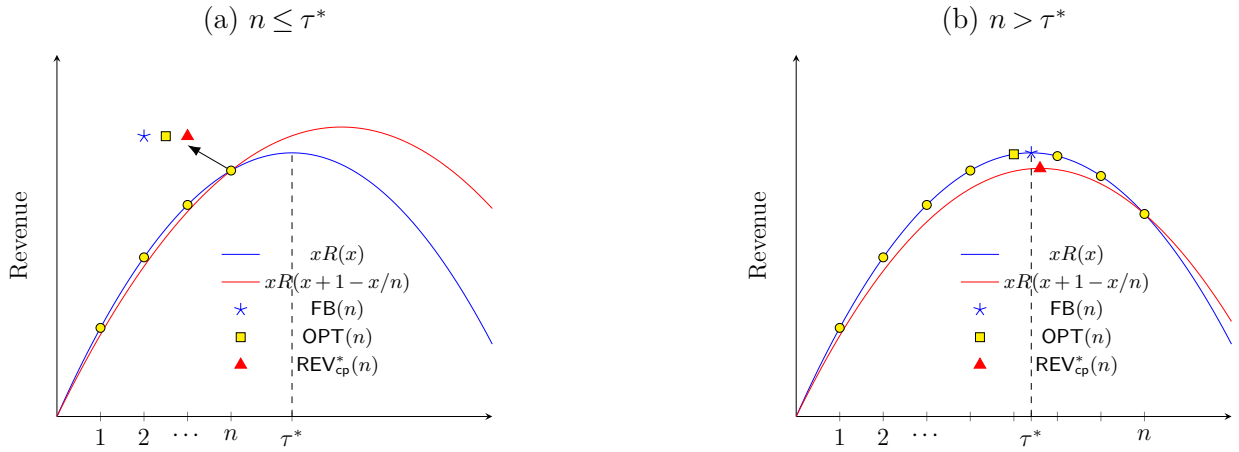
(i) If $n \leq \tau^*$, $\mathsf{FB}(n) = \mathsf{OPT}(n) = \mathsf{REV}^*_{\mathsf{cp}}(n)$.

(ii) If $n > \tau^*$, $\mathsf{FB}(n) \geq \mathsf{OPT}(n) \geq \mathsf{REV}^*_{\mathsf{cp}}(n)$. *The first inequality is strict unless* τ^* *is an integer; the second inequality is strict if and only if* $n > \frac{R^{-1}(0)+1}{2}$.

Proposition 2 shows that, for $n \leq \tau^*$, all three revenues coincide, as shown by Figure 1 (a). For $n > \tau^*$, the three revenues are usually strictly ordered, see Figure 1 (b) for an illustration.

[12]We call it first-best since under a linear reward function, the maximum price that the platform can set while still ensuring $x(n)$ participating providers is $R(x(n))$, which results in a zero utility for every participating provider.

Figure 1: Comparison of $\mathsf{FB}(n)$, $\mathsf{OPT}(n)$ and $\mathsf{REV}^*_{\mathsf{cp}}(n)$ on a Per-State Basis



Note. Consider state $n \in \Omega$. The first-best (blue star) is the maximum of the blue curve within the range $[0, n]$. The second-best (yellow square) is similar to the first-best, but over integers within $[0, n]$, i.e., the yellow dots. The optimal contingent pricing (red triangle) is the maximum of the red curve within the range $[0, n]$.

The dominance of the optimal signaling mechanism with fixed pricing over optimal contingent pricing may seem surprising at first glance, as both policies are able to extract all surplus from participating providers at each state $n \in \Omega$. However, there is one key difference. Under the optimal contingent pricing policy, providers endogenously choose their participation probability in response to the commission fee charged by the platform. It can be observed from Figure 1 (b) that the self-selected participation probability from providers at equilibrium is higher than the optimal participation probability under the first-best policy when $n > \tau^*$. This coordination failure is alleviated by the optimal signaling mechanism with fixed pricing, under which either $\lceil \tau^* \rceil$ or $\lfloor \tau^* \rfloor$ providers would be recommended to participate, resulting in a higher expected reward from participating providers. The upper bound of the ratio between the expected revenues from the optimal signaling mechanism with fixed pricing and the optimal contingent pricing, which we dub *coordination gap*, is fully characterized for linear reward functions in Corollary 1. This coordination issue underpins the "revelation principle" in persuasion settings, as Bergemann and Morris (2016) point out, the set of outcomes that can arise from a Bayes equilibrium, when players may observe additional information, corresponds to the set of Bayes *correlated* equilibria.

6. Extensions

In this section, we explore two extensions of the base model. Section 6.1 studies a two-stage dynamic extension of the base model, and Section 6.2 generalizes the base model to a competitive market in which a pool of potential providers competes for two leads. The extensions to welfare maximization and heterogeneous service providers are provided in Appendix F.

6.1. Dynamic Setting

We now extend the base model to a two-stage dynamic environment. At the beginning of stage 1, there are $N_0 \sim \mu_0 \in \Delta(\Omega_0)$ providers who are interested in the lead. Let $N_1 \sim \mu_1 \in \Delta(\Omega_1)$ denote the number of providers who participate by the end of stage 1, *endogenously* determined by the platform's stage-1 signaling mechanism and pricing policy. To capture heterogeneity in customer patience and job urgency, we assume that with some probability the customer selects one of these providers and the lead closes at the end of stage 1. We refer to this probability as the *close probability*; it may depend on the realized number of first-stage participants. Specifically, let $\gamma(n_1)$ denote the close probability when n_1 providers purchase the lead in stage 1. Naturally, one may expect $\gamma(\cdot)$ to be increasing in n_1, in the sense that the customer is more likely to close the job when more providers purchase the lead; however, our results below do not rely on this monotonicity. Note also that if $\gamma(n_1) = 1$ for all n_1, the model reduces to the base model; whereas if $\gamma(n_1) = 0$ for all n_1, the lead never closes in stage 1 and the model is always two-stage.

In stage 2, there are potentially $N_2 \sim \mu_2 \in \Delta(\Omega_2)$ providers who are interested in the same lead. We assume that N_0 and N_2 are independent, and that providers do not behave strategically in *when* to participate. Specifically, a provider who is present in stage 1 decides only whether to purchase in that stage, and cannot choose to "wait" for a later stage. This is a reasonable assumption for lead-selling platforms, where speed is essential: interested providers typically purchase the lead and contact the customer immediately. Based on discussions with industry practitioners, there is a substantial first-mover advantage; if a provider does not reach out within the first hour after the lead is posted, their chance of winning the job falls markedly.

Let R_i be the stage-dependent reward function for $i \in \{1, 2\}$. As in the base model, assume $R_i(\cdot)$ satisfies Assumption 1 for each i. Given stage prices p_1 and p_2, suppose n_1 providers purchase the lead in stage 1 and, if the game reaches stage 2, n_2 providers purchase the lead in stage 2. The ex-post utility of a provider who appears in stage 1 and purchases the lead is

$$\mathbb{1}\{\text{the lead closes in stage 1}\} \cdot R_1(n_1) \;+\; \mathbb{1}\{\text{the lead does not close in stage 1}\} \cdot R_2(n_1 + n_2) \;-\; p_1,$$

and the ex-post utility of a provider who appears in stage 2 and purchases the lead is

$$\mathbb{1}\{\text{the lead does not close in stage 1}\} \cdot \big[R_2(n_1 + n_2) - p_2\big].$$

That is, if the lead closes in stage i, the realized reward is $R_i(\cdot)$ evaluated at the total number of participants at the time of closing.

The platform chooses both what information to disclose and what prices to charge. The information structure consists of two components:

(i) a signal to stage-1 providers about the total number of potential stage-1 providers N_0; and

(ii) a signal to stage-2 providers about (a) the realized number of stage-1 participants N_1 and (b) the total number of potential stage-2 providers N_2.

Below we specify the platform's decision problem in detail and solve it by backward induction.

6.1.1. Second-Stage Problem. In this subsection, we take the stage-1 mechanism as given and study the platform's stage-2 problem, conditional on the lead not closing in stage 1. The most general stage-2 policy is a mechanism $\kappa_2 : \Omega_1 \times \Omega_2 \to \Delta(\mathbb{R} \times \mathbb{S}^{N_2})$, which, for each state (n_1, n_2), specifies a joint distribution over a uniform price charged to all stage-2 providers and a profile of private signals. Let $\mathcal{M}_2$ denote the set of all such mechanisms.

THEOREM 5 **(Optimal Signaling and Pricing in Stage 2).** *For any (μ_1, Ω_1) describing the distribution of stage-1 participation induced by the platform's stage-1 policy, the following mechanism achieves the optimal stage-2 revenue over the general mechanism class $\mathcal{M}_2$.*

(i) The platform truthfully discloses the realized stage-1 participation $N_1 = n_1$ to all stage-2 providers.

(ii) At state $(n_1, n_2) \in \Omega_1 \times \Omega_2$, the platform recommends participation according to the rule:[13]

$$\sigma_2^*(k \mid n_1, n_2) = \begin{cases} 1, & \text{if } k = k^*(n_1, n_2), \\ 0, & \text{otherwise,} \end{cases}$$

where

$$k^*(n_1, n_2) \in \arg\max_{k \in [n_2]_+} \; k\, R_2(k + n_1), \tag{12}$$

with ties broken arbitrarily.

(iii) The platform sets a deterministic stage-2 price $p_2^*(n_1)$ that depends only on n_1 (and not on N_2). For each $n_1 \in \Omega_1$ with $\mathbb{E}[k^*(n_1, N_2)] > 0$, the optimal price is

$$p_2^*(n_1) \;=\; \frac{\mathbb{E}[k^*(n_1, N_2)\, R_2(k^*(n_1, N_2) + n_1)]}{\mathbb{E}[k^*(n_1, N_2)]}, \tag{13}$$

where the expectation is taken with respect to $N_2 \sim \mu_2$. If $\mathbb{E}[k^*(n_1, N_2)] = 0$, then $p_2^*(n_1)$ can be set arbitrarily high, effectively pricing stage-2 providers out of the market.

Under this mechanism, the expected stage-2 revenue equals

$$\mathbb{E}[k^*(N_1, N_2)\, R_2(k^*(N_1, N_2) + N_1)]\,, \tag{14}$$

where the expectation is over $N_1 \sim \mu_1$ and $N_2 \sim \mu_2$.

Theorem 5 shows that the optimal stage-2 signaling and pricing policy closely parallels the base-model results in Theorem 3 and Theorem 4. The platform should fully disclose realized stage-1

[13] The mechanism is given by $\sigma_2(\cdot \mid n_1, n_2) \in \Delta([n_2]_+)$. Similar to that of the base model, conditional on (n_1, n_2) each stage-2 provider receives a "join" signal with probability $\sum_{k=0}^{n_2} \frac{k}{n_2}\, \sigma_2(k \mid n_1, n_2)$ and a "do not join" signal with probability $\sum_{k=0}^{n_2} \frac{n_2 - k}{n_2}\, \sigma_2(k \mid n_1, n_2)$.

participation $N_1 = n_1$, so the stage-2 problem effectively reduces to the base model with a "shifted" reward function. In particular, when designing the stage-2 signaling rule, the platform recommends $k \in [n_2]_+$ providers to purchase the lead so as to maximize $kR_2(n_1 + k)$. The concavity of $xR_2(x)$ (together with the monotonicity of $R_2(x)$) implies that, for each n_1, the shifted objective $kR_2(n_1+k)$ is unimodal in k, yielding a well-defined target number of stage-2 participants. The platform then sets a stage-2 price to extract the entire expected surplus from recommended participants, namely $\mathbb{E}[k^*(n_1, N_2)\, R_2(k^*(n_1, N_2) + n_1)]$ for any $n_1 \in \Omega_1$ with $\mathbb{E}[k^*(n_1, N_2)] > 0$. As in the base model, the optimal price is independent of the stage-2 market size; however, it *does* depend on the realized stage-1 participation level n_1.

We emphasize that, while this result may appear intuitive ex post, it is far from obvious ex ante. In the dynamic model, the stage-2 information-design problem is inherently two-dimensional: the platform can condition signals on the realized participation levels in both stages. Thus, one might expect the richer dynamic structure to materially alter the optimal policy. Nevertheless, we show that the main insights from the base model continue to apply. In particular, among all feasible mechanisms in $\mathcal{M}_2$, which allow arbitrary signaling and state-dependent randomized pricing, it is without loss of optimality to fully disclose the realized number of stage-1 participants n_1, and set a deterministic stage-2 price that is independent of N_2.

Before turning to the stage-1 analysis, we introduce *stage-2 continuation values* under the optimal stage-2 mechanism in Theorem 5, which will be useful for the stage-1 problem. First, for any realized stage-1 participation level n_1, the *expected stage-2 revenue* is

$$\mathsf{REV}_2^*(n_1) := \mathbb{E}_{N_2 \sim \mu_2}\big[k^*(n_1, N_2)\, R_2\big(k^*(n_1, N_2) + n_1\big)\big]. \tag{15}$$

Second, conditional on $N_1 = n_1$, the *expected continuation payoff* of a provider (whether arriving in stage 1 or stage 2) when the lead does not close in stage 1 and proceeds to stage 2 is

$$u_2^*(n_1) \;:=\; \mathbb{E}_{N_2 \sim \mu_2}\big[\, R_2(n_1 + k^*(n_1, N_2))\big]\,. \tag{16}$$

We can therefore define the *effective stage-1 reward function* as

$$\widetilde{R}_1(n_1) \;:=\; \gamma(n_1)\, R_1(n_1) \;+\; \big(1 - \gamma(n_1)\big)\, u_2^*(n_1), \qquad \forall\, n_1 \in \Omega_1,$$

which equals the expected reward of a provider who purchases the lead in stage 1 when there are n_1 stage-1 participants.

6.1.2. First-Stage Problem. In stage 1, the platform's most general policy is a mechanism $\kappa_1 : \Omega_0 \to \Delta\big(\mathbb{R} \times \mathbb{S}^{N_0}\big)$, which, for each realized number n_0 of interested providers, specifies a joint distribution over the stage-1 price and a profile of private signals sent to the n_0 potential stage-1 providers. Let $\mathcal{M}_1$ denote the set of all such feasible mechanisms. Given $n_0 \in \Omega_0$, define

$$n_1^*(n_0) \in \operatorname*{arg\,max}_{n_1 \in [n_0]_+} \Big\{ n_1\, \widetilde{R}_1(n_1) + \big(1 - \gamma(n_1)\big)\, \mathsf{REV}_2^*(n_1) \Big\}, \tag{17}$$

breaking ties arbitrarily. Let $\Omega_1^* := \{n_1^*(n_0) : n_0 \in \Omega_0\}$.

THEOREM 6 **(Optimal Signaling and Pricing in Stage 1).** *Fix the stage-2 best-response mechanism in Theorem 5. Suppose that the function*

$$n_1 \mapsto n_1 \widetilde{R}_1(n_1) + (1 - \gamma(n_1))\, \mathsf{REV}_2^*(n_1) \tag{18}$$

is unimodal over integers $n_1 \in \Omega_1^*$. Then the following mechanism is optimal among all mechanisms in $\mathcal{M}_1$:

(i) For each $n_0 \in \Omega_0$, the platform sets

$$\sigma_1^*(n_1^*(n_0) \mid n_0) = 1, \qquad \sigma_1^*(n_1 \mid n_0) = 0 \text{ for } n_1 \neq n_1^*(n_0),$$

where $n_1^*(n_0)$ is defined in (17).

(ii) The platform charges a fixed stage-1 price p_1^* that is independent of N_0, given by

$$p_1^* = \frac{\mathbb{E}\Big[n_1^*(N_0)\, \widetilde{R}_1(n_1^*(N_0))\Big]}{\mathbb{E}[n_1^*(N_0)]},$$

where the expectation is taken with respect to $N_0 \sim \mu_0$.

Under this mechanism, the optimal expected revenue (across both stages) is

$$\mathbb{E}\Big[n_1^*(N_0)\, \widetilde{R}_1(n_1^*(N_0)) + \Big(1 - \gamma(n_1^*(N_0))\Big)\, \mathsf{REV}_2^*(n_1^*(N_0))\Big]. \tag{19}$$

When (18) is unimodal, the optimal stage-1 signaling and pricing policy mirrors the structure of the base model. Conditional on the realized number n_0 of potential providers, the platform recommends $n_1^*(n_0)$ participants to maximize the total expected surplus generated by stage-1 participation, including continuation value from stage 2. It then sets a fixed stage-1 price, independent of N_0, to extract the entire expected surplus from providers who are recommended to join in stage 1. In Appendix D.2, we provide three simple and intuitive sufficient conditions under which (18) is unimodal. The first condition corresponds to settings in which provider interest in stage 2 is limited. The second requires R_2 to be (approximately) geometric/exponential and stage-2 interest to be sufficiently strong. The third corresponds to environments in which the lead is highly likely to close after stage 1.

6.2. Two Leads

In this section, we generalize the base model to a competitive market in which a pool of potential providers competes for two leads. The number of potential providers N is a random variable with distribution $\mu_0 \in \Delta(\Omega)$. Providers are ex ante identical, and each provider can purchase *at most* one lead. This feature may reflect budget constraints or capacity limits on the provider side, since providers typically cannot take on many projects simultaneously.[14] If, instead, most providers can

[14] Besbes et al. (2023) provide empirical evidence from a leading lead-selling marketplace in Brazil that most service providers purchase at most one lead per day.

purchase both leads, then the two leads effectively decouple, and the platform's optimal signaling and pricing reduce to those characterized in the base model. With a slight abuse of notation, let $R_i(\cdot)$ denote the reward function associated with lead i, $i \in \{1,2\}$.

Analogous to the base model, let REV^* denote the maximal revenue achievable using a signaling mechanism with fixed prices. Under such a mechanism, let $\sigma(k_1, k_2 \mid n)$ denote the probability of recommending $k_1 \geq 0$ providers to purchase lead 1 and $k_2 \geq 0$ providers to purchase lead 2, subject to $k_1 + k_2 \leq n$, when the realized number of potential providers is n. This signaling mechanism operates as follows: (1) After the total number of providers $n \in \Omega$ is realized, the platform draws a pair (k_1, k_2) according to $\sigma(\cdot \mid n)$. (2) Conditional on (k_1, k_2), the platform selects k_1 providers uniformly at random and sends them the signal "purchase lead 1," and selects k_2 *distinct* providers uniformly at random and sends them the signal "purchase lead 2." The two groups are disjoint; the remaining $n - k_1 - k_2$ providers receive the signal "do not purchase." Thus, conditional on (n, k_1, k_2), each provider receives signal $s_i = 1$ with probability $\frac{k_1}{n} \cdot \sigma(k_1, k_2 \mid n)$, signal $s_i = 2$ with probability $\frac{k_2}{n} \cdot \sigma(k_1, k_2 \mid n)$, and signal $s_i = 0$ with probability $\frac{n-k_1-k_2}{n} \cdot \sigma(k_1, k_2 \mid n)$.

Let OPT denote the optimal revenue when the platform can jointly optimize the signaling rule and randomized, state-dependent prices. Formally, in this general design space, the platform chooses a mechanism that maps each state to a *joint* distribution over (i) the prices charged to providers for each lead and (ii) the profile of private signals received by providers. Let $\mathsf{REV}^*_{\mathsf{cp}}$ denote the optimal revenue under *contingent pricing*, i.e., when the platform fully reveals the state but is allowed to set state-dependent prices for the two leads. We explicitly characterize the optimal contingent-pricing policy in this two-lead setting with linear reward functions in Appendix E.2. By definition, $\mathsf{OPT} \geq \mathsf{REV}^*$ and $\mathsf{OPT} \geq \mathsf{REV}^*_{\mathsf{cp}}$. Our next theorem provides two sufficient conditions under which $\mathsf{REV}^* = \mathsf{OPT}$, implying that a signaling mechanism with fixed prices is optimal even within the most general design space. As an immediate corollary, under these conditions $\mathsf{REV}^* \geq \mathsf{REV}^*_{\mathsf{cp}}$.

THEOREM 7 **(Two Leads).** $\mathsf{REV}^* = \mathsf{OPT}$ *if either of the following conditions holds:*

(i) Symmetric leads: $R_1 = R_2 =: R$. *In this case, an optimal signaling rule* σ^* *is given by*

$$\forall\, n < 2\tau^*: \quad \sigma^*(\lfloor n/2 \rfloor, \lceil n/2 \rceil \mid n) = \tfrac{1}{2}, \qquad \sigma^*(\lceil n/2 \rceil, \lfloor n/2 \rfloor \mid n) = \tfrac{1}{2}; \tag{20}$$

$$\forall\, n \geq 2\tau^*: \quad \sigma^*(\tau^*, \tau^* \mid n) = 1. \tag{21}$$

where $\tau^* \in \arg\max_{k \in \mathbb{Z}_+} kR(k)$. *The corresponding optimal prices* (p_1^*, p_2^*) *can be chosen so that the participation constraints* (J1) *and* (J2) *bind.*

(ii) There exists $(k_1^*(n), k_2^*(n)) \in \arg\max_{k_1, k_2 \in \mathbb{N}_+, k_1 + k_2 \leq n} k_1 R_1(k_1) + k_2 R_2(k_2)$ *for each* $n \in \Omega$, *such that, for* $i \in \{1,2\}$ *and* $-i \in \{1,2\} \setminus \{i\}$,

$$\frac{\mathbb{E}[k_{-i}^*(N) \cdot R_{-i}(k_{-i}^*(N))]}{\mathbb{E}[k_{-i}^*(N)]} \geq \frac{\mathbb{E}[k_i^*(N) \cdot R_{-i}(k_{-i}^*(N) + 1)]}{\mathbb{E}[k_i^*(N)]}, \ \forall\, i \in \{1,2\}, \tag{22}$$

where the expectation is taken with respect to $N \sim \mu_0$. In this case, there exists an optimal deterministic signaling mechanism σ^* satisfying $\sigma^*\big(k_1^*(n), k_2^*(n) \mid n\big) = 1, \forall\, n \in \Omega$. The corresponding optimal prices (p_1^*, p_2^*) can be chosen so that (J1) and (J2) bind.

Condition (22) holds if $n \geq \tau_1^* + \tau_2^*$ for all $n \in \Omega$, where $\tau_i^* \in \arg\max_{k \in \mathbb{Z}_+} kR_i(k)$.

Theorem 7 provides two sets of sufficient conditions under which a signaling mechanism with fixed, state-independent prices is globally optimal. The first condition requires the two leads to share the same reward function. This assumption is often reasonable: providers typically specialize in particular job categories, so the leads that attract their attention may be concentrated within a narrow category that plausibly shares a common reward structure. The second condition guarantees that an optimal mechanism can be implemented via a deterministic signaling rule that, for each realized n, recommends the pair $(k_1^*(n), k_2^*(n))$ maximizing $k_1 R_1(k_1) + k_2 R_2(k_2)$. This condition is satisfied when the leads are sufficiently popular that the platform always faces ample provider interest.

The proof parallels that of Theorem 4 and proceeds in two steps. First, we derive an upper bound on the platform's expected revenue over the most general policy space, which allows arbitrary signaling and state-dependent randomized pricing. Second, we show that under either condition in Theorem 7, the proposed mechanism is feasible for problem (two leads) and attains this upper bound, thereby establishing global optimality. The main analytical difficulty stems from *cross-lead incentive compatibility*: a provider recommended to purchase lead 1 must not prefer purchasing lead 2, and vice versa. These constraints are absent in the base model. Under the corresponding conditions in Theorem 7, we show that these constraints are satisfied by the proposed mechanisms. In Appendix E.3, we report additional numerical results for environments in which the conditions in Theorem 7 are violated. Nevertheless, we find that our main insight, namely, that $\mathsf{REV}^* = \mathsf{OPT}$, remains robust.

7. Concluding Remarks

Our paper investigates the use of information and pricing levers by a lead-selling platform to manage supply-side competition, a topic that is underexplored in the literature. We show that within the general space that simultaneously considering signaling mechanisms and the randomized state-dependent pricing, there exists a signaling mechanism combined with fixed (deterministic) pricing that is optimal. The state-contingent pricing—a commonly adopted policy within the above policy space—is hence outperformed in terms of platform revenue. The optimal signaling mechanism with fixed pricing can also be easily implemented by setting a limit on the maximum number of providers allowed to contact customers. This approach is already utilized by some platforms in practice, even though its optimality is not recognized, to the best of our knowledge.

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Appendices

We provide a microfoundation for the reward function $R(\cdot)$ via participating providers' price competition in Appendix A. Auxiliary results for Section 3 are provided in Appendix B. Auxiliary results for Sections 4 and 5 are collected in Appendix C. Auxiliary results for the dynamic setting are in Appendix D. Auxiliary results for the two-lead setting are in Appendix E. Extensions to welfare maximization and the heterogeneous-provider setting are in Appendix F. All proofs are collected in Appendix G.

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Appendix A A Micro-Foundation of the Reward Function via Price Competition

We now provide a micro-foundation for the reward function $R(\cdot)$ using a standard price-competition model with multinomial logit (MNL) demand, in which $R(\cdot)$ coincides with each provider's equilibrium profit. Specifically, suppose $n \in \mathbb{N}_+$ service providers purchase the same lead. If the customer contracts with provider $i \in \{1, \dots, n\}$ at an off-platform negotiated price p_i, her utility is

$$u_i \;=\; v - p_i + \varepsilon_i,$$

while her outside option (not hiring any provider) yields

$$u_0 \;=\; \varepsilon_0.$$

Here v is the customer's baseline value from having the job completed. The taste shocks $(\varepsilon_0, \varepsilon_1, \dots, \varepsilon_n)$ are i.i.d. type-I extreme value with cdf $F(\varepsilon) = \exp(-e^{-\varepsilon})$. We use the standard normalization: the location parameter is absorbed into v, and the scale parameter is normalized to one.

Each service provider incurs a cost c to complete the job. Given a price vector $p = (p_i)_{i=1}^n$, provider i's expected profit is

$$\pi_i(p) \;=\; (p_i - c)\, s_i(p),$$

where $s_i(p)$ denotes the probability that the customer chooses provider i under p. Under the MNL model, these choice probabilities are

$$s_i(p) := \mathbb{P}\{\text{the customer chooses provider } i\} \;=\; \frac{\exp(v - p_i)}{1 + \sum_{k=1}^n \exp(v - p_k)}, \quad i = 1, \dots, n,$$

$$s_0(p) := \mathbb{P}\{\text{the customer does not choose any provider}\} = \frac{1}{1+\sum_{k=1}^{n}\exp(v-p_k)}.$$

We focus on symmetric Nash equilibria of this pricing game. Let $p^*(n)$ denote the symmetric equilibrium price when there are n providers, and let

$$R^*(n) := (p^*(n)-c)\,s^*(n)$$

be the corresponding per-provider expected profit, where $s^*(n)$ is each provider's equilibrium demand share. We also write $T(n) := n\,R^*(n)$ for the total expected profit across all n providers. Finally, let $\delta := v-c$.

LEMMA A.1 (**Anderson and De Palma 1992**). *Under the logit price-competition model, for every $n \geq 1$ there exists a unique symmetric equilibrium price $p^*(n)$. Let $s^*(n) \in (0, 1/n)$ denote the corresponding equilibrium market share of each provider. Then $s^*(n)$ satisfies the scalar equation*

$$\frac{1}{1-s} = \delta + \log\Big(\frac{1}{s} - n\Big), \tag{A.1}$$

and the equilibrium markup and per-provider profit can be written as

$$p^*(n) - c = \frac{1}{1-s^*(n)}, \qquad R^*(n) = \frac{s^*(n)}{1-s^*(n)}. \tag{A.2}$$

A self-contained proof of Lemma A.1 is provided in Appendix G.4. Using Lemma A.1, we study how the equilibrium share $s^*(n)$, price $p^*(n)$, and profits $R^*(n)$ and total revenue $T(n)$ vary with the number of competitors n.

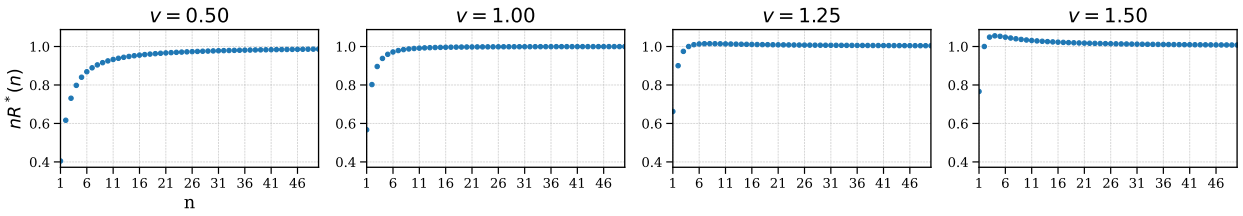
PROPOSITION A.1. *Under the logit price-competition model,*

(i) The equilibrium share $s^*(n)$, the markup $p^*(n)-c$, and the per-provider profit $R^*(n)$ are strictly decreasing in n.

(ii) The total profit $T(n)$ is unimodal *in n.*

The first part of Proposition A.1 is intuitive and straightforward to establish. However, to the best of our knowledge, we have not found an existing proof of the unimodality of the sellers' total profit $T(n)$. Proposition A.1 also continues to hold when n is treated as a continuous variable. The logit price-competition model above provides a concrete micro-foundation for our reward function $R(\cdot)$, which corresponds to the equilibrium per-provider profit $R^*(n)$. This profit exhibits two key properties: (i) it is decreasing in the number of competitors n, reflecting intensified competition; and (ii) the aggregate profit $T(n) = nR^*(n)$ is unimodal in n. Although we impose concavity rather than unimodality in Assumption 1(ii), most of our main results—including Theorem 3, Theorem 4, and all results in Appendix F.2—require only unimodality. We provide numerical results for $T(n)$ in Figure A.1, and find that when $v-c$ is not too large, $T(n)$ is indeed concave.

Figure A.1: Total Profit $T(n)$ under Logit Price Competition



Note. We normalize the provider cost to $c = 0$; the patterns are similar for $c > 0$.

Appendix B Auxiliary Results of Section 3

We now provide auxiliary results of Section 3, and their proofs are deferred to Appendix G.5.

B.1 Binding (join) Constraint in Problem ($\mathcal{P}_{\mathsf{LP}}$)

LEMMA B.2. *Given a price p_0 such that (4) holds, constraint (join) in problem ($\mathcal{P}_{\mathsf{LP}}$) is binding under any optimal signaling mechanism.*

Lemma B.2 suggests that, under an optimal signaling mechanism, the expected utility for any participating provider should be zero. This result may be somewhat expected, as the existence of providers experiencing strictly positive utility under a signaling mechanism implies that the platform could potentially increase the expected supply by recommending more providers to participate under certain state realizations. Consequently, the expected utility for all participating providers is diluted to zero at the point of optimality.

B.2 Algorithm to Find Critical Values in Theorem 1

In this section, we present an efficient algorithm to find the critical values $(\tau,\ k^*,\ \gamma)$ introduced in Theorem 1, which characterize the structure of the optimal signaling mechanism under an exogenous price p_0.

LEMMA B.3. *The critical values (τ, k^*, γ) introduced in Theorem 1 can be found by Algorithm 1 after at most $|\{n \in \Omega : n \geq R^{-1}(p_0)\}|$ iterations, and satisfy (B.3) and (B.4). Furthermore,*

(i) The RHS of (B.3) remains negative during the algorithm's run.

(ii) $\max\{n \in \Omega : n < \tau\} \leq k^* < \min\{n \in \Omega : n \geq \tau\}$.

Algorithm 1 presents an efficient method for finding the threshold τ, the recommended number k^*, and the probability of randomization γ in the optimal signaling mechanism characterized by Theorem 1. The algorithm first checks whether $\mathbb{E}[Nh(N, p_0)] = 0$ (recall by (4) this means checking whether $\mathbb{E}[h(\hat{N}, p_0)] = 0$). If so, the optimal signaling mechanism reduces to the trivial case of *no information*, as all interested providers would find it rational to purchase the lead based on their prior beliefs (so $\sigma^*(n \mid n) = 1$ for all $n \in \Omega$), see (4) and the discussion that follows. Otherwise, the

Algorithm 1 Compute the Critical Values in the Optimal Signaling Mechanism

Require: price p_0, reward function $R(\cdot)$, underlying states Ω, and prior belief μ_0

if $\mathbb{E}[Nh(N,p_0)]=0$ **then**

  return $\tau=+\infty$ ▷ In this case, $\sigma^*(n\mid n)=1$ for all $n\in\Omega$.

else if $\mathbb{E}[Nh(N,p_0)]<0$ **then**

  $\tau \leftarrow \min\{n\in\Omega : n\geq R^{-1}(p_0)\}$

  $k(\tau)\leftarrow$ the unique positive solution to equation (B.3):

$$kh(k,p_0)=\frac{\sum_{n<\tau,n\in\Omega}\mu_0(n)nh(n,p_0)}{-\sum_{n\geq\tau,n\in\Omega}\mu_0(n)}. \tag{B.3}$$

  while $k(\tau)\geq\tau$ **do**

    $\tau \leftarrow \min\{n\in\Omega : n>\tau\}$

    $k(\tau)\leftarrow$ solution k to (B.3)

  end while

  $k^*\leftarrow k(\tau)$

  $\gamma\leftarrow$ solution to the linear equation (B.4):

$$\gamma\cdot\lfloor k^*\rfloor\cdot h\left(\lfloor k^*\rfloor,p_0\right)+(1-\gamma)\cdot\left(\lfloor k^*\rfloor+1\right)\cdot h\left(\lfloor k^*\rfloor+1,p_0\right)=k^*h(k^*,p_0). \tag{B.4}$$

  return τ, k^* and γ

end if

structure of the optimal mechanism follows the description in Theorem 1, and the algorithm proceeds to determine τ, k^*, and γ.

The algorithm initializes τ with the simplest element in set Ω that is greater or equal to the break-even participation $R^{-1}(p_0)$ and progressively increases it until the optimal threshold is found. The corresponding $k^*=k(\tau)$ is determined by solving (B.3), based on the observation that constraint (join) must be binding at optimality, as shown in Lemma B.2. The initialization is based on the observation that the optimal threshold cannot be lower than the break-even participation $R^{-1}(p_0)$ as $h(R^{-1}(p_0),p_0)=0$. At optimality, the platform would recommend all providers to participate for any realized state $n<R^{-1}(p_0)$ due to the positive expected utility for providers to participate. The positive surplus will be redistributed across the unfavorable states when competition is intense, ensuring it is rational for providers to purchase the lead upon receiving a signal to participate. The while loop ensures the feasibility of the resulting mechanism from the algorithm, as the number of providers recommended to participate, $k(\tau)$, can not exceed the total number of interested providers n, for all $n\in\Omega, n\geq\tau$. Lastly, the probability of randomization γ is determined such that a convex combination of recommending $\lfloor k^*\rfloor$ and $\lfloor k^*\rfloor+1$ providers to participate, i.e., the LHS of (B.4), is equivalent to recommending k^* providers.

B.3 Further Discussions of the Structure of the Optimal Signaling Mechanism in Theorem 1

The information structure characterized in Theorem 1 shares some similarities with the "censorship" structure commonly observed in the literature (see Bergemann et al. 2022 and the references therein), where all states above (or below) a threshold are pooled, and all states below (or above) the threshold are fully revealed. In the context of our model, when there is a small pool of interested providers, a provider is more likely to participate and can thus earn a high reward, resulting in high information rents. Consequently, the censorship structure suggests that the platform pools all states when the supply competition is mild. This aligns with Theorem 1, as all providers are recommended to participate when $n < \tau$. However, the optimal signaling mechanism in Theorem 1 would not reveal complete information when the supply competition is intense. Instead, a fixed number of providers are recommended to purchase the lead, with the number being independent of the realized state n.

There are two drivers that collectively shape the structure of the optimal signaling mechanism. The first driver is the commonly observed phenomenon of *information obfuscation*, where favorable states (characterized by mild competition between a small number of interested providers) are pooled with unfavorable states (characterized by intense competition between a large number of interested providers). This pooling redistributes the positive surplus from favorable states in such a way that as many providers as possible are rational to follow the recommendation upon receiving a signal to participate under the optimal information mechanism. This explains why a signal to participate could be associated with any state in Ω, according to a provider's posterior belief.

The second driver is the providers' *inflated prior belief* discussed in Lemma 1. Naturally, one might expect that the number of recommended providers to participate would strictly *increase* with n. This is because, given state n and the recommended number k, a provider would be randomly selected with probability k/n, which strictly decreases as n increases. Thus, upon receiving a signal to participate, states with a greater number of interested providers would receive a lower weight in the provider's posterior belief, making it easier to recommend more providers to participate as n increases while still ensuring it is rational for providers to follow the recommendation. However, this effect is completely nullified by the providers' inflated prior belief, as can be seen from constraint (join)[15], leading to a *constant* number of recommended participants.

B.4 Comparative Statics

We now conduct a comparative static analysis of the threshold τ and the critical value k^*, as outlined in Theorem 1, to understand how these values vary with changes in the underlying states Ω, while keeping the prior distribution μ_0 and the commission price p_0 fixed.

[15]Specifically, refer to the discussion following Lemma 3 on the derivation of (join), where the probability k/n turns to k as n gets canceled by the n in the inflated prior belief $\mu_{\mathsf{P}}(n) = \frac{n\mu_0(n)}{\mathbb{E}_{N\sim\mu_0}[N]}$.

PROPOSITION B.2 **(Comparative Statics).** *Given a prior μ_0 and a price p_0, let $\tau(\Omega)$ and $k^*(\Omega)$ be the threshold and the critical value under the set Ω, respectively. As we change Ω to $\hat{\Omega}$ by increasing one element $n \in \Omega$ to $n+1$ and reassign the probability $\mu_0(n)$ to $\mu_0(n+1)$:*

(i) if $n > \tau(\Omega)$, $\tau(\Omega)$ and $k^*(\Omega)$ remain constant;

(ii) if $n = \tau(\Omega)$, $\tau(\Omega)$ increases and $k^*(\Omega)$ remains constant;

(iii) if $n < \tau(\Omega)$,

(a) *and $xh(x, p_0)$ is increasing on integers at $x = n$, then $\tau(\Omega)$ and $k^*(\Omega)$ increase;*

(b) *and $xh(x, p_0)$ is decreasing on integers at $x = n$, then $\tau(\Omega)$ decreases, and $k^*(\Omega)$ decreases if $n < \max\{n \in \Omega : n < \tau\}$.*

As shown in Theorem 1, the threshold $\tau(\Omega)$ serves as a pivotal element that divides the state space Ω into two mutually exclusive subsets. When the supply competition is mild ($n < \tau(\Omega)$), a provider's expected surplus is generally positive even if all interested providers participate. Consequently, all of them would be recommended to purchase the lead, and their positive surplus will be redistributed to unfavorable states, i.e., $n > \tau(\Omega)$, ensuring it is rational for providers to participate upon receiving a signal to join. For any $n > \tau(\Omega)$, the value of n does not affect $\tau(\Omega)$ and $k^*(\Omega)$; the optimal signaling mechanism remains the same as n increases, as shown in Proposition B.2. The case of $n = \tau(\Omega)$ is similar to that of $n > \tau(\Omega)$, except that the threshold $\tau(\Omega)$ increases as n increases. However, if $n < \tau(\Omega)$ increases, both $\tau(\Omega)$ and $k^*(\Omega)$ will change, and the direction of this change depends on whether $nh(n, p_0)$ increases or decreases with n. The term $nh(n, p_0)$ measures the expected surplus from all participating providers in state n. If $nh(n, p_0)$ increases with n, the platform is able to extract more surplus from participating providers in state n as n increases, allowing it to recommend more providers to participate when competition is intense, leading to greater $\tau(\Omega)$ and $k^*(\Omega)$.

B.5 Algorithm to Find the Critical Values in Theorem 2(iii).(b)

Algorithm 2 finds the critical values in Theorem 2(iii).(b) about the optimal public mechanism.

Algorithm 2 Greedy computation of (ω^*, α^*) for PLC in the Case (iii).(b) in Theorem 2

Require: Prior (Ω, μ_0), threshold r_0; set $K := e_0 - r_0 m_0$

1: $\Omega_+ \leftarrow \{n \in \Omega : \ n > r_0\}$; order $n_{(1)} > \cdots > n_{(L)}$, where $L := |\Omega_+|$

2: $b_i \leftarrow \mu_0(n_{(i)})\, n_{(i)}(n_{(i)} - r_0)$; $\quad C_j \leftarrow \sum_{i=1}^{j} b_i$

3: $j^* \leftarrow \min\{j : \ C_j \geq K\}$

4: Output: $\omega^* \leftarrow n_{(j^*)}$; $\quad \alpha^* \leftarrow (K - C_{j^*-1})/b_{j^*}$; $\quad \pi_H^*(n) = \mathbf{1}\{n > \omega^*\} + \alpha^* \mathbf{1}\{n = \omega^*\}$

B.6 Expected Supply under Full and No Information

In this section, we extend the discussions in Section 3.2 by comparing the expected supply under no and full information with exogenous prices. Given the price p_0, we denote the expected supply under the no-information and full-information scenarios as $\mathsf{Sp}(\text{no info}, p_0)$ and $\mathsf{Sp}(\text{full info}, p_0)$, respectively.

PROPOSITION B.3 **(No and Full Information)**. *Given the price p_0 with $\mathbb{E}[Nh(N,p_0)] \le 0$, for the linear reward function, we have:*

(i) Under no information, the equilibrium participation probability at all states is $f_{\mathsf{no}} = \frac{R^{-1}(p_0)-1}{\mathbb{E}[\hat{N}]-1}$, and the expected supply is $\mathsf{Sp}(\text{no info}, p_0) = f_{\mathsf{no}} \cdot \mathbb{E}[N]$.

(ii) Under full information, the equilibrium participation probability in state $n \in \Omega$ is $f(n) = 1$ if $n \le R^{-1}(p_0)$; otherwise, $f(n) = \frac{R^{-1}(p_0)-1}{n-1}$. The corresponding expected supply is

$$\mathsf{Sp}(\text{full info}, p_0) = \sum_{n\in\Omega, n\le R^{-1}(p_0)} \mu_0(n)\cdot n + \sum_{n\in\Omega, n> R^{-1}(p_0)} \mu_0(n)\cdot n \cdot \frac{R^{-1}(p_0)-1}{n-1}.$$

(iii) If $n \ge R^{-1}(p_0)$ for all $n \in \Omega$, we have $\mathsf{Sp}(\text{full info}, p_0) > \mathsf{Sp}(\text{no info}, p_0)$.

Proposition B.3 (i) characterizes the equilibrium and the expected supply under no information. With no information, all providers, irrespective of the pool size N, adopt the same strategy $f_{\mathsf{no}} \in [0,1]$, which makes them indifferent between participation and the outside option. That is

$$\sum_{n\in\Omega} \mu_{\mathsf{sp}}(n) \cdot \left[\underbrace{(1-f_{\mathsf{no}})}_{\text{not participate}} \cdot 0 + \underbrace{f_{\mathsf{no}}}_{\text{participate}} \cdot \mathbb{E}_{k\sim \mathsf{Bin}(n-1, f_{\mathsf{no}})}[R(1+k) - p_0] \right] = 0, \tag{B.5}$$

where the term $(1-f_{\mathsf{no}})\cdot 0$ represents the utility when the provider does not participate. When the provider participates with probability f_{no}, they gain the expected reward $\mathbb{E}_{k\sim \mathsf{Bin}(n-1,f_{\mathsf{no}})}[R(1+k)-p_0]$ conditional on the rest $n-1$ providers also adopt the strategy f_{no}. Considering $\mathbb{E}_{k\sim \mathsf{Bin}(n-1,f_{\mathsf{no}})}[k] = (n-1)f_{\mathsf{no}}$, by the linearity of $R(\cdot)$, (B.5) yields $\sum_{n\in\Omega} \mu_{\mathsf{p}}(n) \cdot R(1+(n-1)f_{\mathsf{no}}) = p_0$. Using the linear property of $R(\cdot)$ again, we have $R\left(1 + f_{\mathsf{no}} \cdot (\mathbb{E}[\hat{N}]-1)\right) = p_0$, which leads to $f_{\mathsf{no}} = (R^{-1}(0) - 1)/(\mathbb{E}[\hat{N}]-1)$, where $\mathbb{E}[\hat{N}] = \mathbb{E}[N^2]/\mathbb{E}[N]$ by Lemma 1. The corresponding expected supply (with respect to the platform's prior belief) is $\mathsf{Sp}(\text{no info}, p_0) = f_{\mathsf{no}} \cdot \mathbb{E}[N]$.

Proposition B.3 (ii) characterizes providers' equilibrium strategy under full information and the corresponding expected supply. Specifically, in a low state, all providers are willing to join since they have strictly positive utility. In a high state, the equilibrium strategy renders providers indifferent between participation and the outside option. Again, when deriving the equilibrium, the linearity of $R(\cdot)$ plays a key role in avoiding the complicated binomial formula.

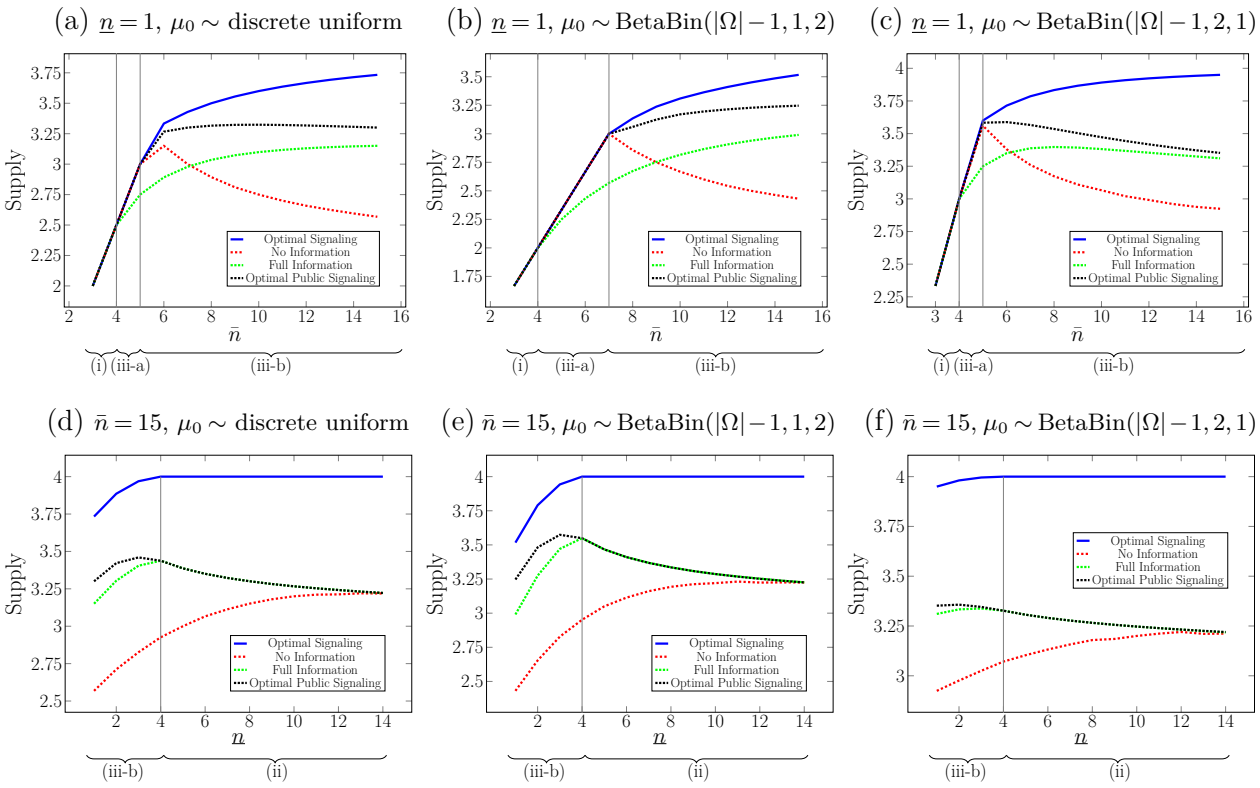
Interestingly, in some cases, no information is inferior to full information,[16] as illustrated in Proposition B.3 (iii). Intuitively, one might expect no information, which pools all states together with

[16]Note that it is possible that no information is better than full information. For example, under the condition $\mathbb{E}[Nh(N,p_0)] \ge 0$, no information is the optimal signal, which yields a higher supply than full information.

zero providers' surplus, to always yield a higher supply than full information, where providers enjoy strictly positive utilities in some states. The key reason for this counterintuitive outcome lies in the biased beliefs, as detailed in Lemma 1. With full information, providers are aware of the exact state, resulting in unbiased posterior beliefs. Conversely, under no information, providers' posterior beliefs align with their biased priors, leading them to anticipate a larger pool size and discouraging them from joining.

B.7 Numerical Comparisons

Figure B.2: Optimal (Private) Signaling Mechanism vs Public Mechanisms



Note. $\Omega = \{\underline{n}, ..., \bar{n}\}$, *Linear Reward* $R(x) = 5 - x$ *and* $p_0 = 1$. *Beta-binomial support is shifted to match* Ω. *The regions (along the horizontal axes) correspond to the cases in Theorem 2.*

Figure B.2 compares the optimal public mechanisms characterized in Theorem 2 with optimal private signaling mechanisms under linear rewards. It also reports two commonly used benchmarks: no information and full information, as derived in Appendix B.6. The patterns are consistent with the theoretical characterization in Theorem 2. In regions (i) and (iii-a), shown in Figure B.2 (a)–(c), the optimal private mechanism coincides with the optimal public mechanism. These regions are essentially the trivial cases: Inequality (4) does not hold, and the first-best expected supply is attainable even without informative signaling. By contrast, the optimal public mechanism and

the optimal (private) mechanism diverge in regions (ii) and (iii-b), where optimal private signaling can substantially outperform the optimal public mechanism. Interestingly, relative to providing no information, full disclosure typically yields higher expected supply when supply competition is more intense (i.e., when $\underline{n}$ and/or $\bar{n}$ is larger). This effect is driven primarily by providers' inflated prior beliefs.

Appendix C Auxiliary Results of Sections 4 and 5

We now provide auxiliary results of Sections 4 and 5.

C.1 Comparative Statics of Reward Functions

The optimal signaling mechanism and pricing depend critically on the reward function $R(\cdot)$, which abstracts the service provider's payoff associated with the job posted on the platform. In practice, job size can vary substantially—from a major project such as renovating a multi-bedroom house to a small task such as painting a door. We now study how the optimal mechanism, pricing, and revenue vary with job size, as captured by the reward function. Given a baseline reward function $R(\cdot)$, consider the additive family of functions $R_a(x) := R(x) + a$, parameterized by a scalar job-size parameter $a \geq 0$. Let $\tau^*(a)$ (resp. $p_0^*(a)$ and $\mathsf{REV}^*(a)$) denote the optimal threshold (resp. the optimal fixed fee and the optimal expected revenue) under the reward function R_a. By Assumption 1(ii), it is straightforward to verify that $\tau^*(a)$ is increasing in a.[17] It then follows that both

$$k^*(a) := \arg \max_{x \in \{\lfloor \tau^*(a) \rfloor, \lceil \tau^*(a) \rceil\}} x R_a(x) \tag{C.6}$$

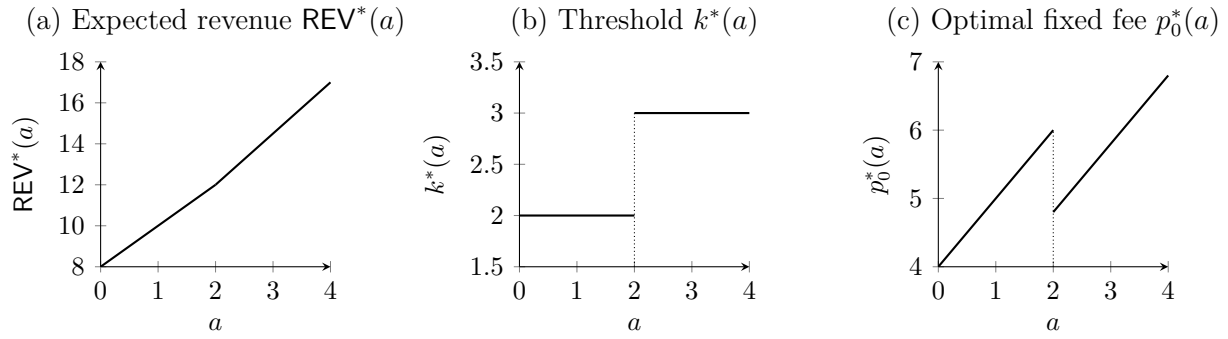
and $\mathsf{REV}^*(a)$ are increasing in a. This implies that for larger jobs it is optimal for the platform to allow more providers to participate, thereby increasing the platform's revenue. However, this does not necessarily mean that the platform should charge a higher fixed fee $p_0^*(a)$ for larger jobs. In general, $p_0^*(a)$ is *not* monotone in a, as the following example shows.

Example C.1. Consider a setting with two possible states for the number of interested providers, $\Omega = \{2, 3\}$, each occurring with probability $\mu_0(2) = \mu_0(3) = 1/2$. At the integer points $k \in \{1, 2, 3\}$, the baseline reward function satisfies $R(1) = 5$, $R(2) = 4$, and $R(3) = 2$. Figure C.3 illustrates how $\mathsf{REV}^*(a)$, $k^*(a)$, and the optimal price $p_0^*(a)$ vary with a. Consistent with the preceding discussion, both $\mathsf{REV}^*(a)$ and $k^*(a)$ are weakly increasing in a. In particular, $k^*(a)$ jumps from 2 to 3 at $a = 2$, which induces a non-monotonic change in $p_0^*(a)$ at that point. The intuition is as follows. When the job is relatively small (i.e., a is small), the platform recommends two providers purchase the lead in both states. In this regime, each provider's expected payoff increases with a, and the platform correspondingly raises the optimal fixed fee. Once the job becomes sufficiently valuable (specifically,

[17] Since $xR_a(x) = xR(x) + ax$, the derivative with respect to x is $R(x) + xR'(x) + a$, which is increasing in a. By strict concavity, the unique (continuous) maximizer $x^*(a)$ therefore increases with a.

when $a \geq 2$), however, the platform prefers to sell the lead to a third provider whenever three providers are interested. This change intensifies competition in the high-supply state, reducing each provider's expected payoff in that state. To maintain participation incentives, the platform must lower the fee at that point, generating the observed downward jump in $p_0^*(a)$ at $a = 2$.

Figure C.3: $\mathsf{REV}^*(a)$, $k^*(a)$, and $p_0^*(a)$ as functions of job size a.



Note. At $a = 2$, $k^*(a)$ defined in (C.6) can take either limiting value, and the same holds for $p_0^*(a)$.

C.2 Contingent Pricing under Linear Rewards

The following result explicitly characterizes the optimal contingent pricing under linear reward functions, complementing Proposition 1.

COROLLARY 2. *For a linear reward function $R(\cdot)$, the optimal contingent price is*

$$p^*(n) = \begin{cases} R(n), & n \leq \frac{R^{-1}(0)+1}{2}, \\ R\left(\frac{R^{-1}(0)+1}{2}\right), & n > \frac{R^{-1}(0)+1}{2}, \end{cases}$$

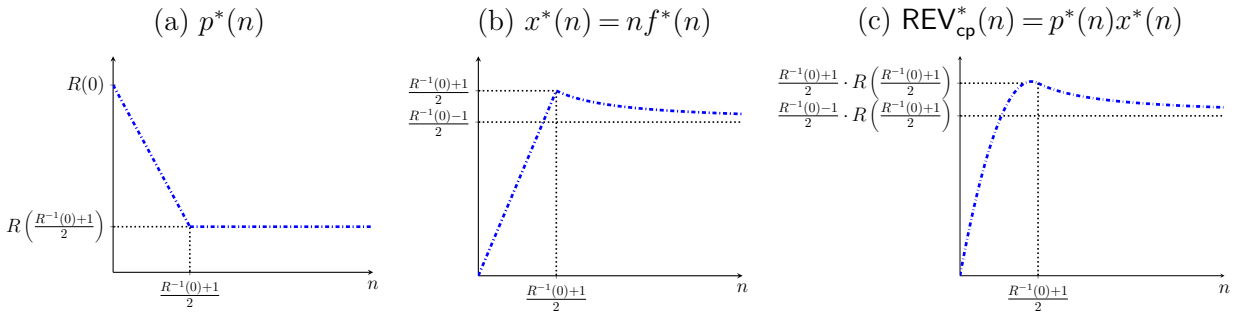
and the corresponding equilibrium participation probability and expected revenue are

$$f^*(n) = \begin{cases} 1, & n \leq \frac{R^{-1}(0)+1}{2}, \\ \frac{R^{-1}(0)-1}{2(n-1)}, & n > \frac{R^{-1}(0)+1}{2}, \end{cases} \quad \mathsf{REV}^*_{\mathsf{cp}}(n) = \begin{cases} R(n) \cdot n, & n \leq \frac{R^{-1}(0)+1}{2}, \\ R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \frac{R^{-1}(0)-1}{2(n-1)} \cdot n, & n > \frac{R^{-1}(0)+1}{2}. \end{cases}$$

Corollary 2 characterizes the optimal contingent pricing policy and the corresponding equilibrium participation probability and expected revenue in each state $n \in \Omega$, under linear reward functions. A geometric illustration is provided in Figure C.4. A few observations are in order: (i) All interested providers will participate surely when the supply competition is mild, i.e., the number of interested providers is below a certain value. This threshold value is $\frac{R^{-1}(0)+1}{2}$ under linear reward functions, which is roughly the midpoint of the numbers of interested providers that yield positive rewards. Above the threshold, the equilibrium participation probability decreases as the number of interested providers increases, and so does the equilibrium expected number of participating providers, see Figure C.4 (b). This is consistent with the empirical observations in Besbes et al. (2023), who demonstrate that providers are discouraged from purchasing the lead as competition intensifies. (ii) The

optimal price $p^*(n)$ (weakly) decreases in the number of interested providers. This is because as supply competition intensifies, any interested provider becomes more reluctant to participate, prompting the platform to lower the price to incentivize provider participation. (iii) The preceding two factors contribute to the pattern of the optimal expected revenue $\mathsf{REV}^*_{\mathsf{cp}}(n)$, as shown in Figure C.4 (c). $\mathsf{REV}^*_{\mathsf{cp}}(n)$ first increases quadratically and then decreases convexly toward a positive limit.

Figure C.4: Optimal Contingent Pricing under Linear Reward Functions



Note. The state n can only take integer values.

C.3 No Information with Fixed Pricing vs. Contingent Pricing

In this section, we first characterize the revenue under no information with the corresponding optimal state-independent price $p_0^*(\mathsf{no\ info})$. Then, we will compare it with the revenue under optimal contingent pricing policy. We find that when the expected states are either very low or uniformly high, the optimal contingent pricing policy outperforms the no-information policy with state-independent pricing. We use $\mathsf{REV}^*_{\mathsf{noinfo}}$ to denote the expected revenue under no information with the corresponding optimal fixed pricing.

PROPOSITION C.4 **(No Information with Fixed Pricing).** *The optimal fixed price under no information and the corresponding optimal revenue are* $p_0^*(\mathsf{no\ info}) = \mathbb{E}[\mathbb{E}_{k\sim\mathsf{Bin}(\hat{N}-1,f^*_{\mathsf{no}})}[R(1+k)]]$ *and* $\mathsf{REV}^*_{\mathsf{noinfo}} = p_0^*(\mathsf{no\ info}) \cdot \mathbb{E}[N] \cdot f^*_{\mathsf{no}}$*, respectively, where*

$$f^*_{\mathsf{no}} \in \arg\max_{f_{\mathsf{no}}\in[0,1]} \ \mathbb{E}[\mathbb{E}_{k\sim\mathsf{Bin}(\hat{N}-1,f_{\mathsf{no}})}[R(1+k)]] \cdot f_{\mathsf{no}}.$$

For the linear reward function, we have

(i) if $\mathbb{E}[\hat{N}] \le \frac{R^{-1}(0)+1}{2}$,

$$p_0^*(\mathsf{no\ info}) = R\left(\mathbb{E}[\hat{N}]\right), \quad \mathsf{REV}^*_{\mathsf{noinfo}} = R\left(\mathbb{E}[\hat{N}]\right) \cdot \mathbb{E}[N].$$

(ii) if $\mathbb{E}[\hat{N}] > \frac{R^{-1}(0)+1}{2}$,

$$p_0^*(\mathsf{no\ info}) = R\left(\frac{R^{-1}(0)+1}{2}\right), \quad \mathsf{REV}^*_{\mathsf{noinfo}} = R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \frac{R^{-1}(0)-1}{2} \cdot \frac{\mathbb{E}[N]}{\mathbb{E}[\hat{N}]-1}.$$

PROPOSITION C.5 **(No Information vs. Contingent Pricing)**. *For the linear reward function,* $\mathsf{REV}^*_{\mathsf{cp}} \geq \mathsf{REV}^*_{\mathsf{noinfo}}$ *if one of the following conditions holds*

(i) $\mathbb{E}[\hat{N}] \leq \frac{R^{-1}(0)+1}{2}$;

(ii) $n \geq \frac{R^{-1}(0)+1}{2}$ *for all* $n \in \Omega$.

C.4 Additional Numerical Results of Section 5

We now numerically compare the second-best revenue, i.e., REV^*, with the expected revenue under optimal contingent pricing, under different reward functions and underlying states. In our comparison, we also include another commonly adopted benchmark: the no-information case, where the platform shares no information about the number of interested providers and charges a fixed price. Its expected revenue is denoted as $\mathsf{REV}^*_{\mathsf{noinfo}}$, which is characterized by Proposition C.4 in Appendix C.3. The comparison results are demonstrated by Figure C.5.

Figure C.5: Revenue Comparison under $\Omega = [\bar{n}]$ and Uniform Prior μ_0

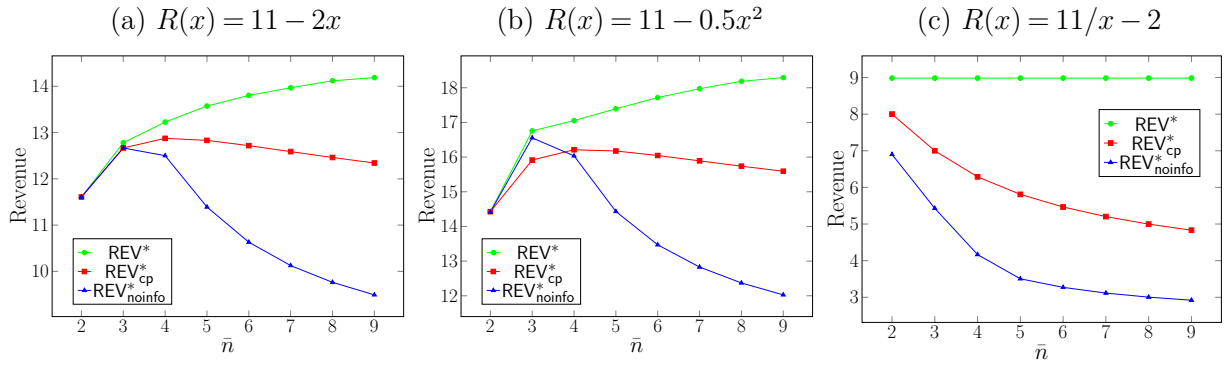


Figure C.5 (a) illustrates the case when the reward function is linear. In particular, the unique maximizer of $xR(x)$ is $\tau^* = 2.75$. Consequently, when the supply competition is mild, with all states being less than or equal to τ^* (e.g. $\bar{n} = 2$), the expected revenues from the optimal signaling mechanism with fixed pricing and optimal contingent pricing are identical. This is consistent with the preceding discussion in Section 5.2.1. Not surprisingly, the expected revenue from the no-information policy is also the same.

As $\bar{n}$ increases, Ω includes a greater number of states that exceed τ^* with higher probability mass. As discussed in Section 5.2.1, this is where the expected revenues from the optimal signaling mechanism with fixed pricing and the optimal contingent pricing diverge. The latter in state n, $\mathsf{REV}^*_{\mathsf{cp}}(n)$, decreases with n, while the former in state n, $\mathsf{REV}^*(n)$, remains constant for $n > \tau^*$. The ratio between the expected revenues from the two policies converges to the bound characterized in Corollary 1, as $\bar{n} \to \infty$.

Interestingly, the discrepancy between the expected revenues from the optimal contingent pricing and the no-information policy also widens with $\bar{n}$. This can be explained by the inflated belief from providers, as established in Lemma 1. As the supply competition intensifies, concealing information from providers, as in the no-information policy, becomes more detrimental. This is because states with a higher number of interested providers are inflated to a greater extent, significantly discouraging provider participation. In such cases, sharing full information, along with adjusting prices in response to the intensity of supply competition, can significantly improve the platform's revenue. We provide further discussion about the relative performance of the optimal contingent pricing policy and the optimal no-information policy in Appendix C.3.

Figure C.5 (b) and (c) show that the preceding observations are robust under alternative functional forms for the reward. One noteworthy point is that the expected revenue from the optimal signaling mechanism with fixed pricing remains constant when the reward function is $R(x) = 11/x - 2$. This is because $xR(x)$ is linear with a negative slope, resulting in $\tau^* = 0$ and constant expected revenue at any state n, by Theorem 3.

Our further numerical experiments demonstrate that the strong performance of our proposed mechanism in Theorem 3 extends well under other reward structures. Across power, root-logarithmic, and logarithmic reward functions (each satisfying unimodality of $xR(x)$) presented in Figure C.6, Figure C.7, and Figure C.8 respectively, our analysis shows that both contingent pricing ($\mathsf{REV}^*_{\mathsf{cp}}$) and no information with single pricing ($\mathsf{REV}^*_{\mathsf{noinfo}}$) indicate pronounced sensitivity to market parameters $(\bar{n}, \mu_0)$. Notably, their performance deteriorates sharply as competition intensifies. In contrast, our signaling mechanism from Theorem 3 remains mostly insensitive to these shifts. Besides, $\mathsf{REV}^*_{\mathsf{noinfo}}$ and $\mathsf{REV}^*_{\mathsf{cp}}$ are always smaller than REV^*. Together, these results highlight the robustness and broad applicability of our policy compared to other benchmarks.

Appendix D Auxiliary Results of Dynamic Setting in Section 6.1

We now provide auxiliary results of Section 6.1 about the dynamic setting.

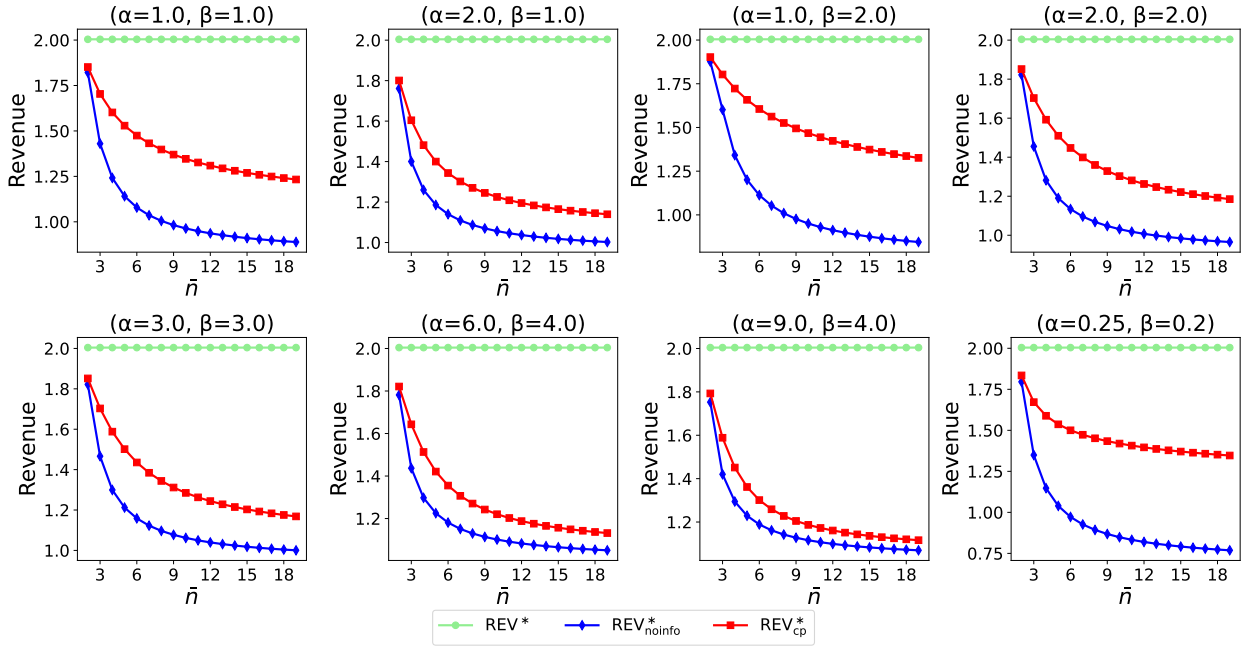
D.1 Second Stage

We now formalize the mechanisms in Theorem 5 as follows: the platform (i) fully discloses the realized number of stage-1 participants $N_1 = n_1$; (ii) charges an n_1-dependent stage-2 price $p_2(n_1)$; and (iii) provides information about the total number of stage-2 potential providers N_2 via private signals.

Because N_1 and N_2 are independent, the same revelation-principle-type argument as in the base model implies that optimizing over this class is equivalent to solving, for each fixed n_1, the following problem:

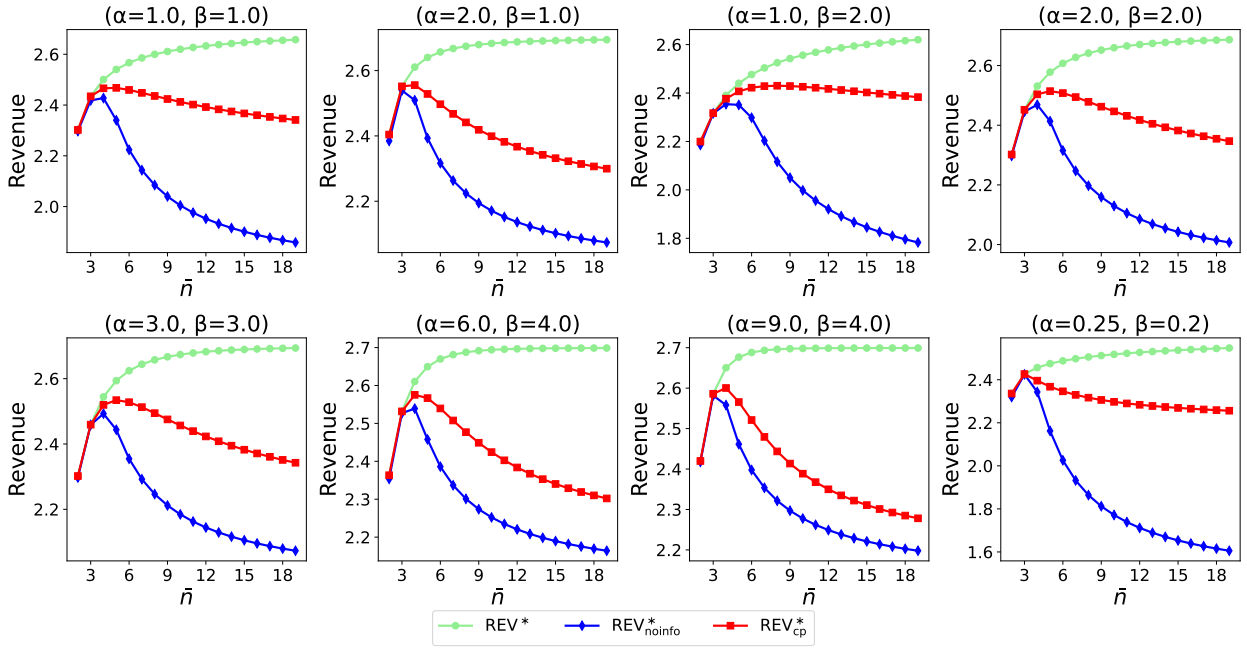
$$\max_{\{\sigma_2(\cdot|n_1,n_2)\},\, p_2(n_1)} p_2(n_1) \cdot \sum_{n_2 \in \Omega_2} \mu_2(n_2) \sum_{k=0}^{n_2} k\, \sigma_2(k \mid n_1, n_2) \qquad \text{(Stage-2 problem given } n_1\text{)}$$

Figure C.6: Revenue Comparisons under Power-Form Reward Functions



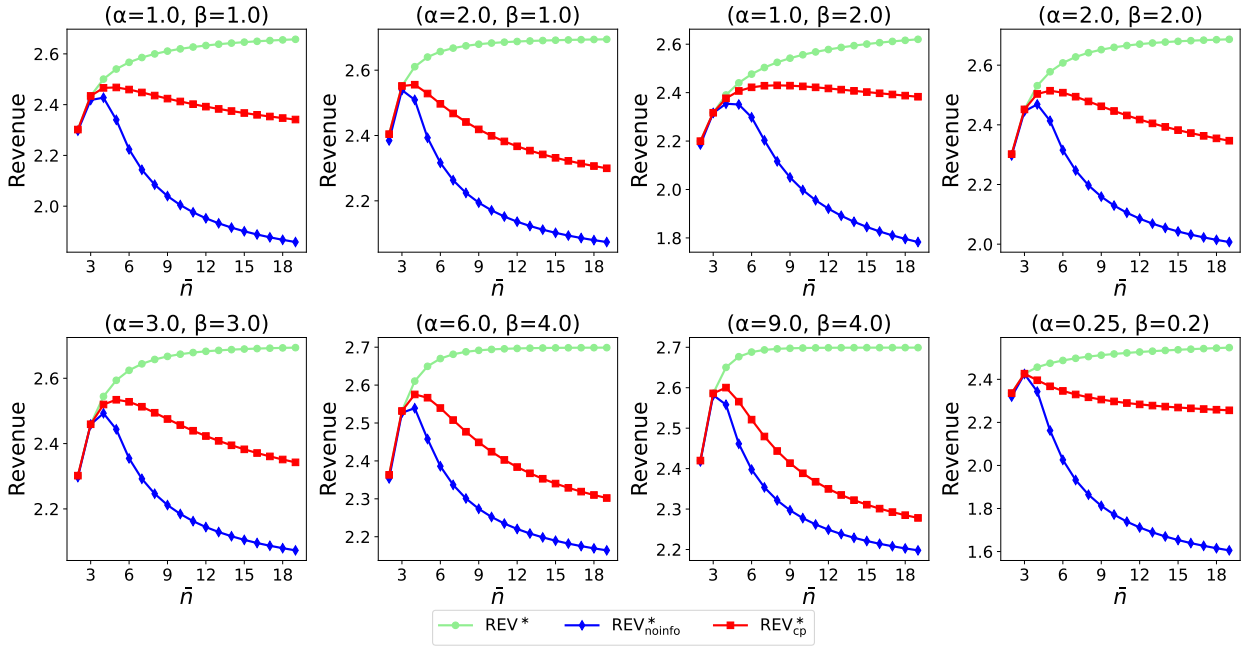
Note. Across panels: $R(x) = \exp(1.6 - 0.5x) - 1$, $\Omega_0 = [\bar{n}]$, $\mu_0 \sim BetaBin(|\Omega| - 1, \alpha, \beta)$.

Figure C.7: Revenue Comparisons under Root-Logarithmic-Form Reward Functions



Note. Across panels: $R(x) = 1.5 - \sqrt{\ln(1+x)}$, $\Omega_0 = [\bar{n}]$, $\mu_0 \sim BetaBin(|\Omega| - 1, \alpha, \beta)$.

Figure C.8: Revenue Comparisons under Logarithmic-Form Reward Functions



Note. Across panels: $R(x) = 2 - \ln(0.005 + x)$, $\Omega_0 = [\bar{n}]$, $\mu_0 \sim BetaBin(|\Omega| - 1, \alpha, \beta)$.

$$\text{s.t.} \quad \sum_{n_2 \in \Omega_2} \mu_2(n_2) \sum_{k=0}^{n_2} \sigma_2(k \mid n_1, n_2)\, k \left[R_2(k + n_1) - p_2(n_1)\right] \geq 0 \tag{J2}$$

$$\sum_{n_2 \in \Omega_2} \mu_2(n_2) \sum_{k=0}^{n_2} \sigma_2(k \mid n_1, n_2)\, (n_2 - k) \left[R_2(k + 1 + n_1) - p_2(n_1)\right] \leq 0 \tag{NJ2}$$

$$\sigma_2(\cdot \mid n_1, n_2) \in \Delta([n_2]_+), \quad \forall\, n_2 \in \Omega_2. \tag{Feas}$$

Formally, Constraint (J2) requires that a supplier who is recommended to join in stage 2 earns a nonnegative expected payoff, while constraint (NJ2) requires that a supplier who is recommended not to join has no profitable deviation to purchasing the lead.

D.2 First Stage

By a similar revelation-principle-type argument, the platform's stage-1 problem within the class of signaling with state-independent pricing is

$$\max_{\{\sigma_1(\cdot|n_0)\},\, p_1} \quad \sum_{n_0 \in \Omega_0} \mu_0(n_0) \sum_{n_1=0}^{n_0} \sigma_1(n_1 \mid n_0) \left[n_1 p_1 + \left(1 - \gamma(n_1)\right) \mathsf{REV}_2^*(n_1)\right]$$

(Stage-1 problem given REV_2^*)

$$\text{s.t.} \quad \sum_{n_0 \in \Omega_0} \mu_0(n_0) \sum_{n_1=0}^{n_0} \sigma_1(n_1 \mid n_0)\, n_1 \left[\widetilde{R}_1(n_1) - p_1\right] \geq 0 \tag{J1}$$

$$\sum_{n_0 \in \Omega_0} \mu_0(n_0) \sum_{n_1=0}^{n_0} \sigma_1(n_1 \mid n_0)\, (n_0 - n_1) \left[\widetilde{R}_1(n_1 + 1) - p_1\right] \leq 0 \tag{NJ1}$$

$$\sigma_1(\cdot \mid n_0) \in \Delta([n_0]_+), \quad \forall\, n_0 \in \Omega_0. \tag{Feas1}$$

The signaling rule σ_1 has the same interpretation as in the base model, and the structure of (Stage-1 problem given REV_2^*) is analogous to ($\mathcal{P}_{\mathsf{LP}}$). The only differences are that the stage-1 reward function $R_1(\cdot)$ is replaced by its effective counterpart $\widetilde{R}_1(\cdot)$, and the objective includes the continuation revenue from stage 2, $(1-\gamma(n_1))\mathsf{REV}_2^*(n_1)$.

For each $n_1 \in \Omega_1$, let $\tau^*(n_1) \in \arg\max_{k\in\mathbb{Z}_+} \; k\,R_2(k+n_1)$ denote the smallest (unconstrained) revenue-maximizing number of stage-2 participants given stage-1 participation level n_1. Let $\underline{n}_2 := \min_{n\in\Omega_2} \; n$ and $\bar{n}_2 := \max_{n\in\Omega_2} \; n$.

Lemma D.4 **(Sufficient conditions for unimodality of (18)).** *Assume that the stage-1 close probability is constant, i.e., $\gamma(n_1) \equiv \gamma \in [0,1]$ for all $n_1 \in \Omega_1^*$. Then (18) is unimodal if any one of the following conditions holds:*

(i) *Supply-constrained stage 2:*

$$\bar{n}_2 \;\le\; \min_{n_1\in\Omega_1^*} \tau^*(n_1). \tag{D.7}$$

(ii) *Supply-sufficient stage 2: There exists an integer $K \ge 0$ such that*

$$\tau^*(n_1) = K, \quad \forall\, n_1 \in \Omega_1^*, \tag{D.8}$$

and stage-2 supply is always at least this target:

$$\underline{n}_2 \;\ge\; K. \tag{D.9}$$

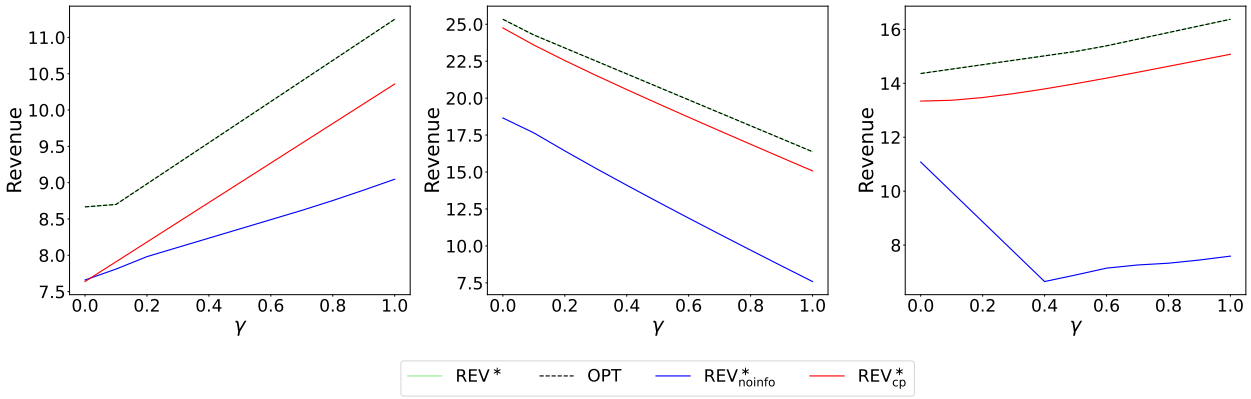
(iii) *High stage-1 close probability: $\gamma \in [\bar{\gamma}, 1]$ for some sufficiently large $\bar{\gamma} \in [0,1)$.*

The first condition corresponds to the case in which provider interest in stage 2 is limited, so it is always optimal for the platform to recommend that all stage-2 providers purchase the lead. The second condition has two parts. First, the maximizer of $kR_2(k+n_1)$ is independent of n_1, which can occur when R_2 is (approximately) geometric/exponential, e.g., $R_2(x) = q^x$ for some $q < 1$. Second, there is sufficient stage-2 interest, i.e., $\underline{n}_2 \ge K$, so that the platform can always identify K providers and recommend them to purchase the lead in stage 2. The third case corresponds to situations in which the lead is highly likely to close after stage 1—for instance, when customers are impatient and/or jobs are urgent. Note that if the close probability equals one, the two-stage model reduces to the base model. Here, we show that the same structural results extend to settings with sufficiently high close probabilities.

Note that the unimodality condition in (18) is sufficient for Theorem 6, but not necessary. To assess the robustness of Theorem 6, we report extensive numerical results below allowing for possibly non-constant close probabilities $\gamma(\cdot)$.

Figure D.9 compares the platform's revenue under different mechanisms as the parameter γ varies. The REV^* corresponds to the (total) revenue under the optimal signaling-and-pricing policy in

Figure D.9: Revenue Comparison in the Dynamic Setting for Different Levels of γ



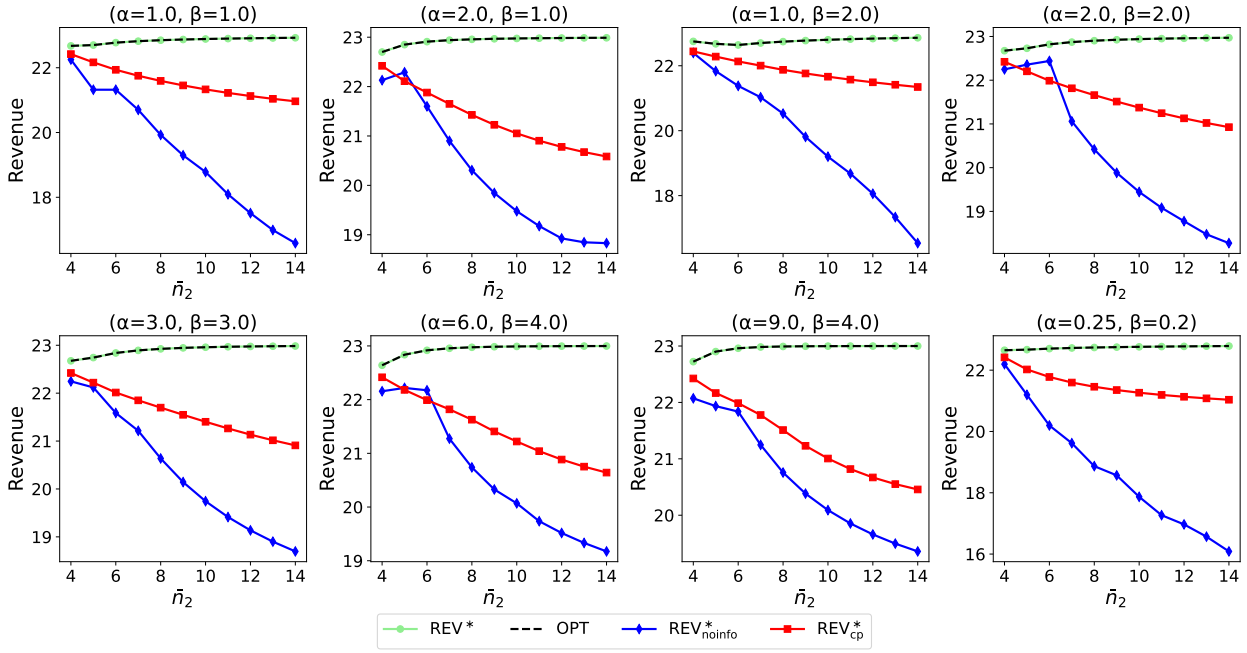
Note. Across panels: we assume that γ is independent of n_1. $\Omega_0 = \{2,3,6,10\}$, $\mu_0 = \{1/4, 2/4, 1/8, 1/8\}$, $\Omega_2 = \{1,2,3\}$, $\mu_2 = \{1/3, 1/3, 1/3\}$. *For panel (a):* $R_1(x) = 10 - 0.5x^2$, $R_2(x) = 6 - 0.5x^2$, *For panel (b):* $R_1(x) = 10 - 0.5x^2$, $R_2(x) = 6 - 0.001x^4$. *For panel (c):* $R_1 = 10 - 0.5x^2$, $R_2 = 6 - 0.01x^4$.

(Stage-1 problem given REV_2^*): in the first stage, the platform uses a signaling mechanism together with a market-size-independent price (i.e., a single posted price); in the second stage, the platform fully reveals the number of first-stage participants and posts prices that depend on the first-stage participants but are independent of the second-stage market size, together with the corresponding signaling mechanism. We contrast this with $\mathsf{REV}^*_{\mathsf{noinfo}}$, which denotes the (total) revenue under a no-information disclosure with optimally chosen but fixed (state-independent) prices in both stages. We also report $\mathsf{REV}^*_{\mathsf{cp}}$, which corresponds to the (total) revenue under the full information disclosure and fully state-dependent prices, where the platform reveals (n_1, n_2) and charges $p_2(n_1, n_2)$ in the second stage, and reveals n_0 and charges $p_1(n_0)$ in the first stage. Finally, OPT represents the second-best revenue, i.e., the platform's optimal revenue under the most general decision spaces $\mathcal{M}_1$ and $\mathcal{M}_2$.

From Figure D.9, we observe that OPT is achieved by REV^*, illustrating the robustness of our result in Theorem 6, even when the reward functions are asymmetric. Also, as shown in Figure D.9, the revenue under REV^* is usually strictly higher than both $\mathsf{REV}^*_{\mathsf{cp}}$ and $\mathsf{REV}^*_{\mathsf{noinfo}}$, highlighting the benefit of information design. Moreover, under each mechanism, the optimal revenue may either decrease or increase with the probability γ, depending on the specific reward functions.

Figure D.10 compares the platform's revenue under different mechanisms when the second-stage state space varies. Specifically, we let $\Omega_2 = [\bar{n}_2]$ and vary $\bar{n}_2$ in the figure. We observe a similar insight: REV^* coincides with OPT, illustrating the robustness of the result in Theorem 6. Figure D.11 further illustrates this robustness when varying the first-stage state space. Finally, Figure D.12 considers cases in which γ depends on the number of service providers that purchase the lead in the first stage, and shows a similar result.

Figure D.10: Revenue Comparison in the Dynamic Setting under Different Second-Stage States



Note. $\gamma = 0.1$, $R_1(x) = \frac{10}{x+1}$, $R_2(x) = 10 - x$, $\Omega_0 = \{1, 2, 3\}$, $\mu_0 = \{1/4, 2/4, 1/4\}$, $\Omega_2 = [\bar{n}_2]$, $\mu_2 = \text{BetaBin}(\alpha, \beta)$.

Appendix E Auxiliary Results of Two-Lead Setting in Section 6.2

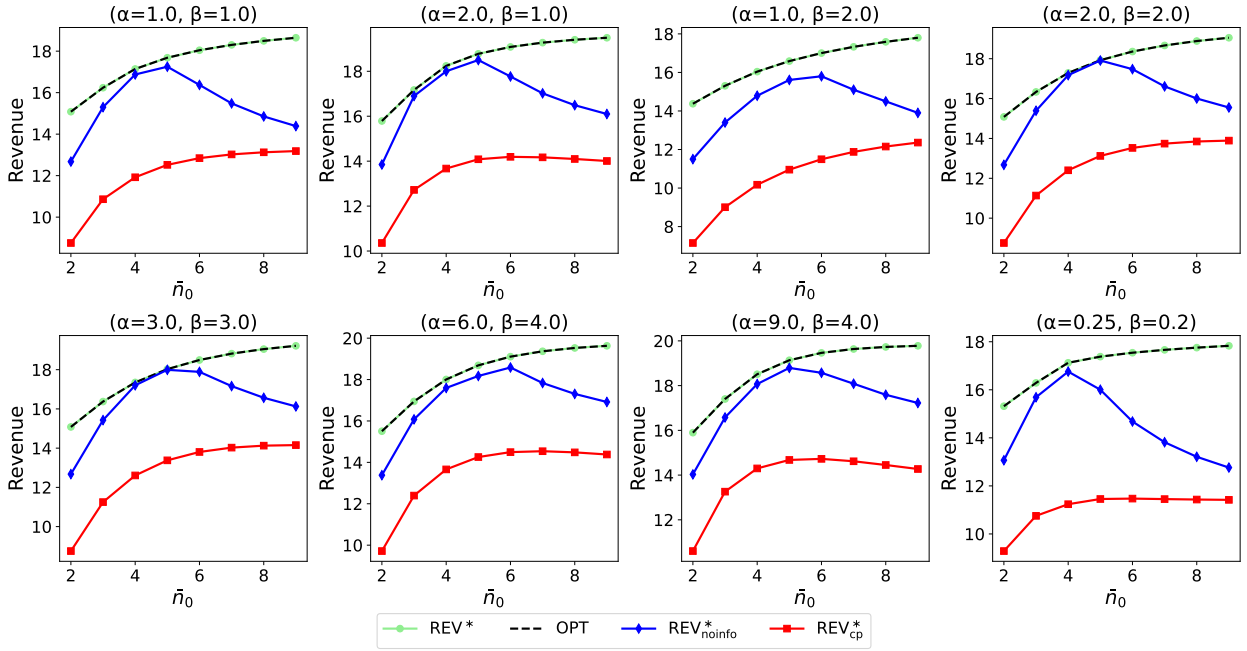
In this section, we provide additional discussion and results about the two-lead setting.

E.1 Problem Formulation

Consider a class of mechanisms that is optimal in the single-lead setting: the platform designs a signaling policy about the number of potential providers N and charges a fixed (state-independent) price vector (p_1, p_2) for the two leads. Theorem 7 provides sufficient conditions under which this restriction is without loss of optimality.

Among this space of mechanisms, by a revelation-principle-type argument analogous to that in the base model, we can, without loss of optimality, restrict attention to obedient recommendation mechanisms. Let $\sigma(k_1, k_2 \mid n)$ denote the probability of recommending $k_1 \geq 0$ providers to purchase lead 1 and $k_2 \geq 0$ providers to purchase lead 2, subject to $k_1 + k_2 \leq n$, when the realized number of potential providers is n. In this two-lead setting, we have new *cross-lead* incentive-compatibility constraints: a provider recommended to purchase lead 1 should not strictly prefer to purchase lead 2 instead, and symmetrically, a provider recommended to purchase lead 2 should not prefer to switch to lead 1. We emphasize that in the two-lead setting (k_1, k_2) are drawn *jointly*. Thus, it is not a priori obvious that the platform's optimization problem can still be cast in a form analogous to the linear

Figure D.11: Revenue Comparison in the Dynamic Setting under Different First-Stage States



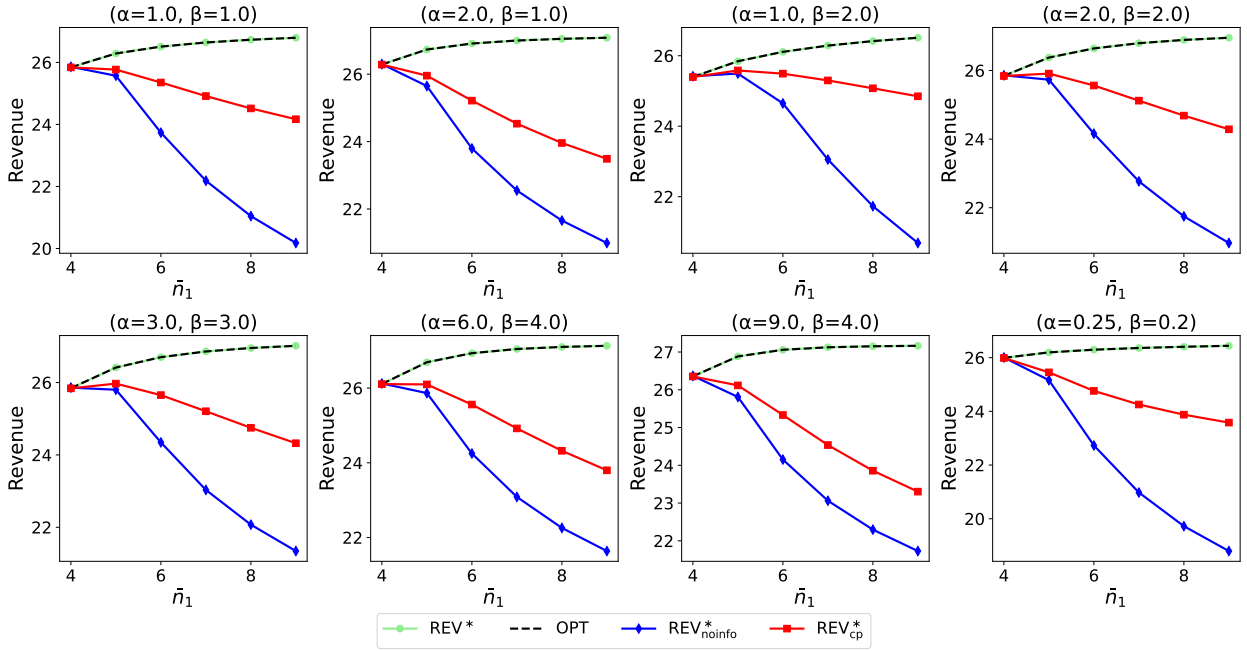
Note. $\gamma = 0.5$, $R_1(x) = 10 - 0.2x^2$, $R_2(x) = 10 - 1.5x$, $\Omega_2 = \{1, 2, 3\}$, $\mu_2 = \{1/4, 2/4, 1/4\}$, $\Omega_0 = [\bar{n}_0]$, $\mu_0 = \text{BetaBin}(\alpha, \beta)$.

program ($\mathcal{P}_{\text{LP}}$) in the base model. We now show that this insight continues to apply in the present setting.

LEMMA E.5. *Solving problem* (two leads) *yields a revenue-maximizing signaling mechanism and state-independent prices* (p_1^*, p_2^*) *in the two-lead setting.*

$$\begin{aligned}
\mathsf{REV}^* := \max_{\sigma, p_1, p_2} \ & \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot [p_1 k_1 + p_2 k_2] && \text{(two leads)} \\
\text{s.t.} \ & \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot k_1 \left[R_1(k_1) - p_1\right] \ge 0, && \text{(J1)} \\
& \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot k_1 \left[R_1(k_1) - p_1 - R_2(k_2 + 1) + p_2\right] \ge 0, && (1 \nrightarrow 2) \\
& \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot k_2 \left[R_2(k_2) - p_2\right] \ge 0, && \text{(J2)} \\
& \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot k_2 \left[R_2(k_2) - p_2 - R_1(k_1 + 1) + p_1\right] \ge 0, && (2 \nrightarrow 1) \\
& \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot (n - k_1 - k_2) \left[R_1(k_1 + 1) - p_1\right] \le 0, && \text{(NJ1)} \\
& \sum_{n \in \Omega} \mu_0(n) \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) \cdot (n - k_1 - k_2) \left[R_2(k_2 + 1) - p_2\right] \le 0, && \text{(NJ2)} \\
& \sum_{k_1 + k_2 \le n} \sigma(k_1, k_2 \mid n) = 1, \quad \forall n \in \Omega. && \text{(Feas)}
\end{aligned}$$

Figure D.12: Revenue Comparison in the Dynamic Setting when $\gamma(n_1)$ is Not Constant



Note. $\gamma(n_1) = 1 - 0.2^{n_1}$, $R_1(x) = 10 - 0.2x^2$, $R_2(x) = 10 - 1.5x$, $\Omega_2 = \{1, 3, 6\}$, $\mu_2 = \{1/4, 2/4, 1/4\}$, $\Omega_0 = \{3, ..., \bar{n}\}$, $\mu_0 = \text{BetaBin}(\alpha, \beta)$.

E.2 Contingent Pricing in the Setting of Two Leads

We now discuss the contingent pricing mechanism in the two-lead setting. Under contingent pricing, the platform posts state-dependent prices $p_i(n)$ and fully reveals n. All interested suppliers observe $(n, p_1(n), p_2(n))$ and adopt a common mixed strategy $(f_1(n), f_2(n))$: with probabilities $f_i(n) \in [0,1]$ they buy lead i, and with probability $1 - f_1(n) - f_2(n)$ they do not buy; hence $f_1(n) + f_2(n) \leq 1$.

To obtain an analytical characterization, we assume a linear reward function in this subsection. Specifically, lead i has decreasing linear reward

$$R_i(m) = r_i - c_i\, m, \qquad r_i > 0,\ c_i > 0, \qquad m \in \mathbb{Z}_{\geq 1}.$$

Assume $R_i(1) > 0$, i.e., $r_i > c_i$ to avoid trivial cases.

When all suppliers follow the common mixed strategy (f_1, f_2), at state n, a supplier who buys lead i faces expected utility

$$u_i(n; f_i(n), p_i(n)) = \mathbb{E}_{K_i \sim \text{Bin}(n-1, f_i(n))}[R_i(1 + K_i)] - p_i(n) = (r_i - c_i) - c_i(n-1) f_i(n) - p_i(n),$$

which is strictly decreasing in $f_i(n)$. In a symmetric equilibrium, if $0 < f_i(n) < 1 - f_{-i}(n)$ then $u_i(\cdot) = 0$, i.e., suppliers are indifferent between purchasing lead i and outside option; if $f_i(n) = 0$ then $u_i(\cdot) \leq 0$; if $f_i(n) = 1 - f_{-i}(n)$ then $u_i(\cdot) \geq 0$.

For a given state n, the platform's expected revenue equals

$$\mathsf{REV}_{\mathsf{cp}}(n) = n\big[p_1(n)f_1(n) + p_2(n)f_2(n)\big].$$

As in the single-lead case, prices outside $[R_i(n), R_i(1)]$ are weakly dominated whenever lead i sells with positive probability. It can be shown that at optimal prices, suppliers must be indifferent between outside option and purchasing lead i; otherwise, the platform can always increase the price without discouraging joining decision. That is, $u_i(n; f_i(n), p_i(n))$ must be zero at optimal contingent pricing. Then, we have the following state-wise optimization problem: for each $n \in \Omega$,

$$\max_{\substack{f_1,f_2\in[0,1]\\ f_1+f_2\le 1}} \; n\sum_{i=1}^{2} f_i \cdot \mathbb{E}_{K_i\sim \mathrm{Bin}(n-1,f_i(n))}\big[R_i(1+K_i)\big] \;=\; \max_{\substack{f_1,f_2\in[0,1]\\ f_1+f_2\le 1}} \; n\sum_{i=1}^{2}\Big[(r_i-c_i)f_i - c_i(n-1)f_i^2\Big]. \quad \text{(E.10)}$$

Given any feasible pair (f_1, f_2), the platform can implement it by setting the prices

$$p_i(n) = \mathbb{E}_{K_i\sim \mathrm{Bin}(n-1,f_i(n))}\big[R_i(1+K_i)\big] = (r_i - c_i) - c_i(n-1)f_i(n), \quad \text{(E.11)}$$

which makes suppliers just indifferent between buying lead i and their outside option. If $f_i(n) = 0$, any price $p_i(n) \ge R_i(1) = r_i - c_i$ deters profitable deviations to lead i. In particular, in a pure corner with $f_i(n) = 1$ and $f_{-i}(n) = 0$, we have $\mathbb{E}_{K_i\sim \mathrm{Bin}(n-1,f_i(n))}[R_i(1+K_i)] = R_i(n) = r_i - c_i n$. (The case $n = 1$ is degenerate: there is a single supplier and the platform simply offers the more profitable lead at price $R_i(1)$, and sets the other price high enough so that the supplier will not deviate; we omit the details.)

Define

$$\lambda(n) := \frac{\frac{r_1}{c_1} + \frac{r_2}{c_2} - 2n}{\frac{1}{c_1} + \frac{1}{c_2}} = \frac{r_1c_2 + r_2c_1 - 2nc_1c_2}{c_1 + c_2}.$$

PROPOSITION E.6 **(Two-lead contingent pricing under linear rewards)**. *Under linear reward functions $R_i(x) = r_i - c_i x$, we have the optimal contingent pricing, the corresponding equilibrium joining probability, and the revenue as follows:*

(i) If $n > \frac{1}{2}\left(\frac{r_1}{c_1} + \frac{r_2}{c_2}\right)$, we have

$$f_i^*(n) = \frac{r_i - c_i}{2c_i(n-1)}, \qquad p_i^*(n) = \frac{r_i - c_i}{2},$$

where the equilibrium joining probability satisfies $f_1^*(n) + f_2^*(n) < 1$. The corresponding revenue is

$$\mathsf{REV}_{\mathsf{cp}}^*(n) = \frac{n}{4(n-1)}\sum_{i=1}^{2}\frac{(r_i - c_i)^2}{c_i}.$$

(ii) If $n \le \frac{1}{2}\left(\frac{r_1}{c_1} + \frac{r_2}{c_2}\right)$ and $\lambda(n) < \min\{r_1 - c_1,\ r_2 - c_2\}$, we have

$$f_i^*(n) = \frac{r_i - c_i - \lambda(n)}{2c_i(n-1)} \in (0,1), \qquad p_i^*(n) = \frac{r_i - c_i + \lambda(n)}{2},$$

where the equilibrium joining probability satisfies $f_1^*(n)+f_2^*(n)=1$. *The corresponding revenue is*

$$\mathsf{REV}^*_{\mathsf{cp}}(n)=\frac{n}{4(n-1)}\left(\sum_{i=1}^{2}\frac{(r_i-c_i)^2}{c_i}\;-\;\lambda^2(n)\Big(\frac{1}{c_1}+\frac{1}{c_2}\Big)\right).$$

(iii) If [18]

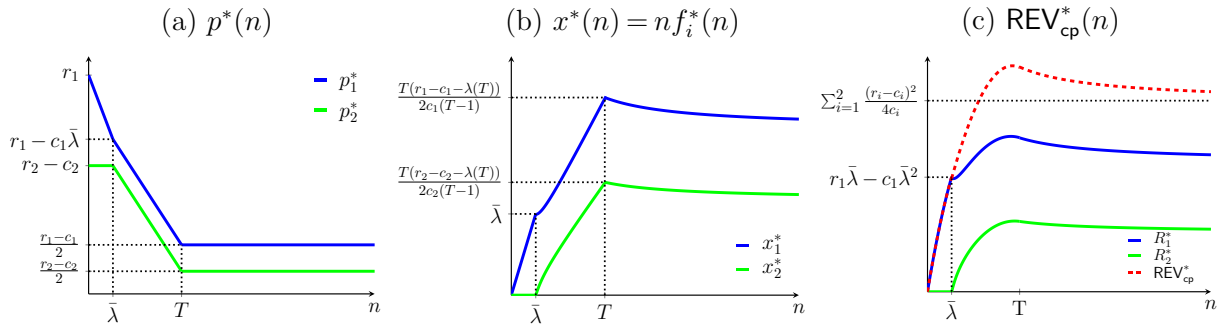
$$n\;<\;\frac{1}{2}\Big(\frac{r_1}{c_1}+\frac{r_2}{c_2}\Big)\quad\textit{and}\quad\lambda(n)\geq\min\{\,r_1-c_1,\;r_2-c_2\,\},$$

then there exists a unique $j\in\{1,2\}$ *with* $\lambda(n)\geq r_j-c_j$. *In the optimal contingent pricing, lead* j *is excluded* ($f_j^*(n)=0$), *and all suppliers buy the other lead* $i\neq j$:

$$f_i^*(n)=1,\qquad p_i^*(n)=R_i(n)=r_i-c_in,$$

while any $p_j^*(n)\geq R_j(1)=r_j-c_j$ *implements* $f_j^*(n)=0$. *The per-state revenue is* $\mathsf{REV}^*_{\mathsf{cp}}(n)=nR_i(n)$.

Figure E.13: Optimal Contingent Pricing under Linear Reward Functions with Two Leads



Note. State n *can only take integer numbers,* $\bar\lambda=\frac{1}{2}(\frac{r_1}{c_1}-\frac{r_2}{c_1}+1+\frac{c_2}{c_1}),T=\frac{1}{2}(\frac{r_1}{c_1}+\frac{r_2}{c_2})$. *In the figure, we assume* $r_1-c_1>r_2-c_2$.

Figure E.13 illustrates Proposition E.6 about how the optimal state-dependent prices $p_i^*(n)$, the equilibrium number of buyers $x_i^*(n)=nf_i^*(n)$, and the resulting revenue $\mathsf{REV}^*_{\mathsf{cp}}(n)$ vary with the realized number of potential suppliers n when $r_1-c_1>r_2-c_2$.

For small markets $n\leq\bar\lambda$ (case (iii)), it is optimal to *exclude* the less profitable lead. In the figure this corresponds to selling only lead 1: the platform sets $p_1^*(n)=R_1(n)$, which decreases linearly in n, and keeps $p_2^*(n)$ at $R_2(1)$ so that no supplier buys lead 2. All suppliers purchase the active lead, so $x_1^*(n)=n$ and $x_2^*(n)=0$, and the revenue curve in panel (c) coincides with the single-lead revenue $nR_1(n)$ up to the level $r_1\bar\lambda-c_1\bar\lambda^2$ at $n=\bar\lambda$.

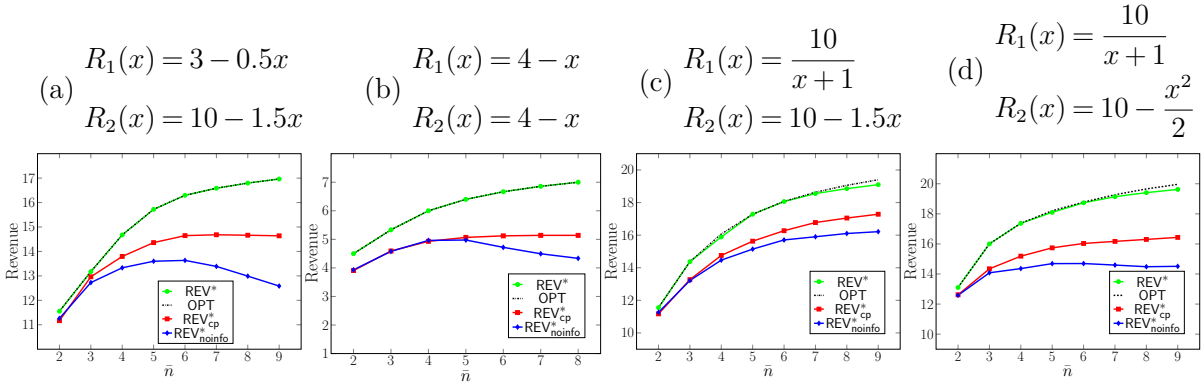
[18] The first condition is implied by the second; however, for completeness, we still state the first condition explicitly. Notice that when $r_1-c_1>r_2-c_2$, the second condition is equivalent to $n\leq\bar\lambda:=\frac{1}{2}\left(\frac{r_1}{c_1}-\frac{r_2}{c_1}+1+\frac{c_2}{c_1}\right)$.

For intermediate market sizes $\bar{\lambda} < n \leq T$ (case (ii)), both leads are active and the participation constraint binds: all suppliers buy some lead, $f_1^*(n) + f_2^*(n) = 1$. The $\lambda(n)$ is positive and decreases in n, so both prices $p_i^*(n) = \frac{r_i - c_i + \lambda(n)}{2}$ fall linearly between $n = \bar{\lambda}$ and $n = T$, as shown in panel (a). The expected purchase-lead-i number $x_i^*(n)$ now increases (panel (b)), and the total revenue in panel (c) continues to increase until the point where $\lambda(T) = 0$.

For large markets $n > T$ (case (i)), some suppliers will not purchase any leads. The optimal prices become a constant, $p_i^*(n) = \frac{r_i - c_i}{2}$. Both $x_i^*(n)$ and each lead's revenue $R_i^*(n)$ decrease slightly as n grows, and total contingent pricing revenue $\mathsf{REV}_{\mathsf{cp}}^*(n)$ is maximized at $n = T$ and declines thereafter, as depicted in panel (c).

E.3 Numerical Comparison in the Setting of Two Leads

Figure E.14: Revenue Comparison with Two Leads



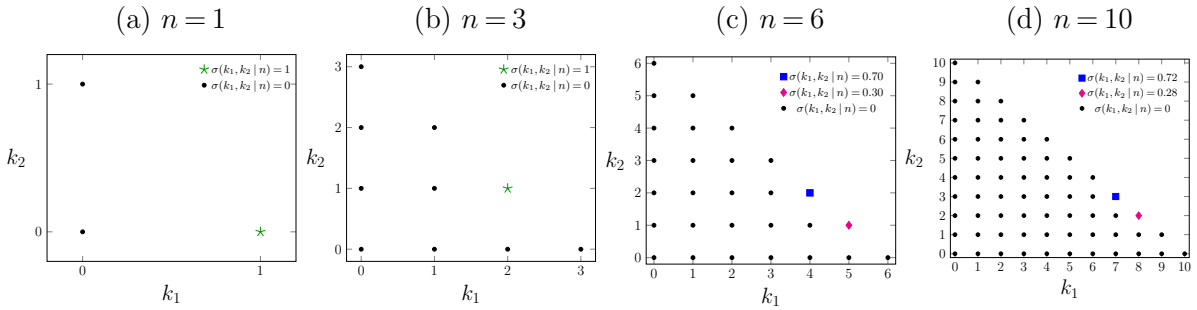
Note. $\Omega = [\bar{n}]$, *and* μ_0 *follows a beta-binomial distribution* $\mathsf{BetaBin}(|\,\Omega\,| - 1, 0.25, 0.2)$.

Figure E.14 reports numerical results for parameter regimes in which one or both conditions in Theorem 7 fail. We compare four classes of mechanisms: (i) OPT, the optimal revenue when jointly optimizing signaling and state-dependent randomized prices; (ii) REV^*, the optimal revenue under signaling with fixed prices; (iii) $\mathsf{REV}_{\mathsf{cp}}^*$, the optimal revenue under contingent pricing; and (iv) $\mathsf{REV}_{\mathsf{noinfo}}^*$, the revenue under no information with the corresponding optimal fixed prices (allowing different prices across the two leads).

Across all numerical experiments, we find that $\mathsf{REV}^* \geq \mathsf{REV}_{\mathsf{cp}}^*$. This highlights the robustness of our main insight that information design with fixed prices outperforms contingent pricing. Unlike in the single-lead setting, however, in the two-lead setting we do find instances in which $\mathsf{REV}^* < \mathsf{OPT}$; see Figure E.14(c) and (d). That said, in all our numerical experiments, the gap is quantitatively small, suggesting that signaling mechanisms with fixed prices typically capture most of the attainable revenue. As a side note, we also find that contingent pricing typically dominates the no-information, fixed-price regime, and that this gap widens as $|\Omega|$ increases. This pattern underscores the growing

value of pricing flexibility, in the absence of appropriate signaling, in environments with greater uncertainty.

Figure E.15: Example of an Optimal Signaling Mechanism with Randomization in the Two-leads Setting



Note. $R_1(x)=10-0.5x$, $R_2(x)=10-1.5x$, $\Omega=\{1,3,6,10\}$, $\mu_0=\{1/4,2/4,1/8,1/8\}$, *with optimal prices charged* $p_1^*=7.916$, $p_2^*=7.382$

In contrast to the base model, where the optimal σ^* suffices to be deterministic under the joint optimality of price and the signaling mechanism (see Theorem 3), the two-lead setting often requires randomization. We illustrate this in Figure E.15 with asymmetric rewards. Note that although σ^* is deterministic in the base model, the final signaling mechanism is not, as the platform sends "join" and "not join" signals to providers uniformly at random.

Appendix F Extensions to Welfare Maximization and Heterogeneous Providers

F.1 Service Providers' Welfare Maximization

Maximizing providers' welfare, defined as the sum of all providers' utilities, as the platform's objective is relevant for various reasons. The platform may be set up by a non-profit organization whose goal is to help self-employed individuals or small and medium-sized enterprises by efficiently matching them with potential customers. Alternatively, the platform may be competing for market share, opting to sacrifice short-term revenue with the hope of attracting more providers to the platform. In what follows, we simply refer to the providers' welfare as welfare.

Following the same revelation-principle-style argument as in the revenue maximization setting, we focus on obedient and symmetric mechanisms. With the goal of maximizing welfare, the commission fee p_0 would be set to zero. Hence the welfare under the obedient signaling mechanism $\sigma(\cdot\mid n)\in\Delta([n]^+)$ is $\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma(k\mid n)k(R(k)-p_0)$. It is important to note that even with a price of zero, not all interested providers would participate due to the negative rewards that can arise from intense supply competition. We formulate the welfare maximization problem as the following LP:

$$\max_{\sigma}\quad \sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma(k\mid n)\cdot kR(k) \tag{$\mathcal{P}_{\text{welfare}}$}$$

$$\text{s.t.} \quad \sigma \text{ satisfies (join), (not-join) and (distribution) with } p_0 = 0.$$

Under $p_0 = 0$, problem ($\mathcal{P}_{\text{welfare}}$) differs from the supply-maximization problem ($\mathcal{P}_{\text{LP}}$) only in the objective function. Although the difference between the two problems may seem small, our next result highlights distinctly different properties when it comes to maximizing welfare.

LEMMA F.6. *In problem ($\mathcal{P}_{\text{welfare}}$), constraint (not-join) is binding, while constraint (join) is non-binding under any welfare-maximizing signaling mechanism.*

Recall that in Section 3.1, we have shown that constraint (join) is binding, while constraint (not-join) is non-binding at optimality when the platform's objective is to maximize the expected supply (equivalently, the expected revenue under given price p_0). In contrast, the exact opposite occurs if the goal is to maximize welfare. The non-binding nature of constraint (join) at optimality implies that any participating provider enjoys a strictly positive surplus. Utilizing these properties, we can characterize the structure of the welfare-maximizing signaling mechanism.

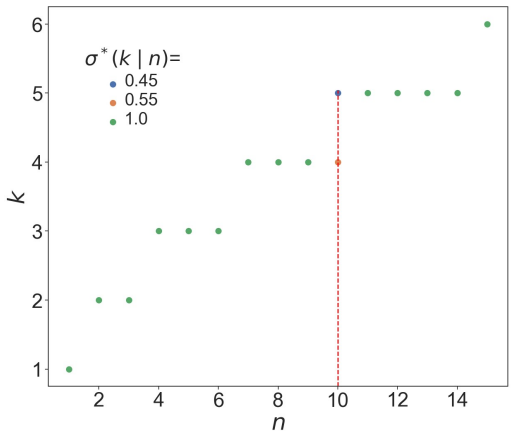
THEOREM F.1 **(Welfare-Maximizing Signaling Mechanism)**. *Under the welfare-maximizing signaling mechanism σ^*:*

(i) There exists at most one state $n \in \Omega$ where the number of providers recommended to join is randomized between two numbers.

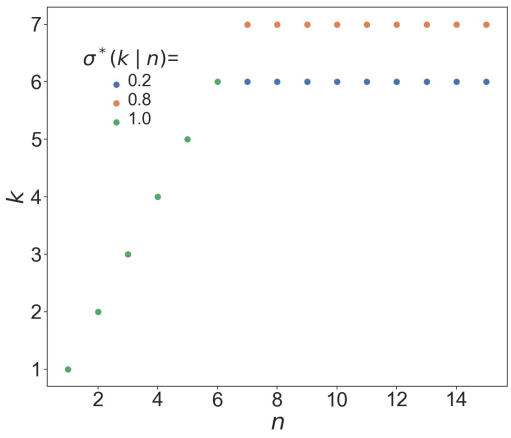
(ii) For all other states, the recommendation is deterministic, i.e., $\exists k(n) \leq n$ such that $\sigma^*(k(n) \mid n) = 1$.

Figure F.16: Welfare-Maximizing vs Supply-Maximizing Signaling Mechanisms

(a) Welfare Maximization



(b) Supply Maximization

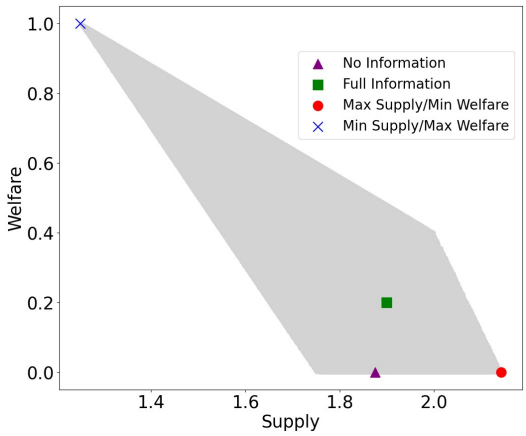


Note. $R(x) = 11/x - 2$, $\Omega = [15]$, $p_0 = 0$, *and prior* μ_0 *is uniform.*

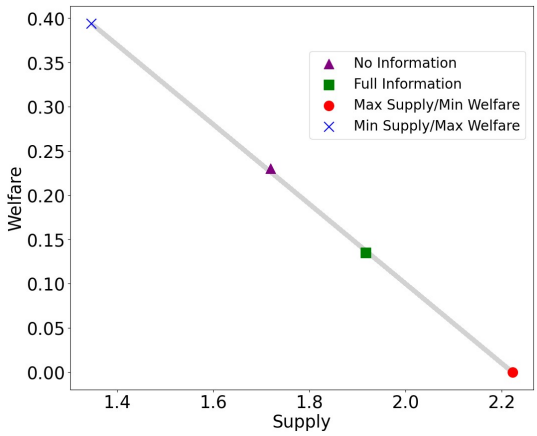
To build intuition for Theorem F.1, we present a numerical example that compares the welfare-maximizing signaling mechanism with the supply-maximizing signaling mechanism, as depicted in

Figure F.16. For a given number of interested providers n along the horizontal axis, each point in the figure indicates the value of $\sigma^*(k \mid n)$ for values of k along the vertical axis, provided it is non-zero. Two notable differences between the two mechanisms include: (i) Under the supply-maximizing mechanism, recommendations are randomized for all states greater than a certain cutoff value, whereas under the welfare-maximizing signaling mechanism, the recommendation to join is randomized for at most one state. (ii) Perhaps unsurprisingly, the number of providers recommended to join at any state n is lower under the welfare-maximizing mechanism than under the supply-maximizing mechanism.

Figure F.17: Welfare and Supply under Various Signaling Mechanisms ($\Omega = \{2, 3\}$, $p_0 = 0$, and Uniform Prior μ_0)



(a) $R(x) = 2.2 - x$



(b) $R(x) = 1/x - 0.45$

To illustrate the intrinsic tradeoff between welfare and supply maximization, we conduct a numerical analysis and depict the set of achievable welfare and supply values by any feasible signaling mechanism. This set is represented by the shaded areas in Figure F.17 for two reward functions.

We have two key observations: (i) The welfare-maximizing mechanism minimizes supply, while the supply-maximizing mechanism minimizes welfare. This inverse relationship occurs because maximizing supply involves extracting all providers' surplus, resulting in minimal welfare, and vice versa. (ii) The feasible region in Figure F.17 (b) forms a straight line. By Lemma F.7 provided in the Appendix, this outcome arises from the specific form of the reward function, $R(x) = r/x - c$. The decreasing linear relation between welfare and supply shows a stark tradeoff between welfare and supply.

We characterize the relation between welfare and supply under reciprocal reward functions, which explains Figure F.17b.

LEMMA F.7. *Assume that $p_0 = 0$, $\mathsf{W}(\sigma)$ and $\mathsf{Sp}(\sigma)$ are welfare and supply under any feasible signaling mechanism σ and p_0. If the reward function takes the form of $R(x) = \frac{r}{x} - c$, where r and c are constants, then the relation between welfare and supply is given by,*

$$\mathsf{W}(\sigma) = r - c \cdot \mathsf{Sp}(\sigma).$$

We now show that full information is the welfare-maximizing public signaling mechanism under linear reward functions.

PROPOSITION F.7 **(Optimal Public Signal Under Welfare Maximization)**. *Under linear rewards, among the public signaling mechanisms, full information is welfare-maximizing.*

While private mechanisms may in principle generate even higher welfare by enabling finer coordination across market participants, our finding establishes a clear benchmark for public mechanisms: any coarsening of the signal that reduces transparency strictly diminishes welfare. In the proof, we show that the interim welfare function is convex in the posterior belief, leading to the optimality of full information.

F.2 Heterogeneous Providers

In this section, we extend our base model to account for providers' heterogeneous utility functions. Providers' expected utilities may be dissimilar as they value each lead differently due to differing skill levels, which affects the amount of effort required to complete the job. Additionally, providers may operate at different scales, with larger service providers potentially being more efficient in communicating with customers due to economies of scale.

Consider L types of providers represented by $\theta_i \in [L]$ for provider i. Different types of providers may have different reward functions, as specified below. We assume that providers of the same type are homogeneous as in our base case. The *state* of the market can then be defined as a vector of the (random) number of interested providers of each type. With slight abuse of notation, we denote the state vector by $N = (N_1, \ldots, N_L) \in \Omega \subseteq \mathbb{Z}_+^L$, where Ω is the finite set of states, and $\mu_0 \in \Delta(\Omega)$ the platform's prior belief (PMF) about the underlying state N. Let $n = (n_1, \ldots, n_L) \in \Omega$ denote the realized state vector, and denote by $[n_\ell]$ (by slight abuse of notation) the set of n_ℓ type-ℓ providers.

Our next result characterizes the prior beliefs of different types of providers.

LEMMA F.8 **(Heterogeneous Service Providers' Prior Belief)**. *The prior belief of a type-ℓ provider μ_{p}^{ℓ} is given by:*

$$\mu_{\mathsf{p}}^{\ell}(n) = \frac{n_\ell \mu_0(n)}{\mathbb{E}[N_\ell]}, \quad \forall\, n \in \Omega,\, \ell \in [L].$$

The structure of providers' prior beliefs is similar to that in the homogeneous setting (see Lemma 1). Providers of the same type still share the same prior belief. However, with heterogeneous providers, the prior beliefs of different types are asymmetric.

Given vector n of the numbers of interested providers and the action profile $a^n \in \{0,1\}^{1^\top n}$, define the number of participating type-ℓ providers as $k_\ell(a^n) = \sum_{i\in[n_\ell]} a_i$ and let $k(a^n) = (k_\ell(a^n))_{\ell\in[L]}$ be the corresponding vector. Let $R_\ell : \mathbb{R}_+^L \to \mathbb{R}$ map from the vector of the number of participating providers of each type to the reward a type-ℓ provider receives. The platform charges a commission fee $p_\ell \geq 0$ to type-ℓ providers. The utility function for a provider i of type ℓ is given by

$$u_i(a_i, a_{-i}^n \mid \theta_i = \ell) = \begin{cases} R_\ell(k(a^n)) - p_\ell, & \text{if } a_i = 1, \\ 0, & \text{if } a_i = 0. \end{cases}$$

The underlying assumption, a common one in the literature, is that the platform has complete knowledge of provider types. This enables the platform to charge different prices based on provider types. If provider type were private information, this would become an information design problem with private information, significantly increasing the analytical complexity (Kolotilin et al. 2017). We also assume negative externality in that $R_\ell(k_1, \ldots, k_L)$ is decreasing in $k_{\ell'}$ for any $\ell' \in [L]$. This aligns with the intuition that more intense competition leads to reduced rewards per provider.

The platform's objective is to maximize the expected revenue by choosing a type-dependent (but state-independent) price vector $p \in \mathbb{R}^L$ and a signaling mechanism. Following a similar proof to Lemma 2, we can, without loss of generality, restrict our attention to obedient mechanisms $\sigma(\cdot \mid n) \in \Delta(\mathcal{N}(n))$, where $\mathcal{N}(n) := [n_1]^+ \times \cdots \times [n_L]^+$. Specifically, for $k = (k_1, \ldots, k_L) \in \mathcal{N}(n)$, $\sigma(k \mid n) \in [0,1]$ represents the probability of recommending k_ℓ providers of type-ℓ to participate, for all $\ell \in [L]$, when the underlying state is $n \in \Omega$. Conditional on $\sigma(k \mid n)$, each type-ℓ provider receives a signal to join with probability $\frac{k_\ell}{n_\ell}$ and a signal not to join with probability $\frac{n_\ell - k_\ell}{n_\ell}$. Utilizing Lemma F.8 and following a similar proof for Lemma 3 regarding the LP formulation in the homogeneous setting, we have the following formulation of the platform's revenue-maximization problem:

$$\begin{aligned}
\max_{p\in\mathbb{R}_+^L,\sigma} \quad & \sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \sum_{\ell\in[L]} k_\ell \cdot p_\ell && (\mathcal{P}_{\mathsf{hetero}}) \\
\text{s.t.} \quad & \sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \cdot k_\ell \cdot [R_\ell(k) - p_\ell] \geq 0, \quad \forall \ell \in [L] && (\textsf{join-type-}\ell) \\
& \sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \cdot (n_\ell - k_\ell) \cdot [R_\ell(k_\ell + 1, k_{-\ell}) - p_\ell] \leq 0, \quad \forall \ell \in [L] && (\textsf{not-join-type-}\ell) \\
& \sigma(\cdot \mid n) \in \Delta(\mathcal{N}(n)), \quad \forall n \in \Omega. && (\textsf{distribution})
\end{aligned}$$

Note that problem ($\mathcal{P}_{\mathsf{hetero}}$) is a bi-linear programming problem, which is NP-hard in general. Additionally, with heterogeneous providers, we face a multidimensional information design problem, which is known to be hard (Dworczak and Kolotilin 2019, Curello and Sinander 2022). For tractability, we impose the following assumption.

ASSUMPTION F.1. *We assume that for the reward function and underlying states:*

(i) (Discrete order-unimodality) Let $f(k) := k^\top R(k)$, where $R(k) = [R_\ell(k)]_{\ell\in[L]}$. There exists $\bar{\tau} \in \arg\max_{k\in\mathbb{Z}_+^L} f(k)$ such that: for any $k \leq k' \leq \bar{\tau}$ (element-wise), $f(k) \leq f(k')$.

(ii) For any $n \in \Omega$, either $n \le \bar{\tau}$ or $n \ge \bar{\tau}$ (element-wise).

Assumption F.1 (i), which holds if $k^\top R(k)$ is joint concave in $k \in \mathbb{R}^L$, is a natural extension of Assumption 1 (ii) for heterogeneous providers. One sufficient condition for the element-wise inequality in Assumption F.1 (ii) is that the numbers of interested providers of different types are highly positively correlated. This can be a reasonable assumption in many situations. For instance, if providers are heterogeneous in their skill levels, Assumption F.1 (ii) requires that if a job attracts a large number of skilled providers, it should also attract a large number of novice providers.

PROPOSITION F.8. *Under Assumption F.1, the optimal solution to problem ($\mathcal{P}_{\mathsf{hetero}}$) is as follows: For any state $n \in \Omega$, $\sigma^*(k^*(n) \mid n) = 1$, where $k^*(n) \in \arg\max_{k \in \mathcal{N}(n)} \sum_{\ell \in [L]} k_\ell \cdot R_\ell(k)$. The optimal price is*

$$p_\ell^* = \frac{\sum_{n \in \Omega} \mu_0(n) \cdot k_\ell^*(n) \cdot R_\ell(k^*(n))}{\sum_{n \in \Omega} \mu_0(n) \cdot k_\ell^*(n)}, \ \forall \ell \in [L].$$

Proposition F.8 implies that under Assumption F.1, when the total number of providers n is small, the platform recommends that all providers join; once n exceeds the threshold, it instead recommends fixed participation levels that no longer depend on the magnitude of n. Moreover, the corresponding prices are pinned down by making the (join-type-ℓ) incentive constraints bind. This result generalizes Theorem 3 of the base model.

The proof of Proposition F.8 is similar to that of Theorem 3. In particular, we show that constraint (not-join-type-ℓ) is non-binding at optimality for all $\ell \in [L]$, while constraints (join-type-ℓ) are binding at optimality and lead to an equivalent maximization of the expected reward for participating providers in each state $n \in \Omega$. The binding constraints (join-type-ℓ) also give rise to the optimal price for each type of provider. Under Assumption F.1, as detailed in the proof, we have $k^*(n) = n$ if $n \le \bar{\tau}$; otherwise $k^*(n)$ is a constant, where k_ℓ^* takes value of $\bar{\tau}_\ell$.

Under Assumption F.1, we can also establish that the optimal signaling mechanism with type-dependent (but state-independent) pricing remains optimal even if we expand the policy space to jointly optimize the signaling mechanism and randomized prices, allowing them to depend on both types and states. The result is formally summarized as follows.

PROPOSITION F.9. *Within the policy space that jointly considers signaling and (possibly randomized) state- and type-dependent pricing, the optimal signaling mechanism with fixed pricing for each type of providers remains optimal under Assumption F.1.*

Appendix G Proofs

G.1 Proofs for Sections 2 and 3

Proof of Lemma 1. We refer to Lemma 1 in McAfee and McMillan (1987) for a formal proof. Here, we adopt a special case as used in Lingenbrink and Iyer (2018) to elaborate on the result.

Assume that we have a pool of N_{max} providers, and we randomly and independently select n providers from this pool to be present with probability $\mu_0(n)$. Note that this independent selection is only for illustration; we do not actually need this assumption. For a given lead, the events that different providers are interested in can be *inter*dependent random variables, as illustrated in Example 2. Fix a focal agent and let B be the event that the focal agent is present. We assume that condition on N, providers are independently and uniformly drawn from N_{max}. Thus, $\mathbb{P}(B \mid N=n) = \frac{n}{N_{max}}$. Then, we have

$$\mu_{\mathsf{p}}(n) = \mathbb{P}(N=n \mid B) = \frac{\mathbb{P}(B \mid N=n)\cdot\mathbb{P}(N=n)}{\sum_{\hat{n}\in\Omega}\mathbb{P}(B \mid N=\hat{n})\cdot\mathbb{P}(N=\hat{n})} = \frac{n\mu_0(n)}{\sum_{\hat{n}\in\Omega}\hat{n}\mu_0(\hat{n})} = \frac{n\mu_0(n)}{\mathbb{E}[N]}.$$

We now proceed to prove the stochastic dominance result. Let $F(x)$ and $G(x)$ be the CDF of μ_0 and μ_{p}, respectively. To prove $\mu_0 \prec_{\mathsf{fsd}} \mu_{\mathsf{p}}$ is equivalent to show $F(x) \geq G(x)$ for all $x \in [0, \bar{n}]$, where $\bar{n}$ is the maximum state. We have

$$F(x) = \sum_{n\in\Omega, n\leq x} \mu_0(n) \quad \text{and} \quad G(x) = \sum_{n\in\Omega, n\leq x} \frac{n}{\mathbb{E}[N]}\mu_0(n).$$

Taking the difference, we have

$$F(x) - G(x) = \sum_{n\in\Omega, n\leq x} \mu_0(n)\cdot\frac{\mathbb{E}[N]-n}{\mathbb{E}[N]}.$$

For $0 \leq x \leq \mathbb{E}[N]$, the above term is positive. For $x \in (\mathbb{E}[N], \bar{n}]$, we can split the above formula into two terms:

$$F(x) - G(x) = \sum_{n\in\Omega, n\leq \mathbb{E}[N]} \mu_0(n)\cdot\frac{\mathbb{E}[N]-n}{\mathbb{E}[N]} + \sum_{n\in\Omega, \mathbb{E}[N]<n\leq x} \mu_0(n)\cdot\frac{\mathbb{E}[N]-n}{\mathbb{E}[N]},$$

which is decreasing in x. Thus, for $x \in (\mathbb{E}[N], \bar{n}]$, we have

$$F(x) - G(x) \geq F(\bar{n}) - G(\bar{n}) = 1 - 1 = 0.$$

This completes the proof. □

Proof of Lemma 2. The proof consists of three steps in establishing the equivalence of any signaling mechanism and an obedient and symmetric one:

$$\begin{aligned} \text{mechanisms with mixed strategy equilibrium} &\overset{\text{(i)}}{\Longleftrightarrow} \text{mechanisms with pure strategy equilibrium} \\ &\overset{\text{(ii)}}{\Longleftrightarrow} \text{obedient mechanisms} \\ &\overset{\text{(iii)}}{\Longleftrightarrow} \text{obedient and symmetric mechanisms} \end{aligned}$$

(i) We first show that for any signaling mechanism that induces a mixed strategy equilibrium, there exists a mechanism inducing a pure strategy equilibrium with the same expected supply.

Consider signaling mechanism $\Sigma_1 = (\pi_1, \mathbb{S})$ and its induced providers' equilibrium mixed strategy $f : \mathbb{S} \to [0,1]$, we construct another mechanism $\Sigma_2 = (\pi_2, \mathbb{U})$, accompanied by a *pure strategy* $d : \mathbb{U} \to \{0,1\}$, that achieves the same objective value for the platform. Let $\mathbb{U} := \mathbb{S} \times \{0,1\}$. For

each provider i that receives a signal $s_i^n \in \mathbb{S}$, the platform now sends a signal $(s_i^n, r_i^n) \in \mathbb{U}$ with $r_i^n \in \{0,1\}$ an independent Bernoulli trial with success probability $f(s_i^n)$, where $f(s_i^n)$ is the equilibrium mixed strategy of provider i under Σ_1. The platform sends signal $(s^n, r^n) \in \mathbb{U}^n$ with probability

$$\pi_2((s^n, r^n) \mid n) = \pi_1(s^n \mid n) \prod_{i=1}^{n} (\mathbb{1}_{\{r_i^n=0\}}(1 - f(s_i^n)) + \mathbb{1}_{\{r_i^n=1\}} f(s_i^n)).$$

We now find the equilibrium pure strategy $d(s_i^n, r_i^n)$ under Σ_2. By the definition of r_i^n, provider i receiving $r_i^n = 1 \implies f(s_i^n) > 0$ (since otherwise a Bernoulli trial with a success probability of zero will not return 1). Since f forms an equilibrium, it follows from Definition 1 that the expected utility of the provider is non-negative. Thus it is optimal for the provider to join, i.e., $d(s_i^n, 1) = 1$. Similarly, provider i receiving $r_i^n = 0 \implies f(s_i^n) < 1$, which implies the non-positive expected utility of the provider. It is thus optimal for the provider to not join, i.e., $d(s_i^n, 0) = 0$. Together, we get the equilibrium pure strategy $d(s_i^n, r_i^n) = r_i^n \in \{0,1\}$.

It is straightforward to see that provider i's joining probability upon receiving signal s_i^n is the same under Σ_1 and Σ_2. The probability of sending signals that contain s^n is also the same under both mechanisms. Therefore, mechanism Σ_2 yields the same expected supply (and hence revenue, under the given price p_0) for the platform as Σ_1.

(ii) Next, we show any mechanism with pure strategy is equivalent to an obedient mechanism.

Consider our signaling mechanism $\Sigma_2 = (\pi_2, \mathbb{U})$ with pure strategy $d : \mathbb{U} \to \{0,1\}$. For each provider, let $U_j := \{u \in \mathbb{U} | d(u) = j\}$ denote the set of signals that lead to action $j \in \{0,1\}$. Now construct a new signaling mechanism $\Sigma_3 = (\pi_3, \mathbb{V})$, where the signal space for each provider is the (binary) action space, $\mathbb{V} = \{0,1\}$, and $\pi_3(v^n|n) = \sum_{u^n \in \mathbb{U}^n} \mathbb{1}_{\{u_i^n \in U_{v_i^n}, \forall i\}} \pi_2(u^n|n)$. Under mechanism Σ_3, it is apparent that each provider i receives signal $v_i^n \in \{0,1\}$ with probability $\sum_{u^n \in \mathbb{U}^n} \mathbb{1}_{\{u_i^n \in U_{v_i^n}\}} \pi_2(u^n|n) = \sum_{u^n \in \mathbb{U}^n} \mathbb{1}_{\{d(u_i^n) = v_i^n\}} \pi_2(u^n|n)$, i.e., the probability of receiving signals that lead to action v_i^n under mechanism Σ_2. Consequently, it is optimal for each provider to follow the signal under mechanism Σ_3. Σ_3 is hence an obedient mechanism by Definition 2. The definition of π_3 also yields identical platform objectives under Σ_2 and Σ_3. We conclude that mechanism Σ_2 with pure strategy is equivalent to obedient mechanism Σ_3.

(iii) Finally, we show that an equivalent obedient and symmetric mechanism exists for any obedient mechanism.

Consider our obedient mechanism $\Sigma_3 = (\pi_3, \mathbb{V})$, where the signal space $\mathbb{V} = \{0,1\}$ is also the action space. Denote $\mathfrak{S}(s^n)$ as the set of all permutations (i.e., the symmetric group) of signal profile s^n. Now construct a new signaling mechanism $\Sigma_4 = (\pi_4, \mathbb{W})$, with $\mathbb{W} = \{0,1\}$ the signal (and action) space where

$$\pi_4(w^n|n) = \frac{\sum_{v^n \in \mathfrak{S}(w^n)} \pi_3(v^n|n)}{|\mathfrak{S}(w^n)|},$$

i.e., the probability of signaling $w^n \in \mathbb{W}^n$ under mechanism Σ_4 is the *average* probability of signaling any permutation of w^n under mechanism Σ_3 (note that the signal spaces under Σ_3 and Σ_4 are identical: $\mathbb{W}^n = \mathbb{V}^n$). It is easy to verify that π_4 is a proper mapping and, furthermore, $\pi_4(w^n|n) = \pi_4(\hat{w}^n|n)$ for any $\hat{w}^n \in \mathfrak{S}(w^n)$. As a result, Σ_4 is symmetric by Definition 3. Because we assume that the indices are randomly assigned to providers after n is realized, π_3 and π_4 are equivalent for any specific provider. Therefore, Σ_4 is also obedient.

Finally, the platform attains the same objective value under Σ_3 and Σ_4 since the expected number of participating providers out of any n providers is the same under both mechanisms:

$$\begin{aligned}
\sum_{w^n \in \mathbb{W}^n} \pi_4(w^n|n) \sum_{i\in[n]} w_i^n &= \sum_{w^n \in \mathbb{W}^n} \frac{\sum_{v^n \in \mathfrak{S}(w^n)} \pi_3(v^n|n)}{|\mathfrak{S}(w^n)|} \sum_{i\in[n]} w_i^n \\
&= \sum_{j=0}^{n} \sum_{w^n \in \mathbb{W}^n : \sum_{i\in[n]} w_i^n = j} \frac{\sum_{v^n \in \mathfrak{S}(w^n)} \pi_3(v^n|n)}{|\mathfrak{S}(w^n)|} j \\
&= \sum_{j=0}^{n} \sum_{v^n \in \mathbb{V}^n : \sum_{i\in[n]} v_i^n = j} \pi_3(v^n|n) j \\
&= \sum_{v^n \in \mathbb{V}^n} \pi_3(v^n|n) \sum_{i\in[n]} v_i^n,
\end{aligned}$$

where obedient mechanisms imply $\sum_{i\in[n]} w_i^n$ and $\sum_{i\in[n]} v_i^n$ in ($\mathcal{P}$), and the third equality follows because the inner summation is over all signal profiles with j 1's (and $n-j$ 0's), which contain exactly all $|\mathfrak{S}(w^n)|$ permutations for any $w^n \in \mathbb{W}^n$ with j elements of 1 (and $n-j$ elements of 0). We may conclude that the obedient and symmetric mechanism Σ_4 is equivalent to the obedient mechanism Σ_3. □

Proof of Lemma 3. We will show that (join) and (not-join) are equivalent to obedience. To do so, it is sufficient to show that upon receiving a signal to join ($s_i = 1$), the expected utility for provider i to join, $\mathbb{E}^{\Sigma,f}[h(1 + 1^\top a_{-i}^N, p_0) \mid s_i = 1]$, is non-negative and upon receiving a signal not to join ($s_i = 0$), $\mathbb{E}^{\Sigma,f}[h(1 + 1^\top a_{-i}^N, p_0) \mid s_i = 1]$ is non-positive.

First of all, if the platform recommends k out of n providers to participate, any provider $i \in [n]$ will receive the recommendation to participate with probability $\frac{k}{n}$ and receive a recommendation not to participate with probability $\frac{n-k}{n}$. Let $p(s_i = 1 \mid n, k) = \frac{k}{n}$ denote the probability that provider i receives a signal to join if there are n providers and the platform asks k of them to participate.

Let $p(s_i = 1)$ be the probability that provider i receives a signal to join, and let $p(n, k) = \mu_{\mathsf{p}}(n)\sigma(k \mid n)$ be provider i's belief that $N = n$ and the platform will ask k to participate. Furthermore, let $p(n, k \mid s_i = 1)$ be provider i's posterior belief that $N = n$ and k providers are asked to participate upon receiving a signal to join. We have

$$p(n, k \mid s_i = 1) = \frac{p(s_i = 1 \mid n, k) \cdot p(n, k)}{p(s_i = 1)} = \frac{\frac{k}{n}\mu_{\mathsf{p}}(n)\sigma(k \mid n)}{p(s_i = 1)}. \tag{G.1}$$

Obedience requires that upon receiving $s_i = 1$, the provider's expected utility from joining should be greater than the outside option, which is 0. That is to say,

$$\begin{aligned}
0 \le \mathbb{E}^{\Sigma,f}[h(1+1^\top a_{-i}^N, p_0) \mid s_i = 1] &= \sum_{n\in\Omega}\sum_{k=1}^{n} p(n,k \mid s_i=1)\cdot h(k,p_0) \\
\text{(by (G.1))} &= \sum_{n\in\Omega}\sum_{k=1}^{n} \frac{\frac{k}{n}\mu_{\mathsf{p}}(n)\sigma(k\mid n)}{p(s_i=1)}\cdot h(k,p_0) \\
\text{(by Lemma 1)} &= \sum_{n\in\Omega}\sum_{k=1}^{n} \frac{n\mu_0(n)}{\mathbb{E}[N]}\cdot\frac{k}{n}\cdot\frac{\sigma(k\mid n)}{p(s_i=1)}\cdot h(k,p_0) \\
&= \frac{1}{\mathbb{E}[N]p(s_i=1)}\sum_{n\in\Omega}\mu_0(n)\sum_{k=1}^{n}\sigma(k\mid n)\cdot k\cdot h(k,p_0),
\end{aligned}$$

which becomes (join). We can show (not-join) in a similar way, which states that upon receiving a recommendation not to participate ($s_i = 0$), if the provider deviates from the recommendation and joins, their expected utility should be lower than their outside option. □

Proof of Theorem 1. Since problem ($\mathcal{P}_{\mathsf{LP}}$) is a linear program, to prove the optimality of σ^* given by Theorem 1, it is sufficient to prove that σ^* and the optimal Lagrangian multipliers satisfy the KKT conditions. In what follows, we will first write the Lagrangian function, give the optimal Lagrangian multipliers, and verify the KKT conditions.

By Lemma B.2, constraint (join) is binding, which implies the following equivalent LP:

$$\begin{aligned}
\min_{\sigma}\quad & -\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n} k\cdot\sigma(k\mid n) && (\mathsf{LP}_{\mathsf{eq}})\\
\text{s.t.}\quad & \sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma(k\mid n)\cdot k\cdot h(k,p_0) = 0 && \text{(G.2a)}\\
& \sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma(k\mid n)\cdot(n-k)\cdot h(k+1,p_0) \le 0 && \text{(G.2b)}\\
& \sum_{k=0}^{n}\sigma(k\mid n) = 1, \quad \forall n\in\Omega && \text{(G.2c)}\\
& -\sigma(k\mid n)\le 0, \quad \forall n\in\Omega, k\in[n]^+. && \text{(G.2d)}
\end{aligned}$$

Assume that (τ, k^*, γ) are obtained by Algorithm 1. Therefore, by the definition of the algorithm, (τ, k^*, γ) satisfy (B.3) and (B.4). Furthermore, by Lemma B.3 (ii), $\max\{n\in\Omega : n<\tau\} \le k^* < \min\{n\in\Omega : n\ge\tau\}$. Additionally by Lemma B.3 (i), the RHS of (B.3), remains negative during algorithm run, which indicates $k^* \ge R^{-1}(p_0)$.

Let λ_0, ν_0, λ_n, $\nu_{k,n}$ be the Lagrange multipliers associated with constraints (G.2a), (G.2b), (G.2c), and (G.2d), respectively. The Lagrangian is:

$$\begin{aligned}
\mathcal{L} = &-\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}k\cdot\sigma(k\mid n) + \lambda_0\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma(k\mid n)\cdot k\cdot h(k,p_0) + \nu_0\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma(k\mid n)\cdot(n-k)\cdot h(k+1,p_0)\\
&+\sum_{n\in\Omega}\left(\lambda_n(\sum_{k=0}^{n}\sigma(k\mid n)-1)\right) - \sum_{n\in\Omega}\sum_{k=0}^{n}\nu_{k,n}\sigma(k\mid n).
\end{aligned}$$

For ease of notation, define $F(k,p_0) := kh(k,p_0)$. We define λ_0^* and ν_0^* as follows:

$$\lambda_0^* = \frac{-1}{F(\lfloor k^* \rfloor, p_0) - F(\lfloor k^* \rfloor + 1, p_0)} \quad \text{and} \quad \nu_0^* = 0.$$

Moreover, for any $n \in \Omega$, define λ_n^* as:

$$\lambda_n^* = \begin{cases} \mu_0(n)(n - \lambda_0^* F(n,p_0)), & \text{if } n < \tau \\ \mu_0(n)\left[\dfrac{F(\lfloor k^* \rfloor, p_0)}{F(\lfloor k^* \rfloor, p_0) - F(\lfloor k^* \rfloor + 1, p_0)} + \lfloor k^* \rfloor\right], & \text{if } n \geq \tau \end{cases}.$$

We further define $\nu_{k,n}^*$ as follows. For any $n \in \Omega$ such that $n < \tau$ and for $k \in [n]$:

$$\nu_{k,n}^* = \begin{cases} 0, & \text{if } k = n \\ \mu_0(n)\left[n - k - \lambda_0(F(n,p_0) - F(k,p_0))\right], & \text{if } k < n \end{cases}.$$

For any $n \in \Omega$ such that $n \geq \tau$, and $k \in [n]$:

$$\nu_{k,n}^* = \begin{cases} 0, & \text{if } k \in \{\lfloor k^* \rfloor, \lfloor k^* \rfloor + 1\} \\ \mu_0(n)\left[\lfloor k^* \rfloor - k - \lambda_0(F(\lfloor k^* \rfloor, p_0) - F(k,p_0))\right], & \text{if } k \notin \{\lfloor k^* \rfloor, \lfloor k^* \rfloor + 1\} \end{cases}.$$

We will show that σ^* along with $(\lambda_0^*, \nu_0^*, \lambda_n^*, \nu_{k,n}^*)$ satisfies the KKT conditions.

Primal Feasibility

First, we show that σ^* is a feasible solution to ($\mathsf{LP}_{\mathsf{eq}}$). Note that by Lemma B.3, $k^* < \min\{n \in \Omega : n \geq \tau\}$. Hence, σ^* clearly satisfies (G.2c) and (G.2d). For (G.2a),

$$\begin{aligned} &\sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma^*(k \mid n) \cdot k \cdot h(k,p_0) \\ &= \sum_{n<\tau, n\in\Omega} \mu_0(n) n h(n,p_0) + \sum_{n\geq\tau, n\in\Omega} \mu_0(n) \underbrace{\left[\gamma \cdot \lfloor k^* \rfloor \cdot h\left(\lfloor k^* \rfloor, p_0\right) + (1-\gamma) \cdot (\lfloor k^* \rfloor + 1) \cdot h\left(\lfloor k^* \rfloor + 1, p_0\right)\right]}_{k^* h(k^*, p_0)} \\ &\overset{(a)}{=} \sum_{n<\tau, n\in\Omega} \mu_0(n) n h(n,p_0) + k^* h(k^*,p_0) \sum_{n\geq\tau, n\in\Omega} \mu_0(n) \overset{(b)}{=} 0, \end{aligned}$$

where (a) and (b) hold because γ satisfies (B.4) and k^* satisfies (B.3). Thus, constraint (G.2a) holds under σ^*. Finally, we argue that constraint (G.2b) is satisfied.

$$\begin{aligned} &\sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma^*(k \mid n) \cdot (n-k) \cdot h(k+1,p_0) \\ &= \sum_{n\geq\tau, n\in\Omega} \mu_0(n)\left[\gamma \cdot (n - \lfloor k^* \rfloor) \cdot h\left(\lfloor k^* \rfloor + 1, p_0\right) + (1-\gamma) \cdot (n - \lfloor k^* \rfloor - 1) \cdot h\left(\lfloor k^* \rfloor + 2, p_0\right)\right] < 0. \end{aligned}$$

Note that the inequality holds because under a non-trivial problem, $\{n : n \in \Omega, n \geq \tau\}$ is a non-empty set and $k^* \geq R^{-1}(p_0)$. Hence, $h\left(\lfloor k^* \rfloor + 1, p_0\right)$ and $h\left(\lfloor k^* \rfloor + 2, p_0\right)$ are both strictly negative. We have shown that constraint (G.2b) is satisfied under σ^* and is non-binding under the non-trivial problem (see Observation 1). As a result, σ^* satisfies all the constraints, and it is a feasible mechanism.

Dual Feasibility

In this part, we show that $(\lambda_0^*, \nu_0^*, \lambda_n^*, \nu_{k,n}^*)$ are dual feasible. Note that constraint (G.2a) and (G.2c) are equality; thus, λ_0^* and λ_n^* could be any real numbers. As a result, we only need to show that $\nu_{k,n}^* \geq 0$. We will analyze $\nu_{k,n}^*$ under two different cases, with $n < \tau$ and $n \geq \tau$.

Case 1: For any $n \in \Omega$ such that $n < \tau$ if $k = n$, we have $\nu_{k,n}^* = 0$ and if $k < n$ we have

$$\nu_{k,n}^* = \mu_0(n) \left[\underbrace{n - k - \lambda_0^*(F(n, p_0) - F(k, p_0))}_{:= H_n(k, p_0)} \right].$$

Note that $F(k, p_0)$ is concave in k and decreasing in k at $R^{-1}(p_0)$. As mentioned before, k^* must be greater than $R^{-1}(p_0)$. Therefore,

$$k^* \geq R^{-1}(p_0) \implies F(\lfloor k^* \rfloor, p_0) > F(\lfloor k^* \rfloor + 1, p_0) \implies \lambda_0^* \leq 0 \implies H_n(k, p_0) \text{ is convex in } k.$$

We now show that $H_n(k, p_0)$ is decreasing in k at $k = n$. Considering its derivative, we have

$$\left. \frac{\mathrm{d}H_n}{\mathrm{d}k} \right|_{k=n} = -1 - \frac{F'(n)}{F(\lfloor k^* \rfloor) - F(\lfloor k^* \rfloor + 1)} \overset{(a)}{\leq} 0,$$

where (a) holds true because F is concave in k and $k^* \geq \max\{n : n \in \Omega, n < \tau\}$. Additionally, $H(n, p_0) = 0$. Thus, for any $k < n$, $H(k, p_0)$ is positive. As a result $\nu_{k,n}^* \geq 0$ for any $n < \tau$ and any $k \in [n]$.

Case 2: For any $n \in \Omega$ such that $n \geq \tau$ if $k = \lfloor k^* \rfloor$ or $k = \lfloor k^* \rfloor + 1$ we have $\nu_{k,n}^* = 0$ and for any other $k \in [n]$:

$$\nu_{n,k}^* = \mu_0(n) [\underbrace{\lfloor k^* \rfloor - k - \lambda_0(F(\lfloor k^* \rfloor, p_0) - F(k, p_0))}_{:= G(k, p_0)}].$$

$G(k, p_0)$ is convex in k and $G(\lfloor k^* \rfloor, p_0)$ and $G(\lfloor k^* \rfloor + 1, p_0)$ are equal to 0. Thus, for any $n \geq \tau$, and any $k \in [n]$, $G(k, p_0)$ is non-negative which implies that $\nu_{k,n}^*$ is non-negative. Putting the above two cases together, we have shown that for any $n \in \Omega$ and any $k \in [n]$, $\nu_{k,n}^* \geq 0$ and our Lagrangian multipliers are dual feasible.

Stationary Conditions

Consider the partial derivative of the Lagrangian:

$$\frac{\partial \mathcal{L}}{\partial \sigma(k \mid n)} = -\mu_0(n) n + \lambda_0 \mu_0(n) k h(k, p_0) + \nu_0 \mu_0(n)(n - k) h(k + 1, p_0) + \lambda_n - \nu_{k,n}.$$

By definition of λ_0^*, ν_0^*, λ_n^*, and $\nu_{k,n}^*$ for any $n \in \Omega$ and any $k \in [n]$ we have

$$-\mu_0(n) n + \lambda_0^* \mu_0(n) k h(k, p_0) + \nu_0^* \mu_0(n)(n - k) h(k + 1, p_0) + \lambda_n^* - \nu_{k,n}^* = 0,$$

which means the stationary conditions are satisfied.

Complementary Slackness

$\sigma^*(k \mid n) = 0$ unless $n < \tau$ and $k = n$ or $n \geq \tau$ and k is either $\lfloor k^* \rfloor$ or $\lfloor k^* \rfloor + 1$. Therefore, the complementary slackness conditions follow from the definition of σ^*, ν_0^*, and $\nu_{k,n}^*$.

As a result, the KKT conditions are satisfied under σ^* along with $(\lambda_0^*, \nu_0^*, \lambda_n^*, \nu_{k,n}^*)$. For an LP, the KKT conditions are necessary and sufficient to prove optimality; thus, σ^* is optimal. □

G.1.1 Discussion and Proof of Optimal Public Mechanisms There are two main challenges arising in identifying the optimal public mechanisms. Firstly, the problem can no longer be formulated as an LP (Bergemann and Morris 2016) and in general, it is NP-hard (Dughmi and Xu 2017). Consequently, we adopt the belief-based approach used in Kamenica and Gentzkow (2011), which involves computing the posterior beliefs of both the platform and providers. Note that the platform commits to a signaling mechanism *before* the state realization. Thus, when deciding the optimal signaling mechanism, the platform seeks to maximize the expected supply, which is calculated based on its posterior belief of the number of interested providers induced by the signaling mechanism, ex ante (Kamenica and Gentzkow 2011, Alonso and Câmara 2016). The posterior belief is obtained via Bayesian updating based on the prior belief, committed signaling mechanism, and the received signal. Since beliefs are sufficient for providers to make decisions so as to maximize their *expected* utility, we will work with beliefs directly in this section instead of signals.[19]

This brings us to the second challenge in deriving the optimal public mechanism, which concerns the relationship between the platform's and providers' posterior beliefs. Recall that due to the private information of a provider being interested in the focal lead, the prior belief of providers differs from that of the platform. We establish the relationship between the two in Lemma 1. Given the heterogeneous prior beliefs, the posterior beliefs of the platform and the providers would naturally diverge. The good news is that the relationship of the prior beliefs, as described in Lemma 1, is preserved in the posterior beliefs.

LEMMA G.1 **(Posterior Beliefs under Public Mechanisms).** *Under the assumption of symmetric prior beliefs among providers, their posterior belief* $\mu_{\mathsf{p}}^{\mathsf{pos}} \in \Delta(\Omega)$ *is given by*

$$\mu_{\mathsf{p}}^{\mathsf{pos}}(n) = \frac{n\mu_0^{\mathsf{pos}}(n)}{\mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N]}, \ \forall\, n \in \Omega,$$

where $\mu_0^{\mathsf{pos}} \in \Delta(\Omega)$ *denotes the platform's posterior belief under some public signaling mechanism. Besides,* $\mu_{\mathsf{p}}^{\mathsf{pos}}$ *first-order stochastically dominates* μ_0^{pos}.

The proof of Lemma G.1 builds on the work of Alonso and Câmara (2016), who show that when the sender and the receiver share heterogeneous prior beliefs in a Bayesian persuasion model, the receiver's posterior belief only depends on the sender's posterior belief and the ratio of the sender's and receiver's prior beliefs.

[19] Note that, in our Definition 1 of providers' equilibrium, it is equivalent to define their strategy as a mapping from belief to joining probability.

Despite Lemma G.1, finding the optimal public mechanism remains elusive for the general case. We hence restrict our focus to the situations where the reward function $R(\cdot)$ is linear. This allows us to have a clear characterization of the optimal public mechanism, which sheds light on the underlying drivers for the superiority of optimal private signaling mechanisms over optimal public ones.

With a slight abuse of notation, denote $f(\mu_{\mathsf{p}}^{\mathsf{pos}})$ as the providers' equilibrium participation probability for a given price p_0 and providers' posterior belief $\mu_{\mathsf{p}}^{\mathsf{pos}}$ under certain public mechanism. If $\mathbb{E}_{\hat{N}\sim\mu_{\mathsf{p}}^{\mathsf{pos}}}[\hat{N}] < R^{-1}(p_0)$, which is equivalent to $\mathbb{E}_{\hat{N}\sim\mu_{\mathsf{p}}^{\mathsf{pos}}}[h(\hat{N},p_0)] > 0$ by the linearity of $R(\cdot)$, it is rational for all interested providers to participate under the posterior belief $\mu_{\mathsf{p}}^{\mathsf{pos}}$. Thus, $f(\mu_{\mathsf{p}}^{\mathsf{pos}}) = 1$. Otherwise, all providers adopt the same strategy $f(\mu_{\mathsf{p}}^{\mathsf{pos}}) \in [0,1]$, such that they are indifferent between participating and their outside option.[20] That is,

$$\sum_{n\in\Omega} \mu_{\mathsf{p}}^{\mathsf{pos}}(n) \cdot \left[\underbrace{[1-f(\mu_{\mathsf{p}}^{\mathsf{pos}})]}_{\text{not participate}} \cdot 0 + \underbrace{f(\mu_{\mathsf{p}}^{\mathsf{pos}})}_{\text{participate}} \cdot \mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(\mu_{\mathsf{p}}^{\mathsf{pos}}))}[R(1+k)-p_0] \right] = 0. \tag{G.3}$$

In this case, their expected utility for participating, conditional on the other $n-1$ providers also adopting the same mixed strategy $f(\mu_{\mathsf{p}}^{\mathsf{pos}})$, is given by $\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(\mu_{\mathsf{p}}^{\mathsf{pos}}))}[R(1+k)-p_0]$. Here, the random variable k represents the number of participating providers among the other $n-1$ ones. Because the prior beliefs are symmetric, and all providers receive the same signal under a public mechanism, k follows a binomial distribution with parameters $n-1$ and $f(\mu_{\mathsf{p}}^{\mathsf{pos}})$. Given that $\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(\mu_{\mathsf{p}}^{\mathsf{pos}}))}[k] = (n-1)f(\mu_{\mathsf{p}}^{\mathsf{pos}})$ and the reward function $R(\cdot)$ is linear, we have $\mathbb{E}_k[R(1+k)] = R(1+\mathbb{E}[k])$ and hence (G.3) implies $R\left(1+f(\mu_{\mathsf{p}}^{\mathsf{pos}})\cdot(\mathbb{E}_{\hat{N}\sim\mu_{\mathsf{p}}^{\mathsf{pos}}}[\hat{N}]-1)\right) = p_0$. Consequently, the providers' equilibrium participation probability for a given price p_0 and providers' posterior belief $\mu_{\mathsf{p}}^{\mathsf{pos}}$ under certain public mechanism is

$$f(\mu_{\mathsf{p}}^{\mathsf{pos}}) = \min\left\{ \frac{R^{-1}(p_0)-1}{\mathbb{E}_{\hat{N}\sim\mu_{\mathsf{p}}^{\mathsf{pos}}}[\hat{N}]-1},\ 1 \right\}, \tag{G.4}$$

where $R^{-1}(p_0)-1>0$ as we assume $R(1)-p_0>0$ in this section with exogenously given price p_0, and $\mathbb{E}_{\hat{N}\sim\mu_{\mathsf{p}}^{\mathsf{pos}}}[\hat{N}]-1>0$ follows from our assumption $|\Omega|\geq 2$.

The providers' equilibrium participation probability under the special cases of no information and full information are summarized in Proposition B.3, which is deferred to Appendix B. Because $\mathbb{E}_{\hat{N}\sim\mu_{\mathsf{p}}^{\mathsf{pos}}}[\hat{N}] = \mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N^2]/\mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N]$ by Lemma G.1, the corresponding expected supply (with respect to the platform's posterior belief μ_0^{pos}) is given by

$$V(\mu_0^{\mathsf{pos}},p_0) := f\left(\mu_{\mathsf{p}}^{\mathsf{pos}}\right)\cdot\mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N] = \min\left\{ \frac{R^{-1}(p_0)-1}{\mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N^2]/\mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N]-1}, 1 \right\}\cdot\mathbb{E}_{N\sim\mu_0^{\mathsf{pos}}}[N]. \tag{G.5}$$

[20] Under public mechanisms, $f(\mu_{\mathsf{p}}^{\mathsf{pos}})=0$ cannot be the equilibrium for any $\mu_{\mathsf{p}}^{\mathsf{pos}}$. This is because, under the public mechanism, all providers have the same posterior beliefs. If $f(\mu_{\mathsf{p}}^{\mathsf{pos}})=0$, it means that no providers choose to join. However, since $R(1)-p_0>0$, any provider would benefit by deviating from the strategy of not joining.

Given a public mechanism, a realized state n and a realized public signal s will induce providers' posterior beliefs. Since the state and the signal are randomly drawn, the posterior beliefs will also be random. Let $\phi \in \Delta(\Delta(\Omega))$ be the distribution over posterior belief μ_0^{pos} under some public mechanism. Then, for a given price p_0, the optimal public mechanism can be found by solving the following problem:

$$\max_{\phi \in \Delta(\Delta(\Omega))} \mathbb{E}_{\mu_0^{\mathsf{pos}} \sim \phi}[V(\mu_0^{\mathsf{pos}}, p_0)] \tag{G.6}$$

$$\text{s.t.} \quad \mathbb{E}_{\mu_0^{\mathsf{pos}} \sim \phi}[\mu_0^{\mathsf{pos}}] = \mu_0, \tag{G.7}$$

where the constraint is to guarantee that ϕ is Bayes-plausible (or in other words, the expected posterior belief is equal to the prior since the belief is a martingale) (Kamenica and Gentzkow 2011).

Proof of Lemma G.1. This result is directly implied by Proposition 1 in Alonso and Câmara (2016) and Lemma 1 in Section 2. □

Proof of Theorem 2. For notational simplicity, let $r_0 := R^{-1}(p_0)$. In what follows, we write the platform's posterior belief μ_0^{pos} simply as μ. Throughout, write, for any posterior μ,

$$m(\mu) := \mathbb{E}_\mu[N], \qquad e(\mu) := \mathbb{E}_\mu[N^2], \qquad x(\mu) := \frac{e(\mu)}{m(\mu)} \in [\underline{n}, \bar{n}].$$

Define

$$\psi(x) := \min\Big\{\frac{r_0 - 1}{x - 1}, 1\Big\}.$$

Note that this is the equilibrium joining probability (G.4), but with a different input $x \in [\underline{n}, \bar{n}]$. Based on the above definition, we can write (G.5), given a posterior belief $\mu \in \Delta(\Omega)$, as follows:

$$V(\mu) = m(\mu)\, \psi(x(\mu)),$$

where we drop p_0 in $V(\mu, p_0)$.

Proof of (i)

For Theorem(i), if $\bar{n} \le r_0$, even if all potential providers purchase the lead, each of them has utility $R(\bar{n}) - p_0 \ge 0$. Thus, under any information structure, all providers will join, leading to the first-best expected supply m_0.

Proof of (ii)

For Theorem (ii), we will show that the interim supply function $V(\mu)$ is convex in $\mu \in \Delta(\Omega)$. As a result, full information is optimal.

Since $r_0 \le \underline{n}$ we have $x(\mu) := e(\mu)/m(\mu) \ge \underline{n} \ge r_0$. Thus, we have

$$V(\mu) = m(\mu)\, \frac{r_0 - 1}{x(\mu) - 1} = (r_0 - 1)\, \frac{m(\mu)^2}{e(\mu) - m(\mu)} := (r_0 - 1)\, F(m(\mu), s(\mu)),$$

where $F : \mathcal{D} \to \mathbb{R}$, $\mathcal{D} := \{(m, s) : m > 0,\ s > 0\}$, is

$$F(m, s) := \frac{m^2}{s}.$$

We prove *joint* convexity of F in (m,s). Since the map $\mu \mapsto (m(\mu), s(\mu)) = \sum_{n\in\Omega} \mu(n)\,(n,\, n(n-1))$ is *linear* in μ, joint convexity of F implies convexity of $V(\mu)$ in μ.

Observe that, on $\mathcal{D}$,

$$\partial_m F = \frac{2m}{s}, \qquad \partial_s F = -\frac{m^2}{s^2}, \qquad \partial^2_{mm} F = \frac{2}{s}, \quad \partial^2_{ms} F = \partial^2_{sm} F = -\frac{2m}{s^2}, \quad \partial^2_{ss} F = \frac{2m^2}{s^3}.$$

Therefore, the Hessian matrix becomes

$$\nabla^2 F(m,s) = \frac{2}{s^3} \begin{bmatrix} s \\ -m \end{bmatrix} \begin{bmatrix} s & -m \end{bmatrix} \;\succeq\; 0 \qquad \text{for all } (m,s) \in \mathcal{D},$$

and for any vector $v = (v_m, v_s)$,

$$v^\top \nabla^2 F(m,s)\, v = \frac{2}{s^3} \left(s\, v_m - m\, v_s\right)^2 \;\geq\; 0.$$

Thus, F is (jointly) convex on $\mathcal{D}$.

Let $\mu_1, \mu_2 \in \Delta(\Omega)$ and $\lambda \in [0,1]$. Because $(m(\cdot), s(\cdot))$ is linear,

$$(m(\lambda\mu_1 + (1-\lambda)\mu_2),\, s(\lambda\mu_1 + (1-\lambda)\mu_2)) = \lambda(m(\mu_1), s(\mu_1)) + (1-\lambda)(m(\mu_2), s(\mu_2)).$$

By convexity of F,

$$\begin{aligned} V(\lambda\mu_1 + (1-\lambda)\mu_2) &= (r_0 - 1)\, F\big(\lambda(m(\mu_1), s(\mu_1)) + (1-\lambda)(m(\mu_2), s(\mu_2))\big) \\ &\leq (r_0 - 1)\big[\lambda F(m(\mu_1), s(\mu_1)) + (1-\lambda) F(m(\mu_2), s(\mu_2))\big] \\ &= \lambda V(\mu_1) + (1-\lambda) V(\mu_2). \end{aligned}$$

Hence $V(\mu)$ is convex in μ.

Proof of (iii).(a)

Under no information, the distribution over posteriors degenerates to μ_0. By (G.5), the expected supply is

$$\mathsf{Sp}(\text{no info}) = V(\mu_0) = \min\left\{\frac{r_0 - 1}{x_0 - 1},\; 1\right\} \cdot m_0 \overset{(a)}{=} 1 \cdot m_0 = m_0,$$

where (a) holds since $x_0 \leq r_0$. That is, no information has achieved the first-best expected supply m_0.

Proof of (iii).(b)

Let $\phi \in \Delta(\Delta(\Omega))$ be the distribution over posteriors induced by a public mechanism, and let $\hat{\mu} \sim \phi$ denote the random posterior. Bayes plausibility is $\mathbb{E}[\hat{\mu}] = \mu_0$ (expectation in the space of beliefs).

Recall

$$m(\mu) := \mathbb{E}_\mu[N], \qquad e(\mu) := \mathbb{E}_\mu[N^2], \qquad x(\mu) := \frac{e(\mu)}{m(\mu)}.$$

Because $m(\cdot)$ and $e(\cdot)$ are linear in μ,

$$\mathbb{E}\big[m(\hat{\mu})\big] = m\big(\mathbb{E}[\hat{\mu}]\big) = m(\mu_0) =: m_0, \qquad \mathbb{E}\big[e(\hat{\mu})\big] = e\big(\mathbb{E}[\hat{\mu}]\big) = e(\mu_0) =: e_0.$$

The ex-ante objective in (G.6) is

$$\mathbb{E}[V(\hat{\mu}, p_0)] = \mathbb{E}\big[m(\hat{\mu})\,\psi(x(\hat{\mu}))\big]\,.$$

Define the size-biased probability measure Q on posteriors by

$$\mathrm{d}Q(\mu) \;=\; \frac{m(\mu)}{m_0}\,\mathrm{d}\phi(\mu). \tag{G.8}$$

Note that $Q \in \Delta(\Delta(\Omega))$ is a well-defined probability measure as long as ϕ satisfies Bayes plausibility. To see this, we have

$$\int_{\mu} \mathrm{d}Q(\mu) = \int_{\mu} \frac{m(\mu)}{m_0}\mathrm{d}\phi(\mu) = \frac{\mathbb{E}_{\hat{\mu}\sim\phi}[m(\hat{\mu})]}{m_0} \overset{(a)}{=} \frac{m\left(\mathbb{E}_{\hat{\mu}\sim\phi}[\hat{\mu}]\right)}{m_0} \overset{(b)}{=} \frac{m(\mu_0)}{m_0} = 1.$$

The equality (a) holds since the first-moment $m(\mu)$ is a linear function in μ, and (b) holds by Bayes plausibility.

We now write the problem (G.6)–(G.7) in terms of Q rather than ϕ. Observe that, given any ϕ satisfying Bayes plausibility (G.7), using Q defined in (G.8), we have the ex-ante expected supply (G.6) as follows:

$$\mathbb{E}_{\hat{\mu}\sim\phi}\left[V(\hat{\mu}, p_0)\right] = \mathbb{E}_{\hat{\mu}\sim\phi}\left[m(\hat{\mu})\,\psi(x(\hat{\mu}))\right] = m_0\,\mathbb{E}_{\hat{\mu}\sim Q}\left[\psi(x(\hat{\mu}))\right].$$

We also have

$$\mathbb{E}_{\hat{\mu}\sim Q}\left[x(\hat{\mu})\right] = \frac{\mathbb{E}_{\hat{\mu}\sim\phi}\left[m(\hat{\mu})\,x(\hat{\mu})\right]}{m_0} \overset{(a)}{=} \frac{\mathbb{E}_{\hat{\mu}\sim\phi}\left[e(\hat{\mu})\right]}{m_0} \overset{(b)}{=} \frac{e\left(\mathbb{E}_{\hat{\mu}\sim\phi}[\hat{\mu}]\right)}{m_0} \overset{(c)}{=} \frac{e(\mu_0)}{m_0} = \frac{e_0}{m_0} \;=:\; x_0.$$

The equality (a) holds by the definition of $e(\mu)$, the equality (b) holds since the second moment $e(\mu)$ is a linear function in μ, and the equality (c) holds by Bayes plausibility.

To summarize, the problem (G.6)–(G.7) for finding the optimal public signaling mechanism is equivalent to

$$\max_{Q\in\Delta(\Delta(\Omega))} \; m_0 \cdot \mathbb{E}_{\hat{\mu}\sim Q}[\psi(x(\hat{\mu}))] \tag{$\mathcal{P}_{\mathsf{public-Q}}$}$$

$$\text{s.t.} \quad \mathbb{E}_{\hat{\mu}\sim Q}\left[x(\hat{\mu})\right] = x_0 \tag{G.9}$$

$$\int \frac{\mu(n)}{m(\mu)}\,\mathrm{d}Q(\mu) = \frac{\mu_0(n)}{m_0},\;\; \forall n \in \Omega. \tag{G.10}$$

Notice that the (G.10) implies (G.9); however, (G.9) does *not* imply (G.10).

Without (G.10), the problem ($\mathcal{P}_{\mathsf{public-Q}}$) reduces to the one-dimensional concavification problem

$$\max_{w\in\Delta([\underline{n},\bar{n}])} \; m_0 \cdot \mathbb{E}_{Y\sim w}[\psi(Y)] \tag{$\mathcal{P}_{\mathsf{public-ub}}$}$$

$$\text{s.t.} \quad \mathbb{E}_{Y\sim w}[Y] = x_0,$$

where we define $Y := x(\hat{\mu})$. The concavification of ($\mathcal{P}_{\mathsf{public-ub}}$) is the dashed red curve $\bar{\psi}$ in Figure G.1. However, because of the Bayes–plausibility constraint (G.10), $\bar{\psi}$ is, in general, not attainable. We will discuss when we can focus on this one-dimensional problem without loss.

Our proof is as follows. We first show that one can, without loss, scale the high side so that the *pooled low posterior* satisfies $x = r_0$ (this makes the budget bind). Then, *conditional on which states are sent high*, revealing the realized state *within* the high block maximizes value because ψ is convex on (r_0, ∞). Finally, choosing *which* states to send high reduces to a *fractional knapsack* with cost-per-weight ratio strictly decreasing in n, solved by taking the largest n first with a lottery at the cutoff.

Step 1: Binding the low side at r_0 without loss.

Given a public mechanism $(\pi, \mathbb{S})$, for each message $s \in \mathbb{S}$ let

$$q_s := \Pr[s] = \sum_{n \in \Omega} \mu_0(n)\, \pi(s \mid n), \qquad \mu^s(n) := \frac{\mu_0(n)\, \pi(s \mid n)}{q_s},$$

and define

$$m_s := \sum_n n\, \mu^s(n), \qquad e_s := \sum_n n^2\, \mu^s(n), \qquad x_s := \frac{e_s}{m_s}.$$

Bayes-plausibility implies

$$\sum_s q_s m_s = m_0, \qquad \sum_s q_s e_s = e_0, \qquad e_0 - r_0 m_0 = \sum_s q_s (e_s - r_0 m_s). \tag{G.11}$$

Partition signals by their size-biased ratio:

$$\mathsf{L} := \{s : \ x_s \le r_0\}, \qquad \mathsf{H} := \{s : \ x_s > r_0\}.$$

Aggregate *high* intensities at the state level and define the prior-weighted aggregates

$$\pi_{\mathsf{H}}(n) := \sum_{s \in \mathsf{H}} \pi(s \mid n), \qquad A_1 := \sum_n \mu_0(n)\, n\, \pi_{\mathsf{H}}(n), \qquad A_2 := \sum_n \mu_0(n)\, n^2\, \pi_{\mathsf{H}}(n). \tag{G.12}$$

Using $q_s\, \mu^s(n) = \mu_0(n)\, \pi(s \mid n)$ and swapping sums gives

$$A_1 = \sum_{s \in \mathsf{H}} q_s m_s, \qquad A_2 = \sum_{s \in \mathsf{H}} q_s e_s. \tag{G.13}$$

By definition, we have, $Q(\{\mu^s\}) = \frac{q_s m_s}{m_0}$. Hence

$$Q(\mathsf{H}) = \sum_{s \in \mathsf{H}} \frac{q_s m_s}{m_0} = \frac{A_1}{m_0} =: w_{\mathsf{H}},$$

and

$$\mathbb{E}_Q[x \mid \mathsf{H}] = \frac{\sum_{s \in \mathsf{H}} Q(\{\mu^s\})\, x_s}{Q(\mathsf{H})} = \frac{\sum_{s \in \mathsf{H}} \frac{q_s m_s}{m_0} \cdot \frac{e_s}{m_s}}{A_1/m_0} = \frac{\sum_{s \in \mathsf{H}} q_s e_s}{A_1} = \frac{A_2}{A_1}, \tag{G.14}$$

which establishes (G.15):

$$w_{\mathsf{H}} := Q(\mathsf{H}) = \frac{A_1}{m_0}, \qquad \mathbb{E}_Q[x \mid \mathsf{H}] = \frac{A_2}{A_1} \quad (A_1 > 0). \tag{G.15}$$

From (G.11),

$$e_0 - r_0 m_0 = \sum_{s \in \mathsf{H}} q_s (e_s - r_0 m_s) \ + \sum_{s \in \mathsf{L}} q_s (e_s - r_0 m_s)$$

$$= \underbrace{(A_2 - r_0 A_1)}_{\text{high block by (G.13)}} + \underbrace{\sum_{s \in \mathsf{L}} q_s m_s (x_s - r_0)}_{\leq 0}, \tag{G.16}$$

hence for *every* mechanism

$$A_2 - r_0 A_1 \;\geq\; e_0 - r_0 m_0. \tag{G.17}$$

We refer to the above inequality as "*budget inequality*".

Pooling all low messages (which does not change value because $\psi(x) = 1$ for $x \leq r_0$) yields a single low posterior with

$$M_{\mathsf{L}} := \sum_{s \in \mathsf{L}} q_s m_s = m_0 - A_1, \qquad E_{\mathsf{L}} := \sum_{s \in \mathsf{L}} q_s e_s = e_0 - A_2,$$

so its size-biased ratio is

$$x_{\mathsf{L}} = \frac{E_{\mathsf{L}}}{M_{\mathsf{L}}} = \frac{e_0 - A_2}{m_0 - A_1}. \tag{G.18}$$

Fix $\lambda \in (0, 1]$ and define a modified mechanism by *uniformly* scaling the intensities of *all* high messages:

$$\pi'_{\mathsf{H}}(n) := \lambda\, \pi_{\mathsf{H}}(n), \qquad \pi'_{\mathsf{L}}(n) := 1 - \pi'_{\mathsf{H}}(n).$$

(Concretely, shrink each $\pi(s \mid n)$ for $s \in \mathsf{H}$ by λ and absorb the residual $(1-\lambda)\pi_{\mathsf{H}}(n)$ into the pooled low message; posteriors μ^s on each s remain unchanged because both numerator and q_s scale by λ.) By linearity in π,

$$A'_1 = \sum_n \mu_0(n)\, n\, \pi'_{\mathsf{H}}(n) = \lambda A_1, \qquad A'_2 = \sum_n \mu_0(n)\, n^2\, \pi'_{\mathsf{H}}(n) = \lambda A_2.$$

Thus, by (G.14), the conditional expectation $\mathbb{E}[x \mid \mathsf{H}]$ under the new signaling mechanism π' does not change.

Since the *total* first/second moments remain (m_0, e_0), the pooled low block under the scaled mechanism has

$$M'_{\mathsf{L}} = \sum_{s \in \mathsf{L}'} q'_s m'_s = m_0 - A'_1, \qquad E'_{\mathsf{L}} = \sum_{s \in \mathsf{L}'} q'_s e'_s = e_0 - A'_2.$$

Therefore its size-biased ratio is

$$x'_{\mathsf{L}} = \frac{E'_{\mathsf{L}}}{M'_{\mathsf{L}}} = \frac{e_0 - A'_2}{m_0 - A'_1} = \frac{e_0 - \lambda A_2}{m_0 - \lambda A_1}. \tag{G.19}$$

The map $\lambda \mapsto x'_{\mathsf{L}}(\lambda) = (e_0 - \lambda A_2)/(m_0 - \lambda A_1)$ is continuous on $[0, 1)$ (the denominator is > 0 there) and

$$\lim_{\lambda \downarrow 0} x'_{\mathsf{L}}(\lambda) = \frac{e_0}{m_0} = x_0.$$

Moreover, $x'_{\mathsf{L}}(1) = x_{\mathsf{L}} \leq r_0$ (it is a size-biased average of $\{x_s\}_{s \in \mathsf{L}} \subset (\underline{n}, r_0]$). Thus, in the non-trivial case $x_0 > r_0$, by the intermediate value theorem there exists $\lambda^* \in (0, 1]$ with $x'_{\mathsf{L}}(\lambda^*) = r_0$.

At such λ^*,

$$x'_{\mathsf{L}} = r_0 \iff \frac{e_0 - A'_2}{m_0 - A'_1} = r_0 \iff A'_2 - r_0 A'_1 = e_0 - r_0 m_0,$$

i.e., the budget holds with equality:

$$A'_2 - r_0 A'_1 = e_0 - r_0 m_0 \qquad \text{(binding when the low support sits at } x'_{\mathsf{L}} = r_0\text{)}. \tag{G.20}$$

The ex-ante value can be written under Q as

$$m_0\, \mathbb{E}_{\hat{\mu}\sim Q}\left[\psi(x(\hat{\mu}))\right] = m_0\Big((1 - w'_{\mathsf{H}}) \cdot 1 + w'_{\mathsf{H}}\, \mathbb{E}[\psi(x) \mid \mathsf{H}]\Big), \qquad w'_{\mathsf{H}} = \frac{A'_1}{m_0} = \lambda \frac{A_1}{m_0}.$$

Here $\mathbb{E}[\psi(x) \mid \mathsf{H}] \le 1$ because $\psi(x) = 1$ on $x \le r_0$ and $\psi(x) = (r_0 - 1)/(x - 1) < 1$ on $x > r_0$. Thus, decreasing λ (while keeping the high conditional mean unchanged) *weakly increases* the value.

To summarize, w.l.o.g., we can restrict to mechanisms with a *single* low message L satisfying $x(\mu^{\mathsf{L}}) = r_0$, which also leads to (G.20).

Step 2: For a fixed high selection, full revelation on the high block is optimal.

Recall that, given a signaling mechanism $(\pi, \mathbb{S})$, we define the *high set of signals* by

$$\mathsf{H} := \{\, s \in \mathbb{S} :\ x_s > r_0 \,\},$$

where $x_s := e_s/m_s$ is the posterior size-biased ratio under signal s. Assume we have fixed the aggregate high intensities $\pi_{\mathsf{H}}(\cdot)$ from Step 1, hence the aggregates

$$A_1 := \sum_{t \in \mathsf{H}} q_t m_t, \qquad A_2 := \sum_{t \in \mathsf{H}} q_t e_t,$$

are fixed and the single low message L satisfies $x(\mu^{\mathsf{L}}) = r_0$.

Fix any $s \in \mathsf{H}$ with posterior μ^s and $x_s > r_0$. Define a refined mechanism $(\pi', \mathbb{S}')$ by *replacing* the label s with state-labelled copies and leaving all other signals unchanged:

$$\mathbb{S}' := (\mathbb{S} \setminus \{s\}) \ \cup\ \{\hat{s}_n :\ n \in \Omega\},$$

$$\pi'(\hat{s}_n \mid m) := \pi(s \mid m)\, \mathbf{1}\{m = n\} \quad \text{for all } n, m \in \Omega, \qquad \pi'(u \mid m) := \pi(u \mid m) \quad \text{for all } u \in \mathbb{S}\setminus\{s\},\ m \in \Omega.$$

Thus, whenever the original mechanism would send s, the refined mechanism sends the state label $\hat{s}_n$ that *reveals* the realized state n.

Under the new signaling mechanism $(\pi', \mathbb{S}')$, for each $n \in \Omega$,

$$q'_{\hat{s}_n} = \sum_{m \in \Omega} \mu_0(m)\, \pi'(\hat{s}_n \mid m) = \mu_0(n)\, \pi(s \mid n), \quad \mu^{\hat{s}_n} = \delta_n, \quad m_{\hat{s}_n} = n, \quad e_{\hat{s}_n} = n^2.$$

Hence

$$\sum_{n \in \Omega} q'_{\hat{s}_n}\, m_{\hat{s}_n} = q_s m_s, \qquad \sum_{n \in \Omega} q'_{\hat{s}_n}\, e_{\hat{s}_n} = q_s e_s.$$

Therefore, we have

$$A'_1 = \sum_{t \in \mathsf{H}'} q'_t m'_t, \qquad A'_2 = \sum_{t \in \mathsf{H}'} q'_t e'_t,$$

are unchanged by construction, where $\mathsf{H}' = (\mathsf{H} \setminus \{s\}) \cup \{\hat{s}_n : \ n \in \Omega\}$ for accounting. In particular $A_1' = A_1$ and $A_2' = A_2$.

Before refinement, under the signaling mechanism $(\pi, \mathbb{S})$, the contribution of s to the expected supply $m_0 \, \mathbb{E}_Q[\psi(x)]$ (recall that $\mathrm{d}Q(\mu) = \frac{m(\mu)}{m_0}\mathrm{d}\phi(\mu)$) is

$$\frac{q_s m_s}{m_0} \, \psi(x_s).$$

Define the size-biased weights (for the posterior under s)

$$w_n^s \ := \ \frac{\mu^s(n)\, n}{m_s}, \qquad \sum_{n\in\Omega} w_n^s = 1, \qquad x_s = \sum_{n\in\Omega} w_n^s \, n.$$

Since $\psi(x) = \min\{\frac{r_0 - 1}{x-1}, 1\}$ is decreasing and convex on (r_0, ∞), Jensen's inequality yields

$$\psi(x_s) \ \leq \ \sum_{n\in\Omega} w_n^s \, \psi(n).$$

After refinement, under the new signaling mechanism $(\pi', \mathbb{S}')$, the contribution $\{\hat{s}_n : n \in \Omega\}$ becomes

$$\sum_{n\in\Omega} \frac{q'_{\hat{s}_n} m_{\hat{s}_n}}{m_0} \, \psi(n) = \frac{q_s m_s}{m_0} \sum_{n\in\Omega} w_n^s \, \psi(n) \ \geq \ \frac{q_s m_s}{m_0} \, \psi(x_s).$$

Thus, refining s (state revelation along s) *weakly increases* the objective while leaving (A_1, A_2) and $x(\mu^{\mathsf{L}})$ unchanged.

Applying the refinement to every $s \in \mathsf{H}$ (one by one) yields a mechanism that, for the fixed aggregate high selection $\pi_{\mathsf{H}}(\cdot)$, *fully reveals the realized state on the "high block"* and attains a (weakly) larger ex-ante value. Hence, w.l.o.g., we can take the "high messages" to be state labels.

Step 3: Choosing the high selection is a fractional knapsack.

After Step 2 we restrict attention to mechanisms with a single low message L and state–revealing "high" messages. To avoid overloading notation used in Step 1 (where in Step 1, $\mathsf{H} = \{s : \ x_s > r_0\}$ denoted a *mechanism–dependent partition* by the realized posterior ratio), we now introduce explicit labels for the high messages.

Let $\Omega_H \subseteq \Omega$ be the set of states that may be announced as "high" with positive probability. For each $n \in \Omega_H$ let s_n denote the signal that *reveals* state n, i.e. $\mu^{s_n} = \delta_n$. The signal alphabet is

$$\mathbb{S} \ = \ \{\mathsf{L}\} \ \cup \ \mathbb{S}_H, \qquad \mathbb{S}_H \ := \ \{s_n : \ n \in \Omega_H\}.$$

Define the high–selection intensity at state n by

$$\pi_H(n) \ := \ \pi(s_n \,|\, n) \ ,$$

with the convention that $\pi(s_n \,|\, n) = 0$ if $n \notin \Omega_H$.

Masses and moments under state revelation. For each $n \in \Omega$, the probability of sending s_n is

$$q_{s_n} \ = \ \sum_{m\in\Omega} \mu_0(m)\, \pi(s_n \,|\, m) = \mu_0(n)\, \pi(s_n \,|\, n) = \mu_0(n)\, \pi_H(n).$$

Since $\mu^{s_n} = \delta_n$, we have $m_{s_n} = n$ and $e_{s_n} = n^2$. Under the size–biased law Q from Step 1,

$$Q(\{s_n\}) = \frac{q_{s_n} m_{s_n}}{m_0} = \frac{\mu_0(n)\, n\, \pi_H(n)}{m_0}. \tag{G.21}$$

Define the aggregate first/second "high" moments

$$A_1 := \sum_{n\in\Omega} \mu_0(n)\, n\, \pi_H(n), \qquad A_2 := \sum_{n\in\Omega} \mu_0(n)\, n^2\, \pi_H(n). \tag{G.22}$$

By Step 1 (low pinned at $x = r_0$), the binding budget is

$$A_2 - r_0 A_1 = \sum_{n\in\Omega} \mu_0(n)\, n\, (n - r_0)\, \pi_H(n) = \Delta := e_0 - r_0 m_0. \tag{G.23}$$

Objective under state revelation. Because $\psi(n) = 1$ for $n \le r_0$ and $\psi(n) = \frac{r_0 - 1}{n-1}$ for $n > r_0$,

$$\begin{aligned} \mathbb{E}[V] &= m_0 \Big((1 - w_H) \cdot 1 + \sum_{n\in\Omega} Q(\{s_n\})\, \psi(n) \Big) \\ &= m_0 - \underbrace{\sum_{n\in\Omega} \mu_0(n)\, n\, \pi_H(n)}_{=A_1} + \sum_{n>r_0} \mu_0(n)\, n\, \pi_H(n)\, \frac{r_0 - 1}{n-1}, \end{aligned} \tag{G.24}$$

where $w_H = Q(\mathbb{S}_H) = A_1/m_0$. Equivalently,

$$\mathbb{E}[V] = m_0 - \sum_{n>r_0} \mu_0(n)\, n\, \pi_H(n)\, \frac{n - r_0}{n-1} - \sum_{n\le r_0} \mu_0(n)\, n\, \pi_H(n). \tag{G.25}$$

Dominance at $n \le r_0$. If $\pi_H(n) > 0$ for some $n \le r_0$, the last term in (G.25) strictly lowers $\mathbb{E}[V]$ while the budget (G.23) weakens (its LHS decreases because $n - r_0 \le 0$). Hence, at optimum

$$\pi_H(n) = 0 \qquad \forall\, n \le r_0. \tag{G.26}$$

Using (G.26), the objective reduces to

$$\mathbb{E}[V] = m_0 - \sum_{n>r_0} \mu_0(n)\, n\, \pi_H(n)\, \frac{n - r_0}{n-1}, \tag{G.27}$$

subject to the binding budget (G.23).

Fractional knapsack formulation and solution. Let

$$c(n) := \mu_0(n)\, n\, \frac{n - r_0}{n-1}, \qquad b(n) := \mu_0(n)\, n\, (n - r_0), \qquad (n > r_0).$$

Maximizing (G.27) is equivalent to

$$\begin{aligned} \min_{\{\pi_H(n)\in[0,1]\}} \ & \sum_{n>r_0} c(n)\, \pi_H(n) \\ \text{s.t.} \ & \sum_{n>r_0} b(n)\, \pi_H(n) = \Delta. \end{aligned} \tag{ContinuousKnapsack}$$

The cost–per–weight ratio satisfies

$$\frac{c(n)}{b(n)} = \frac{1}{n-1},$$

which is strictly decreasing in n. Hence, by Dantzig (1957), the greedy solution is optimal: take states in strictly decreasing n until the weight would exceed Δ, then take a fraction at the cutoff.

Formally, let $\Omega_+ := \{n \in \Omega : \ n > r_0\}$ ordered $n_{(1)} > \cdots > n_{(L)}$, put $b_i := b(n_{(i)})$, $C_j := \sum_{i=1}^{j} b_i$, and

$$j^* := \min\{j : \ C_j \geq \Delta\}, \qquad \alpha := \frac{\Delta - C_{j^*-1}}{b_{j^*}} \in [0,1].$$

Then the optimal selection intensities are

$$\pi_H^*(n) = \begin{cases} 1, & n \in \{n_{(1)}, \ldots, n_{(j^*-1)}\}, \\ \alpha, & n = n_{(j^*)}, \\ 0, & \text{otherwise (in particular for all } n \leq r_0). \end{cases}$$

Let $A_1^* := \sum_n \mu_0(n)\, n\, \pi_H^*(n)$. Plugging π_H^* into (G.24) yields

$$\mathbb{E}[V^*] = m_0 - A_1^* + (r_0 - 1) \sum_{n > r_0} \frac{\mu_0(n)\, n\, \pi_H^*(n)}{n-1}.$$

Step 4: Implementation and optimal value.

We now show that the probabilistic lower censorship mechanism $\mathsf{PLC}(n_{(j^*)}, \alpha)$ achieves the expected supply in Step 3. Let $\pi_H^*(\cdot)$ be the solution from Step 3. Define the signal alphabet and rule

$$\mathbb{S} = \{\mathsf{L}\} \cup \mathbb{S}_H, \qquad \mathbb{S}_H := \{s_n : \ \pi_H^*(n) > 0\}, \qquad \pi(s_n \mid m) = \pi_H^*(n)\, \mathbf{1}\{m = n\}, \quad \pi(\mathsf{L} \mid m) = 1 - \pi_H^*(m). \tag{G.28}$$

Thus, when state n realizes, the mechanism reveals n with probability $\pi_H^*(n)$ and sends L otherwise. Bayes plausibility holds because for each n,

$$q_{s_n} = \sum_m \mu_0(m)\pi(s_n \mid m) = \mu_0(n)\pi_H^*(n), \qquad q_{\mathsf{L}} = \sum_m \mu_0(m)\, \pi(\mathsf{L} \mid m) = 1 - \sum_n \mu_0(n)\pi_H^*(n),$$

and $q_{s_n}\, \mu^{s_n} = q_{s_n}\, \delta_n$, while

$$q_{\mathsf{L}}\, \mu^{\mathsf{L}}(n) = \mu_0(n)\, [1 - \pi_H^*(n)] \quad \Rightarrow \quad \sum_{s \in \mathbb{S}} q_s\, \mu^s = \mu_0.$$

Binding budget and the low posterior. Write

$$A_1^* := \sum_n \mu_0(n)\, n\, \pi_H^*(n), \qquad A_2^* := \sum_n \mu_0(n)\, n^2\, \pi_H^*(n). \tag{G.29}$$

By Step 3, $\sum_n \mu_0(n)\, n(n - r_0)\, \pi_H^*(n) = \Delta = e_0 - r_0 m_0$, i.e.

$$A_2^* - r_0 A_1^* = \Delta. \tag{G.30}$$

Since there is a *single* low message L,

$$q_{\mathsf{L}}\, m_{\mathsf{L}} = \sum_{t = \mathsf{L}} q_t m_t = m_0 - A_1^*, \qquad q_{\mathsf{L}}\, e_{\mathsf{L}} = \sum_{t = \mathsf{L}} q_t e_t = e_0 - A_2^*,$$

hence

$$x(\mu^{\mathsf{L}}) = \frac{e_{\mathsf{L}}}{m_{\mathsf{L}}} = \frac{q_{\mathsf{L}} e_{\mathsf{L}}}{q_{\mathsf{L}} m_{\mathsf{L}}} = \frac{e_0 - A_2^*}{m_0 - A_1^*} = \frac{r_0(m_0 - A_1^*)}{m_0 - A_1^*} = r_0, \tag{G.31}$$

where we used (G.30).

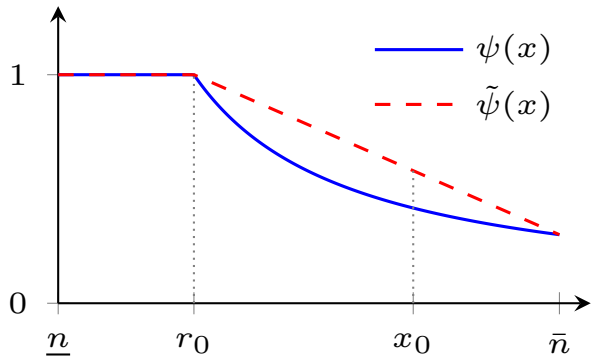
Expected supply. Under the size-biased law Q, $Q(\{s_n\}) = \dfrac{\mu_0(n)\, n\, \pi_H^*(n)}{m_0}$ and $Q(\mathsf{L}) = 1 - \dfrac{A_1^*}{m_0}$, so using $\psi(n) = 1$ for $n \le r_0$ and $\psi(n) = \dfrac{r_0 - 1}{n-1}$ for $n > r_0$,

$$\begin{aligned}
\mathbb{E}[V^*] &= m_0 \Big(Q(\mathsf{L}) \cdot 1 + \sum_n Q(\{s_n\})\, \psi(n) \Big) \\
&= m_0 - \underbrace{\sum_n \mu_0(n)\, n\, \pi_H^*(n)}_{=A_1^*} + \sum_{n > r_0} \mu_0(n)\, n\, \pi_H^*(n)\, \frac{r_0 - 1}{n-1} \\
&= \; m_0 - A_1^* + (r_0 - 1) \sum_{n > r_0} \frac{\mu_0(n)\, n\, \pi_H^*(n)}{n-1}. \qquad \text{(G.32)}
\end{aligned}$$

Special case.

If $j^* = 1$, see Step 4, (equivalently $\Delta \le \mu_0(\bar{n})\, \bar{n}(\bar{n} - r_0)$), which holds under binary states, then only the top state is selected with positive intensity and $\mathbb{S} = \{\mathsf{L}, \sigma_{\bar{n}}\}$. In this case, we have the expected supply under the optimal public signaling mechanism as $m_0 \left(1 - \frac{x_0 - r_0}{\bar{n} - 1}\right)$. This coincides with the $\tilde{\psi}(x_0)$, where $\tilde{\psi}$ is the concave closure of the function ψ in the range $[\underline{n}, \bar{n}]$, see Figure G.1 for an illustration.

Figure G.1: Concave Closure ($\underline{n} < r_0 < \bar{n}$, $r_0 < x_0 < \bar{n}$ and $\frac{e_0 - r_0 m_0}{\mu_0(\bar{n})\bar{n}(\bar{n} - r_0)} \le 1$)



Note. When $\frac{e_0 - r_0 m_0}{\mu_0(\bar{n})\bar{n}(\bar{n} - r_0)} \le 1$, one can drop the constraint (G.10) in problem ($\mathcal{P}_{\mathsf{public-Q}}$) without loss of feasibility. Thus, the problem ($\mathcal{P}_{\mathsf{public-Q}}$) reduces to the one-dimensional problem ($\mathcal{P}_{\mathsf{public-ub}}$). In this case, the optimal expected supply is $m_0 \cdot \tilde{\psi}(x_0)$, where $\tilde{\psi}$ is the concave closure of ψ.

This completes the proof. □

G.2 Proofs for Section 4 and Section 5

Proof of Theorem 3. By the discussion in Section 4 that leads to the theorem, finding the joint optimal pricing and signaling mechanism to ($\mathcal{P}_{\mathsf{rev}}$) is equivalent to solving problem (6). By Assumption 1 (ii), $xR(x)$ is increasing for $x \in [0, \tau^*]$ and decreasing for $x \ge \tau^*$. Hence we get the optimal σ^* in Theorem 3. The corresponding optimal fixed price p_0^* can be solved from (5). □

Proof of Proposition 1 and Corollary 2. Based on the analysis before Proposition 1, we know that ($\mathcal{P}_{\mathsf{cp}}$) can be simplified as (10) for each state $n \in \Omega$. With $f^*(n)$ solved from (10), the middle case of (9) yields the optimal price in state n:

$$p^*(n) = \mathbb{E}_{k\sim \mathsf{Bin}(n-1, f^*(n))}[R(1+k)]. \tag{G.33}$$

The optimal expected number of participating providers in state n is

$$x^*(n) = \mathbb{E}_{k\sim \mathsf{Bin}(n, f^*(n))}[k] = n \cdot f^*(n). \tag{G.34}$$

The optimal expected revenue in state n becomes $\mathsf{REV}^*_{\mathsf{cp}}(n) = p^*(n) \cdot x^*(n)$. This completes the proof of Proposition 1.

For a linear reward function $R(\cdot)$, $\mathbb{E}_k[R(1+k)] = R(1+\mathbb{E}k)$, so problem (10) is simplified to

$$\max_{f(n)\in[0,1]} \; R\big(1+(n-1)\cdot f(n)\big)\cdot f(n).$$

Observe that the above objective is concave and quadratic in $f(n)$. It can be shown that

$$f^*(n) = \min\left\{1,\; \frac{R^{-1}(0)-1}{2(n-1)}\right\} = \begin{cases} 1, & n \le \frac{R^{-1}(0)+1}{2}, \\ \frac{R^{-1}(0)-1}{2(n-1)}, & n > \frac{R^{-1}(0)+1}{2}. \end{cases}$$

With $f^*(n)$, we can derive $p^*(n)$ and $x^*(n)$ by (G.33) and (G.34), respectively; and thus $\mathsf{REV}^*_{\mathsf{cp}} = p^*(n)\cdot x^*(n)$. This completes the proof of Corollary 2. □

Proof of Theorem 4. We first provide the setup and problem formulation where the platform decides on both the (possibly randomized) state-dependent price and signaling mechanism. We then simplify the problem by means of obedience mechanisms. Finally, we show that the signaling mechanism with fixed pricing is optimal.

Setup and problem formulation

Let $P \subseteq \mathbb{R}_+$ be the set of commission fees (prices). A (randomized) state-dependent pricing $\psi : \Omega \to \Delta(P)$ maps each state $n \in \Omega$ to a distribution over the set of prices. For notational simplicity, we assume P contains countably many different prices, so $\psi(p \mid n)$ is the probability that the platform charges price $p \in P$ in state $n \in \Omega$. For P containing uncountably many (e.g. continuous) price levels, the following derivation follows by changing summations to integrals and interpreting $\psi(p \mid n)$ as a probability density function.

The state-dependent pricing ψ has two roles: (i) it influences provider utilities and platform revenue, and (ii) it acts as a *public* signaling mechanism since the platform charges the same price to all n providers in state n.

In addition to the public information revealed to all providers via state-dependent pricing, the platform can also adopt an additional private signaling mechanism $\hat{\pi} : \Omega \times P \to \Delta(\mathbb{S}^n)$, where recall that $\mathbb{S}^n$ represents the nth Cartesian power of some finite signal space $\mathbb{S}$ when the underlying state is $n \in \Omega$. Specifically, $\hat{\pi}(s \mid n, p)$ is the probability that the platform sends signal vector $s^n \in \mathbb{S}^n$ when

the underlying state is n and the price is p. Note that each provider $i \in [n]$ *only privately* observes s_i rather than the whole signal vector s^n.

For example, under the signaling mechanism with fixed pricing introduced in Section 2, we have $\psi(p_0 \mid n) = 1$ for all $n \in \Omega$ and $\hat{\pi}$ degenerates to $\pi : \Omega \to \Delta(\mathbb{S}^n)$. For contingent pricing with full information analyzed in Section 5.1, we have $\psi(p(n) \mid n) = 1$ for state-dependent pricing function $p(\cdot)$, and $\hat{\pi}(n\mathbb{1} \mid n, p(n)) = 1$ for all $n \in \Omega$, where $n\mathbb{1}$ represents the n-dimensional constant vector of n under signal space $\mathbb{S} = \Omega$.

Notice that it is equivalent to represent the platform's decision as a single mechanism

$$\kappa : \Omega \to \Delta(P \times \mathbb{S}^n),$$

which maps each state $n \in \Omega$ to a distribution over prices and private signal profiles. Thus, for each n, the mechanism $\kappa(\cdot \mid n)$ directly specifies a joint distribution over $(p, s^n) \in P \times \mathbb{S}^n$. By contrast, our earlier formulation uses a pair of decision rules

$$\psi : \Omega \to \Delta(P) \quad \text{and} \quad \hat{\pi} : \Omega \times P \to \Delta(\mathbb{S}^n),$$

where $\psi(\cdot \mid n)$ is the distribution over prices in state n, and $\hat{\pi}(\cdot \mid n, p)$ is the distribution over private signal profiles conditional on state n and price p. These two representations are equivalent. Given $(\psi, \hat{\pi})$, we obtain a joint distribution

$$\kappa(p, s^n \mid n) \;=\; \psi(p \mid n)\, \hat{\pi}(s^n \mid n, p).$$

Conversely, given κ, we can recover ψ as the marginal over prices and $\hat{\pi}$ as the conditional distribution of signals given (n, p).

In this situation, providers' joining strategy $\hat{f} : \mathbb{S} \times P \to [0, 1]$ maps from the received signal and price to a joining probability. The equilibrium definition for $\hat{f}$ is similar to that in Definition 1 under fixed pricing.

Definition 5. Given the state-dependent pricing ψ and signaling mechanism $\hat{\Sigma} = (\hat{\pi}, \mathbb{S})$, a joining strategy $\hat{f} : \mathbb{S} \times P \to [0, 1]$ is any equilibrium if and only if for any interested provider i, price $p \in P$ and signal $s_i \in \mathbb{S}$,

$$\hat{f}(s_i, p) = \begin{cases} 1, & \mathbb{E}^{\psi, \hat{\Sigma}, \hat{f}}[h(1 + 1^\top a^N_{-i}, p) \mid s_i, p] > 0, \\ 0, & \mathbb{E}^{\psi, \hat{\Sigma}, \hat{f}}[h(1 + 1^\top a^N_{-i}, p) \mid s_i, p] < 0, \end{cases}$$

otherwise, $\hat{f}(s_i, p) \in [0, 1]$.

The timeline is as follows:

(i) The platform commits to a state-dependent pricing ψ and a signaling mechanism $\hat{\Sigma} = (\hat{\pi}, \mathbb{S})$.

(ii) The state n is realized.

(iii) The price $p \sim \psi(\cdot \mid n)$ is realized, and the signal vector $s^n \sim \hat{\pi}(\cdot \mid n, p)$ is realized.

(iv) All providers observe p. Provider $i \in [n]$ privately observes s_i.

(v) providers update their beliefs from (s_i, p) using Bayes' rule and act according to the equilibrium strategy $\hat{f}$.

(vi) The total number of participating providers $k = \sum_{i=1}^{n} a_i$ is realized, where $a_i \sim \mathsf{Bernoulli}(\hat{f}(s_i, p))$. Each provider obtains utility $h(k, p) = R(k) - p$. The platform receives revenue $p \cdot k$.

The problem of finding the jointly optimal signaling mechanism and (randomized) state-dependent pricing is formulated as

$$\begin{aligned} \max_{\psi, \hat{\Sigma}} \quad & \sum_{n \in \Omega} \mu_0(n) \cdot \sum_{p \in P} \psi(p \mid n) \cdot p \sum_{s^n \in \mathbb{S}^n} \hat{\pi}(s^n \mid n, p) \sum_{i \in [n]} \hat{f}(s_i, p) \\ \text{s.t.} \quad & \hat{f} \text{ forms an equilibrium by Definition 5 under } (\psi, \hat{\Sigma}). \end{aligned}$$

Obedient formulation

By similar argument as for Lemma 2, we can restrict $\hat{\pi}$ to an obedient mechanism $\hat{\sigma} : \Omega \times P \to \Delta([n]^+)$, where $\hat{\sigma}(k \mid n, p) \in [0, 1]$ is the probability that k of the n interested providers are recommended to join when the price is p. Conditional on $\hat{\sigma}(k \mid n, p)$, each provider is recommended to join (i.e., $s_i = 1$) with probability $\frac{k}{n}\hat{\sigma}(k \mid n, p)$ and not to join (i.e., $s_i = 0$) with probability $\frac{n-k}{n}\hat{\sigma}(k \mid n, p)$.

We claim that finding the jointly optimal obedient mechanism and state-dependent pricing is equivalent to:

$$\begin{aligned} \max_{\psi, \hat{\sigma}} \quad & \sum_{n \in \Omega} \mu_0(n) \sum_{p \in P} \psi(p \mid n) \cdot p \sum_{k=0}^{n} \hat{\sigma}(k \mid n, p) \cdot k & (\mathcal{P}_{\mathsf{general}}) \\ \text{s.t.} \quad & \sum_{n \in \Omega} \mu_0(n) \cdot \psi(p \mid n) \sum_{k=0}^{n} k \cdot \hat{\sigma}(k \mid n, p) \cdot (R(k) - p) \geq 0, \ \forall p \in P \\ & \sum_{n \in \Omega} \mu_0(n) \cdot \psi(p \mid n) \sum_{k=0}^{n-1} (n-k) \cdot \hat{\sigma}(k \mid n, p) \cdot (R(k+1) - p) \leq 0, \ \forall p \in P \\ & \psi(\cdot \mid n) \in \Delta(P), \ \forall n \in \Omega \\ & \hat{\sigma}(\cdot \mid n, p) \in \Delta([n]^+), \ \forall n \in \Omega, p \in P. \end{aligned}$$

Similar to formulation ($\mathcal{P}_{\mathsf{LP}}$), the first two constraints ensure providers' incentive compatibility in following the platform's recommendations upon observing the price and the signal. The last two constraints properly define the pricing and signaling mechanism.

We now show the derivation of the first constraint; the second constraint can be derived similarly and will be omitted. In what follows, we denote various probability functions about provider beliefs by μ_{p}. After observing the price $p \in P$, by Bayes' rule, the (homogeneous) providers' belief about the underlying state n is updated from the prior $\mu_{\mathsf{p}}(n)$ to

$$\mu_{\mathsf{p}}(n \mid p) = \frac{\mathbb{P}\{p \mid n\} \cdot \mu_{\mathsf{p}}(n)}{\mathbb{P}\{p\}} = \frac{\psi(p \mid n) \cdot \mu_{\mathsf{p}}(n)}{\sum_{n' \in \Omega} \psi(p \mid n') \cdot \mu_{\mathsf{p}}(n')} = \frac{\psi(p \mid n) \cdot n\mu_0(n)}{\mathbb{E}[N] \sum_{n' \in \Omega} \psi(p \mid n') \cdot \mu_{\mathsf{p}}(n')},$$

where the last equality utilizes Lemma 1. We then have the providers' belief that the platform recommends k out of n interested providers to participate after observing the price p:

$$\mu_{\mathsf{p}}(n,k\,|\,p)=\mu_{\mathsf{p}}(n\,|\,p)\cdot\mathbb{P}\{k\,|\,n,p\}=\mu_{\mathsf{p}}(n\,|\,p)\cdot\hat{\sigma}(k\,|\,n,p)=\frac{\psi(p\,|\,n)\cdot n\mu_0(n)\cdot\hat{\sigma}(k\,|\,n,p)}{\mathbb{E}[N]\sum_{n'\in\Omega}\psi(p\,|\,n')\cdot\mu_{\mathsf{p}}(n')}. \tag{G.35}$$

The providers' posterior belief that there are n interested providers and the platform recommends k of them to join when they observe price p and receive signal $s_i=1$ is given by

$$\begin{aligned}\mu_{\mathsf{p}}(n,k\,|\,p,s_i=1)&=\frac{\mathbb{P}\{s_i=1\,|\,n,k,p\}\cdot\mu_{\mathsf{p}}(n,k\,|\,p)}{\mathbb{P}\{s_i=1\,|\,p\}}\\&\overset{(a)}{=}\frac{\frac{k}{n}\cdot\hat{\sigma}(k\,|\,n,p)}{\mathbb{P}\{s_i=1\,|\,p\}}\cdot\frac{\psi(p\,|\,n)\cdot n\mu_0(n)}{\mathbb{E}[N]\sum_{n'\in\Omega}\psi(p\,|\,n')\cdot\mu_{\mathsf{p}}(n')}\\&=\frac{k\cdot\hat{\sigma}(k\,|\,n,p)\cdot\psi(p\,|\,n)\cdot\mu_0(n)}{\mathbb{P}\{s_i=1\,|\,p\}\cdot\mathbb{E}[N]\sum_{n'\in\Omega}\psi(p\,|\,n')\cdot\mu_{\mathsf{p}}(n')},\end{aligned} \tag{G.36}$$

where (a) holds by noticing that $\mathbb{P}\{s_i=1\,|\,n,k,p\}=\frac{k}{n}$ and by (G.35).

We thus have the providers' expected utility when they observe price p and receive signal $s_i=1$ as

$$\begin{aligned}\mathbb{E}^{\psi,\hat{\Sigma},\hat{f}}[h(1+1^\top a_{-i}^N,p)\,|\,p,s_i=1]&=\sum_{n\in\Omega}\sum_{k=0}^{n}\mu_{\mathsf{p}}(n,k\,|\,p,s_i=1)\cdot h(k,p)\\&=\frac{\sum_{n\in\Omega}\mu_0(n)\cdot\psi(p\,|\,n)\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\cdot h(k,p)}{\mathbb{P}\{s_i=1\,|\,p\}\cdot\mathbb{E}[N]\sum_{n'\in\Omega}\psi(p\,|\,n')\cdot\mu_{\mathsf{p}}(n')}.\end{aligned}$$

Obedience requires $\mathbb{E}^{\psi,\hat{\Sigma},\hat{f}}[h(1+1^\top a_{-i}^N,p)\,|\,p,s_i=1]\geq 0$, i.e.,

$$\sum_{n\in\Omega}\mu_0(n)\cdot\psi(p\,|\,n)\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\cdot h(k,p)\geq 0,$$

which is the first constraint in problem ($\mathcal{P}_{\mathsf{general}}$).

Signaling mechanism with fixed pricing is optimal

We show that the optimal expected revenue of problem ($\mathcal{P}_{\mathsf{general}}$) is no more than the expected revenue under optimal signaling mechanism with fixed pricing characterized in Theorem 3. By the first constraint in problem ($\mathcal{P}_{\mathsf{general}}$), for any $p\in P$, we have

$$\sum_{n\in\Omega}\mu_0(n)\cdot\psi(p\,|\,n)\cdot p\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\;\leq\;\sum_{n\in\Omega}\mu_0(n)\cdot\psi(p\,|\,n)\cdot\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\cdot R(k).$$

Summing over $p\in P$ on both sides gives

$$\sum_{n\in\Omega}\mu_0(n)\sum_{p\in P}\psi(p\,|\,n)\cdot p\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\;\leq\;\sum_{n\in\Omega}\mu_0(n)\sum_{p\in P}\psi(p\,|\,n)\cdot\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\cdot R(k).$$

The left-hand side is the expected revenue under $(\psi,\hat{\sigma})$, as in the objective. For the right-hand side, we have

$$\begin{aligned}\sum_{n\in\Omega}\mu_0(n)\sum_{p\in P}\psi(p\,|\,n)\cdot\sum_{k=0}^{n}k\cdot\hat{\sigma}(k\,|\,n,p)\cdot R(k)&\overset{(a)}{\leq}\sum_{n\in\Omega}\mu_0(n)\sum_{p\in P}\psi(p\,|\,n)\cdot\max_{k\in[n]^+}kR(k)\\&\overset{(b)}{=}\sum_{n\in\Omega}\mu_0(n)\cdot k^*(n)\cdot R(k^*(n))\\&\overset{(c)}{=}\mathsf{REV}^*,\end{aligned}$$

where (a) holds by $\sum_{k=0}^{n} k \cdot \hat{\sigma}(k \mid n,p) \cdot R(k) \le \max_{k\in[n]^+} kR(k)$, (b) holds by the definition $k^*(n) := \arg\max_{k\in[n]^+} kR(k)$, and (c) holds by Theorem 3. Notice that $k^*(n) = n$ if $n \le \tau^*$; otherwise, $k^*(n) \in \arg\max_{x\in\{\lfloor\tau^*\rfloor,\lceil\tau^*\rceil\}}\{xR(x)\}$. Thus, the optimal expected revenue REV^* in Theorem 3 can be written as $\sum_{n\in\Omega} \mu_0(n) \cdot k^*(n) \cdot R(k^*(n))$. We have shown that the expected revenue under any feasible $(\psi, \hat{\sigma})$ is no more than that under the optimal signaling mechanism with fixed pricing. This completes the proof. □

Proof of Corollary 1. The first part of Corollary 1 immediately follows from Theorem 4. In what follows, we prove the second part.

By Theorem 3, the expected revenue under the optimal signaling mechanism with fixed pricing is given by

$$\mathsf{REV}^* = \sum_{n\in\Omega, n\le\tau^*} \mu_0(n) \cdot nR(n) + \sum_{n\in\Omega, n>\tau^*} \mu_0(n) \cdot k^* R(k^*) = \sum_{n\in\Omega} \mu_0(n) \cdot Z(n),$$

where we define $Z(n)$ to be

$$Z(n) := \begin{cases} nR(n), & \text{if } n \le \tau^*, \\ k^* R(k^*), & \text{if } n > \tau^*. \end{cases} \tag{G.37}$$

We use the following result to facilitate our proof.

Lemma G.2. *For a linear reward function $R(\cdot)$, we have*

(i) if $n \le (R^{-1}(0)+1)/2$, $Z(n) = \mathsf{REV}^*_{\mathsf{cp}}(n)$;

(ii) if $n > (R^{-1}(0)+1)/2$, $Z(n) > \mathsf{REV}^*_{\mathsf{cp}}(n)$.

By Corollary 2, when $n > (R^{-1}(0)+1)/2$, $\mathsf{REV}^*_{\mathsf{cp}}(n)$ decreases in n and converges to $\frac{R^{-1}(0)-1}{2} \cdot R\left(\frac{R^{-1}(0)+1}{2}\right)$, see Figure C.4 (c). In other words,

$$\text{for } n > \frac{R^{-1}(0)+1}{2}, \quad \mathsf{REV}^*_{\mathsf{cp}}(n) \ge \frac{R^{-1}(0)-1}{2} \cdot R\left(\frac{R^{-1}(0)+1}{2}\right). \tag{G.38}$$

The inequality is strict unless $n \to \infty$.

Combing this property and Lemma G.2, we have

$$\begin{aligned}
\frac{\mathsf{REV}^*}{\mathsf{REV}^*_{\mathsf{cp}}} = \frac{\sum_{n\in\Omega} \mu_0(n) Z(n)}{\sum_{n\in\Omega} \mu_0(n) \mathsf{REV}^*_{\mathsf{cp}}(n)} &\overset{(a)}{\le} \frac{\sum_{n\in\Omega, n>\frac{R^{-1}(0)+1}{2}} \mu_0(n) Z(n)}{\sum_{n\in\Omega, n>\frac{R^{-1}(0)+1}{2}} \mu_0(n) \mathsf{REV}^*_{\mathsf{cp}}(n)} \\
&\overset{(b)}{\le} \frac{k^* R(k^*) \cdot \sum_{n\in\Omega, n>\frac{R^{-1}(0)+1}{2}} \mu_0(n)}{\frac{R^{-1}(0)-1}{2} \cdot R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \sum_{n\in\Omega, n>\frac{R^{-1}(0)+1}{2}} \mu_0(n)} \\
&= \frac{k^* R(k^*)}{\frac{R^{-1}(0)-1}{2} \cdot R\left(\frac{R^{-1}(0)+1}{2}\right)} \\
&\le \frac{\tau^* R(\tau^*)}{\frac{R^{-1}(0)-1}{2} \cdot R\left(\frac{R^{-1}(0)+1}{2}\right)} \\
&\overset{(c)}{=} \frac{1}{\left(1 - \frac{1}{R^{-1}(0)}\right)^2}.
\end{aligned}$$

Inequality (a) holds by Lemma G.2. Inequality (b) holds by (G.37) about $Z(n)$, where notice that $\tau^* = R^{-1}(0)/2$ under the linear reward function, and by (G.38) about $\mathsf{REV}^*_{\mathsf{cp}}(n)$. The equality (c) can be shown using any linearly decreasing function $R(\cdot)$ satisfying Assumption 1. Note that the gap in the inequalities (a) and (b) shrinks to 0 when all the n's in Ω grow to infinity (i.e., $\underline{n} \to \infty$), and the last inequality is tight if $\tau^* = R^{-1}(0)/2$ is an integer. This completes the proof. $\square$

Proof of Lemma G.2. For a decreasing linear reward function $R(\cdot)$, we have $\tau^* = \arg\max_{x\geq 0}\{xR(x)\} = \frac{R^{-1}(0)}{2}$. By the definition of $Z(n)$, we have

$$Z(n) = \begin{cases} nR(n), & \text{if } n \leq \tau^* = R^{-1}(0)/2, \\ k^*R(k^*), & \text{if } n > \tau^* = R^{-1}(0)/2, \end{cases} \tag{G.39}$$

where $k^* \in \arg\max_{k\in\{\lfloor\tau^*\rfloor,\lceil\tau^*\rceil\}}\{kR(k)\}$. Using the $\mathsf{REV}^*_{\mathsf{cp}}(n)$ given by Corollary 2, we have

$$\mathsf{REV}^*_{\mathsf{cp}}(n) = \begin{cases} nR(n), & \text{if } n \leq (R^{-1}(0)+1)/2, \\ R\left(\frac{R^{-1}(0)+1}{2}\right)\cdot\frac{R^{-1}(0)-1}{2(n-1)}\cdot n, & \text{if } n > (R^{-1}(0)+1)/2. \end{cases} \tag{G.40}$$

Comparing (G.39) and (G.40), we find that when $n \leq \tau^* = R^{-1}(0)/2$, $Z(n) = \mathsf{REV}^*_{\mathsf{cp}}(n) = nR(n)$. We now show that when $n \in (\tau^*, \tau^*+0.5]$ (notice that $(R^{-1}(0)+1)/2 = \tau^*+0.5$), $Z(n) = \mathsf{REV}^*_{\mathsf{cp}}(n)$; when $n > \tau^*+0.5$, $Z(n) > \mathsf{REV}^*_{\mathsf{cp}}(n)$.

(i) For $n \in (\tau^*, \tau^*+0.5]$, i.e., there exists an integer in the interval $(\tau^*, \tau^*+0.5]$, which must be$n = \lceil\tau^*\rceil$. For this to happen, we must have $\lceil\tau^*\rceil - \tau^* \in (0, 0.5]$. This implies $\lceil\tau^*\rceil - \tau^* \leq \tau^* - \lfloor\tau^*\rfloor$. Since $xR(x)$ is a concave quadratic function, we have $\lceil\tau^*\rceil \cdot R(\lceil\tau^*\rceil) \geq \lfloor\tau^*\rfloor \cdot R(\lfloor\tau^*\rfloor)$. Thus,

$$Z(n) = k^*R(k^*) = \lceil\tau^*\rceil \cdot R(\lceil\tau^*\rceil) \overset{(a)}{=} nR(n) = \mathsf{REV}^*_{\mathsf{cp}}(n),$$

where (a) holds since $\lceil\tau^*\rceil = n$ in this case.

(ii) For $n > \tau^*+0.5$, we have

$$\begin{aligned} Z(n) = k^*R(k^*) &\overset{(a)}{\geq} R\left(\frac{R^{-1}(0)+1}{2}\right)\cdot\frac{R^{-1}(0)+1}{2} \\ &\overset{(b)}{>} R\left(\frac{R^{-1}(0)+1}{2}\right)\cdot\frac{R^{-1}(0)-1}{2(n-1)}\cdot n = \mathsf{REV}^*_{\mathsf{cp}}(n). \end{aligned}$$

Inequality (a) follows from the quadratic concavity of $xR(x)$ and the fact that $|k^* - \tau^*| \leq \frac{1}{2}$ while $|\frac{R^{-1}(0)+1}{2} - \tau^*| = \frac{1}{2}$. Inequality (b) holds because

$$\frac{R^{-1}(0)+1}{2} - \frac{R^{-1}(0)-1}{2}\cdot\frac{n}{n-1} = 1 - \frac{R^{-1}(0)-1}{2}\cdot\frac{1}{n-1} > 1 - \frac{R^{-1}(0)-1}{2}\cdot\frac{2}{R^{-1}(0)-1} = 0,$$

where the inequality simply uses $n > \frac{R^{-1}(0)+1}{2}$. This completes the proof. $\square$

Proof of Proposition 2. For a linear and decreasing (by Assumption 1 (i)) reward function $R(\cdot)$, one can verify that $\tau^* = \arg\max_{x\geq 0}\{xR(x)\} = \frac{R^{-1}(0)}{2}$. Then, it follows immediately from the definitions that

$$\text{if } n \leq \tau^*, \quad \mathsf{FB}(n) = \mathsf{OPT}(n);$$

$$\text{if } n > \tau^*, \quad \mathsf{FB}(n) \geq \mathsf{OPT}(n).$$

When $n > \tau^*$, we have $\mathsf{FB}(n) = \tau^* R(\tau^*)$ and $\mathsf{OPT}(n) = \max_{k \in \{\lfloor \tau^* \rfloor, \lceil \tau^* \rceil\}} kR(k)$ by the definition of $\mathsf{OPT}(n)$ and concavity of $xR(x)$. Since τ^* is the unique maximizer of the concave function $xR(x)$, $\mathsf{FB}(n) > \mathsf{OPT}(n)$ unless τ^* is an integer. This completes the proof of the relationship between $\mathsf{FB}(n)$ and $\mathsf{OPT}(n)$, see Figure 1 for graphical illustrations.

The relationship between $\mathsf{OPT}(n)$ and $\mathsf{REV}^*_{\mathsf{cp}}(n)$ directly follows from Lemma G.2 by observing that $\mathsf{OPT}(n) = Z(n)$ for all $n \in \Omega$ with $Z(n)$ defined in (G.37). This completes the proof. □

G.3 Proofs for Section 6

G.3.1 Proofs for Section 6.1

Proof of Theorem 5 Following a similar argument to the proof of Theorem 4 in our base model, one can show that, given (μ_1, Ω_1), the expected second-stage revenue under any mechanism in $\mathcal{M}_2$ is upper bounded by

$$\mathbb{E}\big[k^*(N_1, N_2)\, R_2\big(k^*(N_1, N_2) + N_1\big)\big], \tag{G.41}$$

where the expectation is over $N_1 \sim \mu_1$ and $N_2 \sim \mu_2$. It is straightforward to verify that the mechanism described in Theorem 5 attains this upper bound: conditional on $(N_1, N_2) = (n_1, n_2)$, the mechanism implements $k^*(n_1, n_2)$ second-stage participants and charges price $p_2^*(n_1)$, so the realized second-stage revenue is $k^*(n_1, n_2)\, p_2^*(n_1)$; taking expectations over (N_1, N_2) and substituting (13) gives exactly (G.41). Thus, it remains to show that the proposed mechanism is feasible for problem (Stage-2 problem given n_1) under the concavity assumption on $xR_2(x)$.

Fix $n_1 \in \Omega_1$. For brevity, write $R(\cdot) := R_2(\cdot)$ and consider the continuous functions

$$f(x) := xR(x), \qquad g_{n_1}(x) := xR(x + n_1), \qquad x \geq 0.$$

By assumption, f is concave on $[0, \infty)$, and R is weakly decreasing.

Step 1: Concavity and unimodality of $k \mapsto kR(k + n_1)$. We first show that g_{n_1} is concave, and therefore its restriction to the integers,

$$k \mapsto g_{n_1}(k) = kR(k + n_1), \qquad k \in \mathbb{Z}_+,$$

is discretely concave and unimodal.

We have

$$f'(x) = R(x) + xR'(x), \qquad f''(x) = 2R'(x) + xR''(x),$$

and concavity of f implies

$$f''(x) = 2R'(x) + xR''(x) \leq 0, \qquad \forall x \geq 0. \tag{G.42}$$

Similarly,

$$g'_{n_1}(x) = R(x+n_1) + xR'(x+n_1), \qquad g''_{n_1}(x) = 2R'(x+n_1) + xR''(x+n_1), \qquad x \geq 0.$$

Fix any $x \geq 0$ and set

$$u := x + n_1 \quad (\text{so } u \geq n_1 \text{ and } x = u - n_1).$$

Then we can rewrite the second derivative of g_{n_1} at x as

$$g''_{n_1}(x) = 2R'(x+n_1) + xR''(x+n_1) = 2R'(u) + (u-n_1)R''(u). \tag{G.43}$$

Define

$$A := f''(u) = 2R'(u) + uR''(u), \qquad B := 2R'(u).$$

By (G.42), we have $A \leq 0$ for all $u \geq 0$, and since R is decreasing, $R'(u) \leq 0$ and hence $B \leq 0$ as well.

For $u \geq n_1$, write

$$\lambda := \frac{n_1}{u} \in [0,1],$$

so that $u - n_1 = (1-\lambda)u$. Using (G.43), we obtain

$$\begin{aligned} g''_{n_1}(x) &= 2R'(u) + (u-n_1)R''(u) \\ &= 2R'(u) + uR''(u) - n_1 R''(u) \\ &= (1-\lambda)\big[2R'(u) + uR''(u)\big] + \lambda \cdot 2R'(u) \\ &= (1-\lambda)A + \lambda B. \end{aligned}$$

Since $A \leq 0$, $B \leq 0$, and $\lambda \in [0,1]$, it follows that $g''_{n_1}(x) \leq 0$ for all $x \geq 0$. Thus, g_{n_1} is concave on $[0,\infty)$. It follows that the discrete sequence $k \mapsto g_{n_1}(k)$ is discretely concave and thus unimodal in k. Thus, there exists at least one index $\bar{k}_{n_1} \in \mathbb{Z}_+$ such that

$$g_{n_1}(0) \leq g_{n_1}(1) \leq \cdots \leq g_{n_1}(\bar{k}_{n_1}), \qquad g_{n_1}(\bar{k}_{n_1}) \geq g_{n_1}(\bar{k}_{n_1}+1) \geq g_{n_1}(\bar{k}_{n_1}+2) \geq \cdots.$$

Consequently, for any $n_2 \in \Omega_2$, the maximizer of $k \mapsto g_{n_1}(k)$ over $k \in [n_2]_+$ is

$$k^*(n_1, n_2) = \begin{cases} n_2, & \text{if } n_2 \leq \bar{k}_{n_1}, \\ \bar{k}_{n_1}, & \text{if } n_2 > \bar{k}_{n_1}. \end{cases} \tag{G.44}$$

Step 2: Feasibility of (σ_2^*, p_2^*) *for problem* (Stage-2 problem given n_1). We now check that, for this fixed n_1, the pair

$$(\sigma_2^*(\cdot \mid n_1, \cdot), p_2^*(n_1))$$

specified in Theorem 5 satisfies the feasibility, join, and not-join constraints.

(a) Feasibility. By construction,

$$\sigma_2^*(k \mid n_1, n_2) = \begin{cases} 1, & \text{if } k = k^*(n_1, n_2), \\ 0, & \text{otherwise,} \end{cases}$$

so $\sigma_2^*(\cdot \mid n_1, n_2) \in \Delta([n_2]_+)$ for all (n_1, n_2). Thus constraint (Feas) holds.

(b) Join constraint (J2). Under $(\sigma_2^*, p_2^*(n_1))$, the left-hand side of (J2) is

$$\sum_{n_2 \in \Omega_2} \mu_2(n_2)\, k^*(n_1, n_2)\, [R(k^*(n_1, n_2) + n_1) - p_2^*(n_1)].$$

Using the definition of $p_2^*(n_1)$ in (13), this expression is exactly zero whenever $\sum_{n_2} \mu_2(n_2)\, k^*(n_1, n_2) > 0$. If $\sum_{n_2} \mu_2(n_2)\, k^*(n_1, n_2) = 0$, then no second-stage supplier is ever recommended to join for this n_1, and (J2) holds trivially. Hence the join constraint is satisfied (and binding whenever the second stage is active).

(c) Not-join constraint (NJ2). Under σ_2^*, the inner sum in (NJ2) for a given n_2 simplifies to

$$(n_2 - k^*(n_1, n_2))\, [R(k^*(n_1, n_2) + 1 + n_1) - p_2^*(n_1)].$$

Using the threshold representation (G.44), we consider two cases.

Case 1: $n_2 \le \bar{k}_{n_1}$. Then $k^*(n_1, n_2) = n_2$, so $n_2 - k^*(n_1, n_2) = 0$ and the inner term is zero. Hence these states do not contribute to the left-hand side of (NJ2).

Case 2: $n_2 > \bar{k}_{n_1}$. Then $k^*(n_1, n_2) = \bar{k}_{n_1}$, and

$$\begin{aligned}
&(n_2 - \bar{k}_{n_1})\, [R(\bar{k}_{n_1} + 1 + n_1) - p_2^*(n_1)] \\
&\quad = (n_2 - \bar{k}_{n_1}) \left[R(\bar{k}_{n_1} + 1 + n_1) - \frac{\sum_{\tilde{n}_2 \in \Omega_2} \mu_2(\tilde{n}_2)\, k^*(n_1, \tilde{n}_2)\, R(k^*(n_1, \tilde{n}_2) + n_1)}{\sum_{\tilde{n}_2 \in \Omega_2} \mu_2(\tilde{n}_2)\, k^*(n_1, \tilde{n}_2)} \right] \\
&\quad = \frac{n_2 - \bar{k}_{n_1}}{\sum_{\tilde{n}_2} \mu_2(\tilde{n}_2)\, k^*(n_1, \tilde{n}_2)} \sum_{\tilde{n}_2 \in \Omega_2} \mu_2(\tilde{n}_2)\, k^*(n_1, \tilde{n}_2)\, [R(\bar{k}_{n_1} + 1 + n_1) - R(k^*(n_1, \tilde{n}_2) + n_1)].
\end{aligned}$$

For any $\tilde{n}_2$, we have $k^*(n_1, \tilde{n}_2) \le \bar{k}_{n_1}$ by (G.44), and hence

$$k^*(n_1, \tilde{n}_2) + n_1 \;\le\; \bar{k}_{n_1} + n_1 \;\le\; \bar{k}_{n_1} + n_1 + 1.$$

Since $R(\cdot)$ is weakly decreasing, it follows that

$$R(\bar{k}_{n_1} + 1 + n_1) - R(k^*(n_1, \tilde{n}_2) + n_1) \;\le\; 0 \quad \text{for all } \tilde{n}_2.$$

The denominator $\sum_{\tilde{n}_2} \mu_2(\tilde{n}_2)\, k^*(n_1, \tilde{n}_2)$ is nonnegative, and $n_2 - \bar{k}_{n_1} > 0$ in this case. Therefore, the entire expression is weakly nonpositive. Thus the inner term in (NJ2) is nonpositive for each n_2, and taking expectations over N_2 implies that the not-join constraint (NJ2) holds.

This completes the proof. □

Proof of Theorem 6. The proof is parallel to that of Theorem 5. We briefly sketch the main steps. First, it can be shown that (19) is an upper bound of the revenue that can be achieved in the space $\mathcal{M}_1$. Second, it is straightforward to show that the mechanism in Theorem 6 achieves this upper bound. Finally, under the unimodality assumption, the mechanism in Theorem 6 is indeed feasible. □

G.3.2 Proofs for Section 6.2

Proof of Theorem 7 Following an argument analogous to the proof of Theorem 4, it can be shown that, in this two-lead setting, even when the platform chooses from the most general decision space—namely a mapping $\kappa : \Omega \to \Delta(\mathbb{R}^2 \times \mathbb{S}^n)$, which assigns to each state a distribution over prices and (private) signals, the revenue is upper bounded by

$$\mathbb{E}_{N\sim\mu_0}\left[k_1^*(N)\, R_1\big(k_1^*(N)\big) + k_2^*(N)\, R_2\big(k_2^*(N)\big)\right]. \tag{G.45}$$

Here, for each $n \in \Omega$,

$$(k_1^*(n), k_2^*(n)) \in \arg\max_{k_1+k_2\le n}\ k_1 R_1(k_1) + k_2 R_2(k_2).$$

We will show that the mechanism in Theorem 7 is feasible and, under each set of conditions, attains this upper bound.

Proof of (i)

When two leads have the same reward functions, we have: when $n \ge 2\tau^*$,

$$(\tau^*, \tau^*) = \arg\max_{k_1+k_2\le n}\ k_1 R(k_1) + k_2 R(k_2).$$

Besides, when $n < 2\tau^*$, it can be shown that

$$\{(\lfloor n/2\rfloor, \lceil n/2\rceil), (\lceil n/2\rceil, \lfloor n/2\rfloor)\} \in \arg\max_{k_1+k_2\le n}\ k_1 R(k_1) + k_2 R(k_2).$$

Since p_i^* makes (Ji) binding, the revenue under the mechanism in Theorem 7 (i) exactly equals to the upper bound (G.45). We now show that the mechanism in Theorem 7 (i) is feasible.

(a) *Constraint (Ji).*

Since p_i^* makes (Ji) binding, this constraint is obviously satisfied.

(b) *Constraint* ($1 \nrightarrow 2$).

Let K_i^* be the random variable induced by $N \sim \mu_0(n)$ and $\sigma^*(\cdot, \cdot \mid n)$ in (20)–(21). By the construction, we know that $p_1^* = p_2^* := p^*$, the left-hand side of ($1 \nrightarrow 2$) simplifies to

$$\begin{aligned}\text{LHS of } (1 \nrightarrow 2) &= \mathbb{E}\Big[K_1^*\big(R(K_1^*) - p^* - R(K_2^*+1) + p^*\big)\Big] \\ &= \mathbb{E}\Big[K_1^*\big(R(K_1^*) - R(K_2^*+1)\big)\Big].\end{aligned}$$

We evaluate this state by state, conditioning on $N = n$.

- If $n \ge 2\tau^*$, then $(K_1^*, K_2^*) = (\tau^*, \tau^*)$ with probability one, so

$$\mathbb{E}\big[K_1^*(R(K_1^*) - R(K_2^*+1)) \,\big|\, N = n\big] = \tau^*\big(R(\tau^*) - R(\tau^*+1)\big) \ge 0,$$

since R is decreasing.

- If $n < 2\tau^*$ is even, write $n = 2q$. Then $(K_1^*, K_2^*) = (q, q)$ deterministically, so

$$\mathbb{E}\big[K_1^*(R(K_1^*) - R(K_2^*+1)) \,\big|\, N = n\big] = q\big(R(q) - R(q+1)\big) \ge 0.$$

- If $n < 2\tau^*$ is odd, write $n = 2q+1$. Then

$$(K_1^*, K_2^*) = \begin{cases} (q, q+1), & \text{with prob. } 1/2, \\ (q+1, q), & \text{with prob. } 1/2. \end{cases}$$

  Hence

$$\begin{aligned} &\mathbb{E}\big[K_1^*(R(K_1^*) - R(K_2^*+1)) \,\big|\, N = n\big] \\ &= \tfrac{1}{2}\, q\big(R(q) - R(q+2)\big) + \tfrac{1}{2}\,(q+1)\big(R(q+1) - R(q+1)\big) \\ &= \tfrac{1}{2}\, q\big(R(q) - R(q+2)\big) \;\geq\; 0, \end{aligned}$$

  again because R is decreasing.

In all cases, the conditional expectation given $N = n$ is nonnegative; taking expectations over $N \sim \mu_0$ shows that $(1 \nrightarrow 2)$ holds. By symmetry, the same argument (with K_1^* and K_2^* interchanged) shows that the left-hand side of $(2 \nrightarrow 1)$ is also nonnegative. Thus both cross-lead constraints are satisfied.

(c) *Constraint* (NJ1).

We show that for any n, the second summation of the left-hand side in constraint (NJ1) is non-positive.

- When $n < 2\tau^*$, since $\lfloor n/2 \rfloor + \lceil n/2 \rceil = n$,

$$\sum_{k_1+k_2 \leq n} \sigma^*(k_1, k_2 \,|\, n) \cdot (n - k_1 - k_2)\,[R_1(k_1+1) - p_1^*] = 0.$$

- When $n \geq 2\tau^*$,

$$\begin{aligned} &\sum_{k_1+k_2 \leq n} \sigma^*(k_1, k_2 \,|\, n) \cdot (n - k_1 - k_2)\,[R_1(k_1+1) - p_1^*] \\ &= (n - 2\tau^*) \cdot [R_1(\tau^*+1) - p_1^*] \\ &= (n - 2\tau^*)[R_1(\tau^*+1) \\ &\quad - \frac{\sum_{\tilde{n}<2\tau^*} (\mu_0(\tilde{n}) \cdot [0.5 \cdot \lfloor \tilde{n}/2 \rfloor R_1(\lfloor \tilde{n}/2 \rfloor) + 0.5 \cdot \lceil \tilde{n}/2 \rceil R_1(\lceil \tilde{n}/2 \rceil)]) + \sum_{\tilde{n} \geq 2\tau^*} \mu_0(\tilde{n}) \cdot \tau^* R_1(\tau^*)}{\sum_{\tilde{n}<2\tau^*} (\mu_0(\tilde{n}) \cdot [0.5 \cdot \lfloor \tilde{n}/2 \rfloor + 0.5 \cdot \lceil \tilde{n}/2 \rceil]) + \sum_{\tilde{n} \geq 2\tau^*} \mu_0(\tilde{n}) \cdot \tau^*}] \\ &\propto \sum_{\tilde{n}<2\tau^*} \frac{\mu_0(\tilde{n})}{2} \cdot \lfloor \tilde{n}/2 \rfloor \cdot (R_1(\tau^*+1) - R_1(\lfloor \tilde{n}/2 \rfloor)) + \sum_{\tilde{n}<2\tau^*} \frac{\mu_0(\tilde{n})}{2} \cdot \lceil \tilde{n}/2 \rceil \cdot (R_1(\tau^*+1) - R_1(\lceil \tilde{n}/2 \rceil)) \\ &\quad + \sum_{\tilde{n} \geq 2\tau^*} \mu_0(\tilde{n}) \cdot \tau^* \cdot [R_1(\tau^*+1) - R_1(\tau^*)] \\ &< 0, \end{aligned}$$

  where the last equality holds because R_1 is decreasing, and for $\tilde{n} < 2\tau^*$:

$$\tau^* + 1 > \frac{\tilde{n}}{2} + 1 > \lceil \tilde{n}/2 \rceil.$$

Thus, the constraint (NJ1) is satisfied. Similar logic applies to the constraint (NJ2).

This completes the proof of the first part.

Proof of (ii).

We now move to the second part. Consider the deterministic signaling mechanism σ^* defined by

$$\sigma^*(k_1^*(n), k_2^*(n) \mid n) = 1, \qquad \forall n \in \Omega,$$

and $\sigma^*(k_1, k_2 \mid n) = 0$ for all other (k_1, k_2). Let $K_i^* := k_i^*(N)$, $i = 1, 2$, be the random number of providers recommended to purchase lead i when $N \sim \mu_0$. In what follows, fix $i \in \{1, 2\}$ and write $-i$ for the other lead.

Step 1: Prices and revenue. Define prices p_i^* by making (Ji) binding:

$$p_i^* := \frac{\mathbb{E}\big[K_i^* R_i(K_i^*)\big]}{\mathbb{E}[K_i^*]}, \qquad i = 1, 2, \tag{G.46}$$

where expectations are taken with respect to $N \sim \mu_0$. By construction, constraints (J1) and (J2) hold with equality.

The revenue generated by (σ^*, p_1^*, p_2^*) is

$$\begin{aligned}
\mathbb{E}\big[p_1^* K_1^* + p_2^* K_2^*\big] &= p_1^* \mathbb{E}[K_1^*] + p_2^* \mathbb{E}[K_2^*] \\
&= \mathbb{E}\big[K_1^* R_1(K_1^*)\big] + \mathbb{E}\big[K_2^* R_2(K_2^*)\big] && \text{(by (G.46))} \\
&= \mathbb{E}_{N \sim \mu_0}\Big[k_1^*(N) R_1\big(k_1^*(N)\big) + k_2^*(N) R_2\big(k_2^*(N)\big)\Big],
\end{aligned}$$

which coincides with the upper bound (G.45). Therefore, once we verify that (σ^*, p_1^*, p_2^*) is feasible for problem (two leads), it must be optimal.

Feasibility of (Feas) is immediate from the definition of σ^*. We already argued that (J1) and (J2) hold with equality, so it remains to check the deviation constraints $(1 \nrightarrow 2)$–$(2 \nrightarrow 1)$ and the non-join constraints (NJ1)–(NJ2).

Step 2: Cross deviations $(1 \nrightarrow 2)$ *and* $(2 \nrightarrow 1)$. We start with $(1 \nrightarrow 2)$. Under (σ^*, p_1^*, p_2^*), its left-hand side can be written as

$$\begin{aligned}
L_{1 \nrightarrow 2} &:= \mathbb{E}\Big[K_1^* \big(R_1(K_1^*) - p_1^* - R_2(K_2^* + 1) + p_2^*\big)\Big] \\
&= -\mathbb{E}\big[K_1^* R_2(K_2^* + 1)\big] + \mathbb{E}[K_1^*] \cdot \frac{\mathbb{E}[K_2^* R_2(K_2^*)]}{\mathbb{E}[K_2^*]},
\end{aligned}$$

where we used the definition of p_2^* in (G.46). Thus

$$L_{1 \nrightarrow 2} = \frac{\mathbb{E}[K_1^*]\mathbb{E}[K_2^* R_2(K_2^*)] - \mathbb{E}[K_2^*]\mathbb{E}[K_1^* R_2(K_2^* + 1)]}{\mathbb{E}[K_2^*]}. \tag{G.47}$$

By condition (22) for $i = 1$, the numerator in (G.47) is nonnegative, and the denominator is strictly positive, hence $L_{1 \nrightarrow 2} \geq 0$. This is exactly constraint $(1 \nrightarrow 2)$.

The proof of $(2 \nrightarrow 1)$ is completely analogous: exchanging the roles of leads 1 and 2 in the above computation and applying (22) with $i = 2$ yields the desired inequality.

Step 3: Non-join constraints. We now verify (NJ1) and (NJ2).For each $n \in \Omega$, optimality of $(k_1^*(n), k_2^*(n))$ for

$$\max_{k_1,k_2\in\mathbb{N}_+,\, k_1+k_2\le n} k_1 R_1(k_1) + k_2 R_2(k_2)$$

implies that, holding $k_{-i}^*(n)$ fixed, $k_i^*(n)$ solves the one-dimensional problem

$$k_i^*(n) \in \operatorname*{arg\,max}_{0\le k_i \le n - k_{-i}^*(n)} k_i R_i(k_i). \tag{G.48}$$

Define the *effective capacity* for lead i in state n as

$$\tilde{N}_i(n) := n - k_{-i}^*(n).$$

Then (G.48) is

$$k_i^*(n) \in \operatorname*{arg\,max}_{0\le k_i \le \tilde{N}_i(n)} k_i R_i(k_i).$$

Let $\tau_i^* \in \arg\max_{k\in\mathbb{Z}_+} kR_i(k)$; by the concavity of $k \mapsto kR_i(k)$, the sequence $kR_i(k)$ is increasing on $\{0,\dots,\tau_i^*\}$ and decreasing on $\{\tau_i^*, \tau_i^*+1, \dots\}$. Hence, for any integer $M \ge 0$,

$$\operatorname*{arg\,max}_{0\le k\le M} kR_i(k) = \begin{cases} \{M\}, & \text{if } M \le \tau_i^*, \\ \{\tau_i^*\}, & \text{if } M \ge \tau_i^*. \end{cases}$$

Applying this with $M = \tilde{N}_i(n)$ and using (G.48), we obtain the explicit form

$$k_i^*(n) = \min\{\tilde{N}_i(n), \tau_i^*\}, \qquad \forall n \in \Omega. \tag{G.49}$$

Recall that the number of non-joining providers in state n is $n - k_1^*(n) - k_2^*(n)$; for lead i this can be rewritten as

$$n - k_1^*(n) - k_2^*(n) = \tilde{N}_i(n) - k_i^*(n).$$

Thus the left-hand side of constraint (NJi), under the deterministic mechanism σ^* and price p_i^*, is

$$\text{LHS of (NJ}i\text{)} = \sum_{n\in\Omega} \mu_0(n) \left(\tilde{N}_i(n) - k_i^*(n)\right)\left[R_i\left(k_i^*(n)+1\right) - p_i^*\right]. \tag{G.50}$$

Using (G.49), we can separate states according to whether $\tilde{N}_i(n) \le \tau_i^*$ or $\tilde{N}_i(n) > \tau_i^*$. If $\tilde{N}_i(n) \le \tau_i^*$, then $k_i^*(n) = \tilde{N}_i(n)$ and $\tilde{N}_i(n) - k_i^*(n) = 0$, so the corresponding term in (G.50) is zero. If $\tilde{N}_i(n) > \tau_i^*$, then $k_i^*(n) = \tau_i^*$ and $\tilde{N}_i(n) - k_i^*(n) = \tilde{N}_i(n) - \tau_i^* > 0$, and

$$R_i\left(k_i^*(n)+1\right) = R_i(\tau_i^*+1), \qquad \forall n \text{ with } \tilde{N}_i(n) > \tau_i^*.$$

Hence (G.50) simplifies to

$$\text{LHS of (NJ}i\text{)} = \sum_{n:\, \tilde{N}_i(n) > \tau_i^*} \mu_0(n) \left(\tilde{N}_i(n) - \tau_i^*\right)\left[R_i(\tau_i^*+1) - p_i^*\right].$$

Notice that

$$p_i^* = \frac{\mathbb{E}[K_i^* R_i(K_i^*)]}{\mathbb{E}[K_i^*]} \overset{(a)}{\ge} \frac{\mathbb{E}[K_i^* R_i(\tau_i^*)]}{\mathbb{E}[K_i^*]} = \frac{\mathbb{E}[K_i^*] \cdot R_i(\tau_i^*)}{\mathbb{E}[K_i^*]} = R_i(\tau_i^*),$$

where the equality (a) holds since $k_i^*(n) \le \tau_i^*$ for all $n \in \Omega$ by (G.49). Therefore

$$\text{LHS of } (\mathsf{NJ}i) \;\le\; 0,$$

which proves both (NJ1) and (NJ2).

Step 4: The sufficient condition $n \ge \tau_1^* + \tau_2^*$. Finally, suppose that $n \ge \tau_1^* + \tau_2^*$ for all $n \in \Omega$, where $\tau_i^* = \arg\max_{k \in \mathbb{Z}_+} kR_i(k)$. In this case we may take

$$k_i^*(n) = \tau_i^*, \qquad i = 1, 2, \quad \forall n \in \Omega,$$

which clearly maximizes $k_1 R_1(k_1) + k_2 R_2(k_2)$ subject to $k_1 + k_2 \le n$. Then $K_i^* := k_i^*(N) = \tau_i^*$ almost surely for each i, and hence

$$\frac{\mathbb{E}[K_{-i}^* R_{-i}(K_{-i}^*)]}{\mathbb{E}[K_{-i}^*]} = R_{-i}(\tau_{-i}^*), \qquad \frac{\mathbb{E}[K_i^* R_{-i}(K_{-i}^* + 1)]}{\mathbb{E}[K_i^*]} = R_{-i}(\tau_{-i}^* + 1).$$

Since R_{-i} is decreasing, $R_{-i}(\tau_{-i}^*) \ge R_{-i}(\tau_{-i}^* + 1)$, so condition (22) is automatically satisfied.

Combining all the steps above, (σ^*, p_1^*, p_2^*) is feasible for (two leads) and attains the upper bound (G.45). Therefore, it is optimal, which proves part (ii) of the theorem. □

G.4 Proofs for Appendix A

Proof of Lemma A.1. Take any fixed n and consider a symmetric candidate $p_i = p$ for all i. Let $s(p)$ denote the common demand share at such a profile. Under the logit specification,

$$s(p) = \frac{\exp(v - p)}{1 + n \exp(v - p)}.$$

The derivative of the share with respect to own price is the standard logit expression

$$\frac{\partial s}{\partial p} = -s(p)(1 - s(p)).$$

Provider i's profit at a symmetric profile is $\pi(p) = (p - c)s(p)$, with first–order condition (FOC)

$$0 = \frac{\mathrm{d}\pi}{\mathrm{d}p} = s(p) + (p - c)\frac{\partial s}{\partial p} = s(p)\big[1 - (p - c)(1 - s(p))\big]. \tag{G.51}$$

Since $s(p) > 0$, the FOC reduces to

$$p - c = \frac{1}{1 - s(p)}.$$

Substituting this back into the share expression yields

$$s = \frac{\exp(v - p)}{1 + n \exp(v - p)} = \frac{\exp(\delta - (p - c))}{1 + n \exp(\delta - (p - c))} = \frac{\exp(\delta - 1/(1 - s))}{1 + n \exp(\delta - 1/(1 - s))}.$$

Rearranging,

$$\frac{1}{s} - n = \exp\Big(\frac{1}{1 - s} - \delta\Big).$$

Taking logs and rearranging yields (A.1). Conversely, any $s \in (0, 1/n)$ that solves (A.1) generates a symmetric candidate price $p = c + 1/(1 - s)$ that satisfies the FOC (G.51).

Anderson and De Palma (1992), see their Proposition 1, shows that the logit profit function is strictly concave in own price, so the FOC is both necessary and sufficient for optimality; this implies the uniqueness of the symmetric equilibrium price. Equation (A.2) follows directly from the FOC and the definition of $R^*(n)$. □

Proof of Proposition A.1. We will prove a stronger result by treating n as a continuous variable on $[1,\infty)$.

Step 1: Comparative statics of the share $s^*(n)$. Define

$$F(s,n) := \frac{1}{1-s} - \delta - \log\Big(\frac{1}{s} - n\Big)$$

on the domain $\mathcal{D} = \{(s,n) : n \geq 1,\ 0 < s < 1/n\}$. Lemma A.1 shows that the equilibrium share $s^*(n)$ is characterized by the implicit equation $F(s^*(n),n) = 0$.

We first compute the partial derivatives of F. For $s \in (0,1/n)$,

$$\frac{\partial F}{\partial s}(s,n) = \frac{1}{(1-s)^2} - \frac{1}{\frac{1}{s}-n}\cdot\Big(-\frac{1}{s^2}\Big) = \frac{1}{(1-s)^2} + \frac{1}{s(1-ns)} > 0,$$

$$\frac{\partial F}{\partial n}(s,n) = -\frac{1}{\frac{1}{s}-n}\cdot(-1) = \frac{s}{1-ns} > 0.$$

The inequalities use $0 < s < 1/n$, which implies $1 - ns > 0$. By the implicit–function theorem, $s^*(n)$ is continuously differentiable and

$$s'(n) = -\frac{F_n(s^*(n),n)}{F_s(s^*(n),n)} = -\frac{\frac{s^*(n)}{1-ns^*(n)}}{\frac{1}{(1-s^*(n))^2} + \frac{1}{s^*(n)(1-ns^*(n))}} < 0.$$

Hence the equilibrium share $s^*(n)$ is strictly decreasing in the number of providers n.

Step 2: Monotonicity of price and per–firm profit. From (A.2),

$$p^*(n) - c = \frac{1}{1-s^*(n)}.$$

Because $s^*(n)$ decreases in n, the denominator $1 - s^*(n)$ is strictly increasing, and therefore the markup $p^*(n) - c$ (and the price $p^*(n)$ itself) is strictly decreasing in n.

Using again (A.2), the per–firm profit can be written as

$$R^*(n) = \frac{s^*(n)}{1-s^*(n)}.$$

Differentiating with respect to n,

$$\frac{\mathrm{d}R^*}{\mathrm{d}n} = \frac{\partial}{\partial s}\Big(\frac{s}{1-s}\Big)\Big|_{s=s^*(n)} \cdot s'(n) = \frac{1}{(1-s^*(n))^2}\cdot s'(n) < 0,$$

because $s'(n) < 0$. Hence $R^*(n)$ is strictly decreasing in n, proving part (i).

Step 3: An explicit representation of total profit. We now study $T(n) = nR^*(n)$. It is convenient to express T as a function of the share s. From the derivation in the proof of Lemma A.1,

$$\frac{1}{s} - n = \exp\Big(\frac{1}{1-s} - \delta\Big).$$

Define

$$E(s) := \exp\Big(\frac{1}{1-s} - \delta\Big), \qquad n(s) := \frac{1}{s} - E(s).$$

Then $n(s)$ is exactly the number of providers that is compatible with a symmetric equilibrium share s. Substituting $n(s)$ into $T(n) = n\,s/(1-s)$ yields a reduced–form expression

$$\widetilde{T}(s) := T(n(s)) = \frac{n(s)\,s}{1-s} = \frac{1 - sE(s)}{1-s}, \qquad s \in (0,1). \tag{G.52}$$

Step 4: Unimodality of $\widetilde{T}(s)$. Differentiating (G.52) with respect to s and using $E'(s) = E(s)/(1-s)^2$, we obtain

$$\begin{aligned} \widetilde{T}'(s) &= \frac{-E(s) - sE'(s)}{1-s} + \frac{-sE(s)+1}{(1-s)^2} \\ &= \frac{1 - s - E(s)}{(1-s)^3}. \end{aligned} \tag{G.53}$$

The denominator is strictly positive for $s \in (0,1)$. Thus the sign of $\widetilde{T}'(s)$ is the sign of

$$q(s) := 1 - s - E(s) = 1 - s - \exp\Big(\frac{1}{1-s} - \delta\Big).$$

We now show that $q(s)$ is strictly decreasing in s. Differentiating,

$$q'(s) = -1 - E'(s) = -1 - \frac{E(s)}{(1-s)^2} < 0 \quad \text{for all } s \in (0,1).$$

Hence $q(s)$ can cross zero at most once. Equivalently, $\widetilde{T}'(s)$ changes sign at most once: either $\widetilde{T}'(s)$ is nonnegative for all s (in which case $\widetilde{T}$ is increasing), or it is nonpositive for all s, or there exists a unique $s^\dagger \in (0,1)$ such that $\widetilde{T}'(s) > 0$ for $s < s^\dagger$ and $\widetilde{T}'(s) < 0$ for $s > s^\dagger$. In all three cases, $\widetilde{T}(s)$ is unimodal in s.

Step 5: Unimodality of $T(n)$. The equilibrium share $s^*(n)$ constructed in Step 1 is strictly decreasing and continuously differentiable in n, and $T(n) = \widetilde{T}(s^*(n))$ by definition. By the chain rule,

$$T'(n) = \widetilde{T}'(s^*(n))\, s'(n).$$

We have $\widetilde{T}'(s^*(n))$ given by (G.53) and $s'(n) < 0$ from Step 1. Therefore the sign of $T'(n)$ is the opposite of the sign of $q(s^*(n))$. Since $q(s)$ is strictly decreasing in s and $s^*(n)$ is strictly decreasing in n, the composition $q(s^*(n))$ is strictly *increasing* in n and crosses zero at most once. Consequently, $T'(n)$ changes sign at most once: $T(n)$ is either monotone in n or increases up to a unique peak and then decreases afterwards. This proves the unimodality claimed in part (ii).

Finally, because $T(\cdot)$ is continuous and unimodal on $[1,\infty)$, its restriction to the discrete set $\mathbb{N}_+$ is unimodal in the usual sense. $\square$

G.5 Proofs for Appendix B

Proof of Lemma B.2. We first provide an auxiliary lemma to facilitate the proof. The proof of this lemma is provided immediately after the main proof.

LEMMA G.3. *Given p_0, for any feasible signaling mechanism σ, if there exist $\hat{n}\in\Omega$ and $\hat{k}<\hat{n}$ such that $\sigma(\hat{k}\mid\hat{n})>0$ and $\hat{k}h(\hat{k},p_0)\le\hat{n}h(\hat{n},p_0)$, then σ cannot be optimal.*

First note that by Lemma 1,

$$\mathbb{E}[h(\hat{N},p_0)]=\mathbb{E}[R(\hat{N})-p_0]=\sum_{n\in\Omega}R(n)\mu_{\mathsf{p}}(n)-p_0=\frac{\sum_{n\in\Omega}R(n)n\mu_0(n)}{\mathbb{E}[N]}-p_0=\frac{\mathbb{E}[Nh(N,p_0)]}{\mathbb{E}[N]},$$

which implies that $\mathbb{E}[h(\hat{N},p_0)]$ and $\mathbb{E}[Nh(N,p_0)]$ have the same sign.

For the case $\mathbb{E}[Nh(N,p_0)]=\mathbb{E}[h(\hat{N},p_0)]=0$, recommending all providers to join, i.e., $\sigma^*(n\mid n)=1$ for all $n\in\Omega$, is incentive compatible as they will receive a non-negative expected utility upon joining; it is also optimal in achieving the maximum possible supply. Thus, constraint (join) writes

$$\sum_{n\in\Omega}\mu_0(n)\sum_{k=1}^{n}\sigma(k\mid n)kh(k,p_0)=\sum_{n\in\Omega}\mu_0(n)nh(n,p_0)=\mathbb{E}[Nh(N,p_0)]=0,$$

which is binding. In what follows, we will focus on the case $\mathbb{E}[Nh(N,p_0)]<0$ and prove by contradiction.

Suppose σ_1 is optimal with constraint (join) non-binding, i.e.,

$$\sum_{n\in\Omega}\mu_0(n)\sum_{k=1}^{n}\sigma_1(k\mid n)kh(k,p_0)=\epsilon>0. \tag{G.54}$$

Since $\mathbb{E}[Nh(N,p_0)](=\sum_{n\in\Omega}\mu_0(n)nh(n,p_0))<0$, (G.54) implies that there exists at least an $\hat{n}\in\Omega$ such that $\sigma_1(\hat{n}\mid\hat{n})\neq 1$. (If there is more than one, pick any one $\hat{n}$ for the following analysis.) As a result, there exists $\hat{i}<\hat{n}$ such that $\sigma_1(\hat{i}\mid\hat{n})>0$. (We will specify $\hat{i}$ and $\hat{n}$ later.) Besides, we claim that there exists an integer $\hat{j}\in(\hat{i},\hat{n}]$ such that $\hat{j}h(\hat{j},p_0)<\hat{i}h(\hat{i},p_0)$. If such $\hat{j}$ does not exist, then $\hat{n}h(\hat{n},p_0)\ge\hat{i}h(\hat{i},p_0)$, which by Lemma G.3 contradicts the optimality of σ_1.

In what follows, we construct a feasible signaling mechanism σ_2 and show it achieves a higher expected supply than σ_1. Specifically, for $n\in\Omega$, if $n\neq\hat{n}$, let $\sigma_2(\cdot\mid n)=\sigma_1(\cdot\mid n)$; if $n=\hat{n}$, let

$$\sigma_2(k\mid\hat{n})=\begin{cases}\sigma_1(\hat{i}\mid\hat{n})-\delta, & \text{if } k=\hat{i}\\ \sigma_1(\hat{j}\mid\hat{n})+\delta, & \text{if } k=\hat{j}\\ \sigma_1(k\mid\hat{n}), & \text{otherwise,}\end{cases}$$

where

$$\delta=\min\left\{\sigma_1(\hat{i}\mid\hat{n}),\frac{\epsilon}{\mu_0(\hat{n})\cdot(\hat{i}h(\hat{i},p_0)-\hat{j}h(\hat{j},p_0))}\right\}.$$

In other words, comparing σ_2 with σ_1, we only transfer δ from the probability of recommending $\hat{i}$ of $\hat{n}$ providers to the probability of recommending $\hat{j}$ of $\hat{n}$ providers. Note that $\delta > 0$ since $\hat{j}h(\hat{j}, p_0) < \hat{i}h(\hat{i}, p_0)$.

Clearly, σ_2 generates a higher expected supply than σ_1 since $\hat{j} > \hat{i}$. We are left to show that σ_2 is feasible to problem ($\mathcal{P}_{\mathsf{LP}}$). By construction, σ_2 satisfies (distribution). For (join), we have

$$\begin{aligned}\sum_{n\in\Omega}\mu_0(n)\sum_{k=1}^{n}\sigma_2(k\mid n)\cdot kh(k,p_0) &\overset{(a)}{=} \left[\sum_{n\in\Omega}\mu_0(n)\sum_{k=1}^{n}\sigma_1(k\mid n)\cdot kh(k,p_0)\right] - \mu_0(\hat{n})\cdot\delta\cdot(\hat{i}h(\hat{i},p_0)-\hat{j}h(\hat{j},p_0))\\ &\overset{(b)}{=} \epsilon - \mu_0(\hat{n})\cdot\delta\cdot(\hat{i}h(\hat{i},p_0)-\hat{j}h(\hat{j},p_0))\\ &\overset{(c)}{\geq} 0,\end{aligned}$$

where (a) holds by the construction of σ_2, (b) follows from (G.54) and (c) holds by the definition of δ. For (not-join), we have the following two cases:

(i) $h(k+1,p_0)\cdot\sigma_1(k\mid n)\leq 0$, for all $n\in\Omega$ and $k\in[n]$. In this case, we have $h(\hat{i}+1,p_0)\leq 0$. Since $h(k,p_0)$ is decreasing with k, we have $h(\hat{j}+1,p_0) < h(\hat{i}+1,p_0)\leq 0$. This leads to $\sigma_2(k\mid n)\cdot(n-k)\cdot h(k+1,p_0)\leq 0$ for all $n\in\Omega$ and all $k\in[n]$. Therefore, constraint (not-join) under σ_2 would be the sum of non-positive terms and will hold.

(ii) There exists $n'\in\Omega$ and $i'\in[n']$ such that $(n'-i')\cdot h(i'+1,p_0)\cdot\sigma_1(i'\mid n') > 0$. In this case, we choose $\hat{i}=i'$ and $\hat{n}=n'$. Therefore, $h(\hat{i}+1,p_0) > 0$. The left-hand side of (not-join) under σ_2 becomes:

$$\begin{aligned}&\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n-1}\sigma_2(k\mid n)\cdot(n-k)h(k+1,p_0)\\ &=\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n-1}\sigma_1(k\mid n)\cdot(n-k)h(k+1,p_0)+\big(\mu_0(\hat{n})\cdot\delta\cdot\underbrace{\Big[(\hat{n}-\hat{j})h(\hat{j}+1,p_0)-(\hat{n}-\hat{i})h(\hat{i}+1,p_0)\Big]}_{:=A}\big)\\ &\overset{(a)}{\leq}\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n-1}\sigma_1(k\mid n)\cdot(n-k)h(k+1,p_0)\overset{(b)}{\leq}0,\end{aligned}$$

where (a) holds since $A<0$. This is because $h(\hat{i}+1,p+0) > 0$ and $h(k,p_0)$ is decreasing in k. The inequality (b) holds by the feasibility of σ_1.

Together, σ_2 satisfies (not-join). This completes the proof. □

Proof of Lemma G.3. We prove Lemma G.3 by contradiction. Suppose the signaling mechanism σ_1 is optimal and there exists $\hat{n}\in\Omega$ and $\hat{k}<\hat{n}$ such that $\sigma_1(\hat{k}\mid\hat{n}) > 0$ and $\hat{k}h(\hat{k},p_0)\leq\hat{n}h(\hat{n},p_0)$. By the concavity of $xh(x,p_0)$, $\hat{k}$ must lie in the increasing range of $xh(x,p_0)$ and thus $h(\hat{k},p_0)\geq 0$. Besides, it can be shown that $h(\hat{k}+1,p_0)\geq 0$.

We construct a new signaling mechanism σ_2 as follows: for $n \in \Omega$, if $n \neq \hat{n}$, $\sigma_2(\cdot \mid n) = \sigma_1(\cdot \mid n)$; if $n = \hat{n}$, let

$$\sigma_2(k \mid \hat{n}) = \begin{cases} \sigma_1(\hat{k} \mid \hat{n}) + \sigma_1(\hat{n} \mid \hat{n}) & \text{if } k = \hat{n} \\ 0 & \text{if } k = \hat{k} \\ \sigma_1(k \mid \hat{n}) & \text{if } k \notin \{\hat{k}, \hat{n}\} \end{cases}.$$

In other words, for σ_2, we only transfer the probability $\sigma_1(\hat{k} \mid \hat{n})$ to $\sigma_1(\hat{n} \mid \hat{n})$. Clearly, the expected supply under σ_2 is higher than the one under σ_1. To conclude the proof, we will show that σ_2 is indeed feasible. By construction, σ_2 satisfies the constraint (distribution).

Regarding constraint (join), we have

$$\sum_{n \in \Omega} \mu_0(n) \sum_{k=0}^{n} \sigma_2(k \mid n) \cdot kh(k, p_0) \overset{(a)}{\geq} \sum_{n \in \Omega} \mu_0(n) \sum_{k=0}^{n} \sigma_1(k \mid n) \cdot kh(k, p_0) \overset{(b)}{\geq} 0,$$

where (a) holds since $\hat{k}h(\hat{k}, p_0) \leq \hat{n}h(\hat{n}, p_0)$ and (b) holds by the feasibility of σ_1. Regarding constraint (not-join), we have

$$\sum_{n \in \Omega} \mu_0(n) \sum_{k=0}^{n} \sigma_2(k \mid n) \cdot (n-k)h(k+1, p_0) \overset{(c)}{\leq} \sum_{n \in \Omega} \mu_0(n) \sum_{k=0}^{n} \sigma_1(k \mid n) \cdot (n-k)h(k+1, p_0) \overset{(d)}{\leq} 0,$$

where (c) holds by $h(\hat{k}+1, p_0) \geq 0$ and thus $(\hat{n} - \hat{n})h(\hat{n}+1, p_0) = 0 \leq (\hat{n} - \hat{k})h(\hat{k}+1, p_0)$, and (d) holds by the feasibility of σ_1. This completes the proof. □

Proof of Proposition B.2. Let us assume n is one element in Ω. First, define $\hat{\Omega}$ as described in the proposition, i.e., we increase n to $n+1$ and reassign the probability of $\mu_0(n)$ to $\mu_0(n+1)$. Assume that τ^* and k^* are the outputs of Algorithm 1 when we input Ω, and $\hat{\tau}^*$ and $\hat{k}^*$ are the outputs of the algorithm when we input $\hat{\Omega}$. Before we begin the proof, we first define some new notation. Let $\tau_t(\Omega)$ and $\tau_t(\hat{\Omega})$ be the τ at the t'th iteration of Algorithm 1 when we input Ω and $\hat{\Omega}$, respectively. Similarly, with some abuse of notation, define $k_t(\Omega)$ and $k_t(\hat{\Omega})$ be the solution to equation (B.3) at the t'th iteration of the algorithm when we input Ω and $\hat{\Omega}$ respectively. Finally, define t_f and $\hat{t}_f$ as the iterations when the algorithm terminates under inputs Ω and $\hat{\Omega}$, respectively.

Part (i): If $n > \tau^*$, then by definition of the algorithm for any $t \leq t_f$, $\tau_t(\Omega) = \tau_t(\hat{\Omega})$ and $k_t(\Omega) = k_t(\hat{\Omega})$. Because the algorithm for Ω stops at t_f, then we have

$$k_{t_f}(\Omega) < \tau_{t_f}(\Omega) \Longrightarrow k_{t_f}(\hat{\Omega}) < \tau_{t_f}(\hat{\Omega}). \tag{G.55}$$

As a result, if the algorithm stops at iteration t_f for Ω, it should also stop at iteration t_f for $\hat{\Omega}$ and $t_f = \hat{t}_f$. Hence, in this case, $k^* = \hat{k}^*$ and $\tau^* = \hat{\tau}^*$.

Part (ii): If $n = \tau^*$, then by definition of the algorithm for any $t < t_f$, $\tau_t(\Omega) = \tau_t(\hat{\Omega})$ and $k_t(\Omega) = k_t(\hat{\Omega})$. Hence, $\hat{t}_f \geq t_f$. Because the algorithm for Ω stops at t_f, we have

$$k_{t_f}(\Omega) < \tau_{t_f}(\Omega) \Longrightarrow k_{t_f}(\hat{\Omega}) \overset{(a)}{<} \tau_{t_f}(\hat{\Omega}), \tag{G.56}$$

where inequality (a) holds because $\tau_{t_f}(\Omega) = n < n+1 = \tau_{t_f}(\hat{\Omega})$ and $k_{t_f}(\Omega) = k_{t_f}(\hat{\Omega})$ (by equation (B.3)). As a result, if the algorithm stops at iteration t_f for Ω, it should also stop at iteration t_f for $\hat{\Omega}$ and $t_f = \hat{t}_f$. Hence, in this case, $k^* = \hat{k}^*$ and $\tau^* < \hat{\tau}^*$.

Part (iii-a): If $n < \tau^*$ and $nh(n, p_0) \leq (n+1)h(n+1, p_0)$, then $n_i < n+1 < R^{-1}(p_0)$. This is due to the fact that $xh(x, p_0)$ is decreasing in x for any $x \geq R^{-1}(p_0)$. Thus, at any iteration $t \leq t_f$, we have $n < n+1 < \tau_t(\Omega) = \tau_t(\hat{\Omega})$. This is because Ω and $\hat{\Omega}$ only differ in n and $n+1$ and because we initialize the algorithm by setting $\tau = \min\{n' \in \Omega : n' \geq R^{-1}(p_0)\}$. Moreover, since in this part we assumed $nh(n, p_0) \leq (n+1)h(n+1, p_0)$, then for any $t \leq t_f$, by equation (B.3) we have $k_t(\Omega) \leq k_t(\hat{\Omega})$. Note that for Ω, we stop at iteration t_f. Therefore for any $t < t_f$ we have

$$k_t(\hat{\Omega}) \geq k_t(\Omega) \geq \tau_t(\Omega) = \tau_t(\hat{\Omega}).$$

This means that the algorithm does not stop before t_f for $\hat{\Omega}$. Therefore, $\hat{t}_f \geq t_f$ and $\hat{\tau}^* \geq \tau^*$. Note that if $\hat{t}_f = t_f$ we have already established that, $k^* = k_{t_f}(\Omega) \leq k_{\hat{t}_f}(\hat{\Omega}) = \hat{k}^*$. Additionally, if $t_f < \hat{t}_f$ (meaning $\tau^* < \hat{\tau}^*$), then by definition of the algorithm $\tau^* \in \{n' \in \hat{\Omega} : n' < \hat{\tau}^*\}$; hence we have

$$\hat{k}^* \overset{(a)}{\geq} \max\{n' \in \hat{\Omega} : n' < \hat{\tau}^*\} \overset{(b)}{\geq} \tau^* \overset{(c)}{=} \min\{n' \in \Omega : n' \geq \tau^*\} \overset{(d)}{>} k^*,$$

where inequalities (a) and (d) hold by Lemma B.3 (ii) and inequality (b) holds because $\tau^* \in \{n' \in \hat{\Omega} : n' < \hat{\tau}^*\}$ and equality (c) is true by the definition of the algorithm.

Part (iii-b): If $n < \tau^*$ and $nh(n, p_0) > (n+1)h(n+1, p_0)$ then by equation (B.3), for any iteration t, $k_t(\Omega) \geq k_t(\hat{\Omega})$. Let us assume that the algorithm does not stop for $\hat{\Omega}$ at any iteration before t_f. At iteration $t = t_f$, we will have

$$k_{t_f}(\hat{\Omega}) < k_{t_f}(\Omega) < \tau_{t_f}(\Omega) \leq \tau_{t_f}(\hat{\Omega}).$$

This means that at iteration t_f, by definition, the algorithm will finish for $\hat{\Omega}$. Hence, if we input $\hat{\Omega}$, the algorithm must stop at most after t_f iteration. As a result $\hat{\tau}^* \leq \tau^*$.

Now, we will show that if $n < \max\{n' \in \Omega : n' < \tau\}$ then $\hat{k}^* < k^*$. First, if $t_f = \hat{t}_f$ then we have already established that $k^* = k_{t_f}(\Omega) > k_{\hat{t}_f}(\hat{\Omega}) = \hat{k}^*$. If $t_f > \hat{t}_f$ (meaning $\tau^* > \hat{\tau}^*$), then by the definition of the algorithm, $\hat{\tau}^* \in \{n' \in \Omega : n' < \tau^*\}$ and we have

$$\hat{k}^* \overset{(a)}{<} \min\{n' \in \hat{\Omega} : n' \geq \hat{\tau}^*\} \overset{(b)}{=} \hat{\tau}^* \overset{(c)}{\leq} \max\{n' \in \Omega : n' < \tau^*\} \overset{(d)}{\leq} k^*,$$

where (a) and (d) hold by Lemma B.3 (ii), (b) holds by the definition of the algorithm, and (c) holds because $\hat{\tau}^* \in \{n' \in \Omega : n' < \tau^*\}$. □

Proof of Lemma B.3. It is clear that the algorithm will terminate after at most $|\{n \in \Omega : n > R^{-1}(p_0)\}|$ iterations because the algorithm effectively searches over the elements in Ω that are greater than the break-even participation, see the while loop. (B.3) and (B.4) are satisfied by construction.

Before proving (i), note that by we in Section 3 we assumed that $R(1) > p_0$, (hence $1h(1,p_0) = R(1) - p_0 > 0$) and the LHS of (B.3) is concave in k. It thus follows from (i) that at each iteration (B.3) has a unique positive solution.

We now prove (i). Since the denominator of the RHS of (B.3) is always negative, we just need to show the numerator of the RHS of (B.3), $\sum_{n\in\Omega, n<\tau} \mu_0(n) n h(n,p_0)$, remains positive before the algorithm terminates. Note that we initialize τ with $\min\{n \in \Omega : n \geq R^{-1}(p_0)\}$. Thus in the initial step, $\sum_{n\in\Omega, n<\tau} \mu_0(n) n h(n,p_0) > 0$. As we move forward at each iteration of the while loop, τ is increased. As a result, $\sum_{n\in\Omega, n<\tau} \mu_0(n) n h(n,p_0)$ decreases. We argue that the algorithm must terminate before $\sum_{n\in\Omega, n<\tau} \mu_0(n) n h(n,p_0)$ becomes negative. Assume that we are at iteration $t > 1$ with threshold τ, and $\sum_{n\in\Omega, n<\tau} \mu_0(n) n h(n,p_0)$ becomes negative for the first time. Let $\bar{n} = \max\{n \in \Omega : n < \tau\}$. By definition of the algorithm, $\bar{n} > R^{-1}(p_0)$ and we have

$$\begin{aligned}
&\sum_{n\in\Omega, n<\tau} \mu_0(n) n h(n,p_0) < 0 \\
\implies &\sum_{n\in\Omega, n<\bar{n}} \mu_0 n h(n,p_0) + \underbrace{\mu_0(\bar{n})\bar{n}h(\bar{n},p_0)}_{\text{negative}} < 0 \\
\implies &\sum_{n\in\Omega, n<\bar{n}} \mu_0 n h(n,p_0) + \sum_{n\in\Omega, n\geq\bar{n}} \mu_0(n)\bar{n}h(\bar{n},p_0) \overset{(a)}{<} 0.
\end{aligned}$$

Note that $xh(x,p_0)$ is concave and decreasing at $x = \bar{n}$ because $\bar{n} > R^{-1}(p_0)$. Thus, inequality (a) means that at iteration $t-1$ when $\bar{n}$ was the threshold, $k(\tau)$ must have been smaller than $\bar{n}$ and at iteration $t-1$, the algorithm would have had stopped. As a result, the RHS of (B.3) is negative before the algorithm terminates.

Next, we show (ii). Since the algorithm does not stop while $k(\tau) \geq \tau$, $k^* < \min\{n \in \Omega : n \geq \tau\}$. Assume that at some iteration like t, the threshold is $\tau^{(t)}$, and assume $k^{(t)} > \bar{n}$ where $k^{(t)}$ is the solution to (B.3) and $\bar{n} = \max\{n \in \Omega : n < \tau^{(t)}\}$. By definition of the algorithm, $\bar{n} > R^{-1}(p_0)$. So

$$\begin{aligned}
&\sum_{n\in\Omega, n<\tau^{(t)}} \mu_0(n) n h(n,p_0) + \sum_{n\in\Omega, n\geq\tau^{(t)}} \mu_0(n) k^{(t)} h(k^{(t)},p_0) \overset{(a)}{=} 0 \\
\implies &\sum_{n\in\Omega, n<\bar{n}} \mu_0 n h(n,p_0) + \mu_0(\bar{n})\bar{n}h(\bar{n},p_0) + \sum_{n\in\Omega, n\geq\tau^{(t)}} \mu_0(n) k^{(t)}(\tau) h(k^{(t)}(\tau),p_0) = 0 \\
\implies &\sum_{n\in\Omega, n<\bar{n}} \mu_0 n h(n,p_0) + \sum_{n\in\Omega, n\geq\bar{n}} \mu_0(n)\bar{n}h(\bar{n},p_0) \overset{(b)}{<} 0.
\end{aligned}$$

Equality (a) holds by definition of $k^{(t)}$, and inequality (b) holds because we assumed $k^{(t)} > \bar{n}$. Furthermore, $xh(x,p_0)$ is concave and decreasing at $x = \bar{n}$ (because $\bar{n} > R^{-1}(p_0)$). Hence, inequality (b) means that at iteration $t-1$ when $\bar{n}$ was the threshold (by definition of the algorithm), $k^{(t-1)}$ must have been smaller than $\bar{n}$ and at iteration $t-1$, the algorithm would have had stopped. As a result, $k^* \geq \max\{n \in \Omega : n < \tau^{(t)}\}$. □

Proof of Proposition B.3. We will prove each result separately.

(i) No Information

At equilibrium, providers should be indifferent between participation and outside options. That is to say

$$\sum_{n\in\Omega}\mu_{\mathsf{sp}}(n)\cdot\left[\underbrace{(1-f_{\mathsf{no}})}_{\text{not participate}}\cdot 0+\underbrace{f_{\mathsf{no}}}_{\text{participate}}\cdot\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f_{\mathsf{no}})}[h(1+k,p_0)]\right]=0,$$

By the linearity of $R(\cdot)$, the above equality leads to

$$R\left(1+(\mathbb{E}[\hat{N}]-1)\cdot f_{\mathsf{no}}\right)=p_0.$$

Thus, we have

$$f_{\mathsf{no}}=\frac{R^{-1}(p_0)-1}{\mathbb{E}[\hat{N}]-1}.$$

The expected supply, where the expectation is taken over the platform' prior belief, is

$$\mathsf{Sp}(\text{no info},p_0)=\frac{R^{-1}(p_0)-1}{\mathbb{E}[\hat{N}]-1}\cdot\mathbb{E}[N].$$

(ii) Full Information

When $n\le R^{-1}(p_0)=(r-p_0)/\alpha$, we have $h(n,p_0)\ge 0$. Thus, in these states, the equilibrium participation probability is $f(n)=1$. When $n>(r-p_0)/\alpha$, the equilibrium participation probability $f(n)$ should make providers indifferent between participation and the outside option. That is to say,

$$(1-f(n))\cdot 0+f(n)\cdot\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f(n))}[h(1+k,p_0)]=0.$$

By the linearity of $R(\cdot)$, the above equation leads to $f(n)=\frac{R^{-1}(p_0)-1}{n-1}$.

The corresponding expected supply is

$$\mathsf{Sp}(\text{full info},p_0)=\sum_{n\in\Omega}\mu_0(N)\cdot n\cdot f(n)=\sum_{n\in\Omega,n\le R^{-1}(p_0)}\mu_0(n)\cdot n+\sum_{n\in\Omega,n>R^{-1}(p_0)}\mu_0(n)\cdot\frac{R^{-1}(p_0)-1}{n-1}\cdot n.$$

(iii) Comparison

If $n\ge R^{-1}(p_0)$ for all $n\in\Omega$, we have

$$\mathsf{Sp}(\text{full info},p_0)=\sum_{n\in\Omega}\mu_0(n)\cdot\frac{R^{-1}(p_0)-1}{n-1}\cdot n=(R^{-1}(p_0)-1)\cdot\mathbb{E}\left[\frac{N}{N-1}\right]\overset{(1)}{\ge}\frac{(R^{-1}(p_0)-1)\cdot\mathbb{E}[N]}{\mathbb{E}[N]-1}=\mathsf{Sp}(\text{no info},p_0),$$

where (1) holds by Jensen's inequality. □

G.6 Proofs for Appendix C

Proof of Proposition C.4. We claim that given no information, under the optimal fixed price, providers must have zero expected utility at equilibrium. Otherwise, a slight increase in the price would not reduce the supply, implying that such an adjustment would enhance revenue. Thus, without

loss of optimality, assume that, under price p_0, providers have zero expected utility at equilibrium, i.e.,

$$\sum_{n\in\Omega}\mu_{\mathsf{p}}(n)\cdot\left[(1-f_{\mathsf{no}})\cdot 0+f_{\mathsf{no}}\cdot\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f_{\mathsf{no}})}[R(1+k)-p_0]\right]=0.$$

This leads to

$$p_0=\sum_{n\in\Omega}\mu_{\mathsf{p}}(n)\cdot\mathbb{E}_{k\sim\mathsf{Bin}(n-1,f_{\mathsf{no}})}[R(1+k)]=\mathbb{E}[\mathbb{E}_{k\sim\mathsf{Bin}(\hat{N}-1,f_{\mathsf{no}})}[R(1+k)]].$$

To find the optimal fixed price under no information, the platform is to solve the following problem:

$$\max_{f_{\mathsf{no}}\in[0,1]}\ \mathbb{E}[\mathbb{E}_{k\sim\mathsf{Bin}(\hat{N}-1,f_{\mathsf{no}})}[R(1+k)]]\cdot f_{\mathsf{no}}\cdot\mathbb{E}[N].$$

For the linear reward function $R(\cdot)$, the above problem is simplified as follows:

$$\max_{f_{\mathsf{no}}\in[0,1]}\ R\left(1+(\mathbb{E}[\hat{N}]-1)\cdot f_{\mathsf{no}}\right)\cdot\mathbb{E}[N]\cdot f_{\mathsf{no}}.$$

Observe that the above objective function is quadratic and concave in f_{no}, we have

$$\begin{aligned}f^*_{\mathsf{no}}&=\min\left\{1,\ \frac{R^{-1}(0)-1}{2\left(\mathbb{E}[\hat{N}]-1\right)}\right\}\\&=\begin{cases}1, & \mathbb{E}[\hat{N}]\le\frac{R^{-1}(0)+1}{2}\\ \frac{R^{-1}(0)-1}{2(\mathbb{E}[\hat{N}]-1)}, & \mathbb{E}[\hat{N}]>\frac{R^{-1}(0)+1}{2}\end{cases}.\end{aligned}$$

Plugging the above equation into $p_0=R\left(1+(\mathbb{E}[\hat{N}]-1)\cdot f_{\mathsf{no}}\right)$, we have $p_0^*(\mathsf{no\ info})$ stated in Proposition C.4. The corresponding revenue can be derived by the relation $\mathsf{REV}^*_{\mathsf{noinfo}}=p_0^*(\mathsf{no\ info})\cdot\mathbb{E}[N]\cdot f^*_{\mathsf{no}}$. This completes the proof. □

Proof of Proposition C.5. We will show that under the linear reward function, each condition listed in Proposition C.5 is sufficient for $\mathsf{REV}^*_{\mathsf{cp}}\ge\mathsf{REV}^*_{\mathsf{noinfo}}$.

(i) $\mathbb{E}[\hat{N}]\le\frac{R^{-1}(0)+1}{2}$

In this case, by Proposition C.4, we have $\mathsf{REV}^*_{\mathsf{noinfo}}=R(\mathbb{E}[\hat{N}])\cdot\mathbb{E}[N]=\mathbb{E}[NR(N)]$, where the second equality holds by utilizing the relation that $\mathbb{E}[\hat{N}]=\mathbb{E}[N^2]/\mathbb{E}[N]$. By Proposition 1, we have the expected revenue under the optimal contingent pricing as follows:

$$\begin{aligned}\mathsf{REV}^*_{\mathsf{cp}}&=\sum_{n\in\Omega,n\le\frac{R^{-1}(0)+1}{2}}\mu_0(n)\cdot nR(n)+\sum_{n\in\Omega,n>\frac{R^{-1}(0)+1}{2}}\mu_0(n)\cdot R\left(\frac{R^{-1}(0)+1}{2}\right)\cdot\frac{R^{-1}(0)-1}{2}\cdot\frac{n}{n-1}\\&\overset{(a)}{\ge}\sum_{n\in\Omega,n\le\frac{R^{-1}(0)+1}{2}}\mu_0(n)nR(n)=\mathbb{E}[NR(N)]=\mathsf{REV}^*_{\mathsf{noinfo}},\end{aligned}$$

where (a) holds by the fact that for any $n>\frac{R^{-1}(0)+1}{2}$,

$$nR(n)\ge R\left(\frac{R^{-1}(0)+1}{2}\right)\cdot\frac{R^{-1}(0)-1}{2}\cdot\frac{n}{n-1}.$$

(ii) $n\ge\frac{R^{-1}(0)+1}{2}$ *for all* $n\in\Omega$

In this case, we have $\mathbb{E}[N] \geq \frac{R^{-1}(0)+1}{2}$. This implies that $\mathbb{E}[\hat{N}] = \frac{\mathbb{E}[N^2]}{\mathbb{E}[N]} \geq \mathbb{E}[N] \geq \frac{R^{-1}(0)+1}{2}$, by Proposition C.4, we have

$$\mathsf{REV}^*_{\mathsf{noinfo}} = R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \frac{R^{-1}(0)-1}{2} \cdot \frac{\mathbb{E}[N]}{\mathbb{E}[\hat{N}]-1}.$$

By Proposition 1, the expected revenue under optimal contingent pricing is

$$\begin{aligned}
\mathsf{REV}^*_{\mathsf{cp}} &= \sum_{n\in\Omega} \mu_0(n) \cdot R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \frac{R^{-1}(0)-1}{2} \cdot \frac{n}{n-1} \\
&\overset{(a)}{\geq} R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \frac{R^{-1}(0)-1}{2} \cdot \frac{\mathbb{E}[N]}{\mathbb{E}[N]-1} \\
&\overset{(b)}{\geq} R\left(\frac{R^{-1}(0)+1}{2}\right) \cdot \frac{R^{-1}(0)-1}{2} \cdot \frac{\mathbb{E}[N]}{\mathbb{E}[\hat{N}]-1} = \mathsf{REV}^*_{\mathsf{noinfo}},
\end{aligned}$$

where (a) holds by Jensen's inequality and (b) holds since $\mathbb{E}[\hat{N}] \geq \mathbb{E}[N]$. □

G.7 Proofs for Appendix D

Proof of Lemma D.4 We show that under each of the three conditions, the function in (18) is unimodal in n_1 on the finite set Ω_1^*. Since only values $n_1 \in \Omega_1^*$ can arise under $n_1^*(\cdot)$, unimodality on Ω_1^* is sufficient for Theorem 6.

Preliminaries. Under $\gamma(n_1) \equiv \gamma$, the expression in (18) can be written as

$$F(n_1;\gamma) := n_1 \widetilde{R}_1(n_1) + (1-\gamma)\,\mathsf{REV}^*_2(n_1), \qquad n_1 \in \Omega_1^*.$$

Using the definitions of $\widetilde{R}_1(\cdot)$ and $\mathsf{REV}^*_2(\cdot)$ in (15)–(16), we obtain

$$\begin{aligned}
F(n_1;\gamma) &= n_1\left[\gamma R_1(n_1) + (1-\gamma)u_2^*(n_1)\right] + (1-\gamma)\,\mathsf{REV}^*_2(n_1) \\
&= \gamma\, f_1(n_1) + (1-\gamma)\left[n_1 u_2^*(n_1) + \mathsf{REV}^*_2(n_1)\right],
\end{aligned} \tag{G.57}$$

where

$$f_1(x) := xR_1(x), \qquad f_2(x) := xR_2(x), \qquad x \geq 0.$$

Define

$$B(n_1) := n_1 u_2^*(n_1) + \mathsf{REV}^*_2(n_1), \qquad n_1 \in \Omega_1^*,$$

so that

$$F(n_1;\gamma) = \gamma\, f_1(n_1) + (1-\gamma)\, B(n_1), \qquad n_1 \in \Omega_1^*. \tag{G.58}$$

Because $f_i(\cdot)$ is strictly concave on $[0,\infty)$ $(i=1,2)$, its restriction to integers is also *strictly* discretely concave.

For each fixed $n_1 \in \Omega_1^*$, define

$$g_{n_1}(x) := xR_2(x+n_1), \qquad x \geq 0.$$

As shown in the proof of Theorem 5, strict concavity of $f_2(x) = xR_2(x)$ and monotonicity of R_2 imply that g_{n_1} is concave on $[0,\infty)$, and in fact *strictly* concave on $(0,\infty)$. Let

$$\tau^*(n_1) \in \arg\max_{k\in\mathbb{Z}_+} g_{n_1}(k),$$

be the smallest maximizer, as defined before. Then, we know that, for any $\bar{k} \in \mathbb{Z}_+$, the constrained problem

$$\max_{k\in\{0,1,\ldots,\bar{k}\}} g_{n_1}(k)$$

is solved by

$$k = \begin{cases} \bar{k}, & \text{if } \bar{k} \le \tau^*(n_1), \\ \tau^*(n_1), & \text{if } \bar{k} \ge \tau^*(n_1). \end{cases}$$

We will use this property in the analysis of cases (i) and (ii).

Step 1: Case (i) – supply-scarce second stage

Condition (D.7) states that

$$\bar{n}_2 \;\le\; \tau^*(n_1), \qquad \forall n_1 \in \Omega_1^*,$$

where $\bar{n}_2 = \max \Omega_2$. Fix any $n_1 \in \Omega_1^*$ and any $n_2 \in \Omega_2$. Then $n_2 \le \bar{n}_2 \le \tau^*(n_1)$, and by unimodality of $g_{n_1}(k)$, the constrained maximizer of $g_{n_1}(k)$ over $k \in \{0,1,\ldots,n_2\}$ is $k = n_2$. Hence, in every state (n_1,n_2) with $n_1 \in \Omega_1^*$ and $n_2 \in \Omega_2$, the second-stage optimal number of participants is

$$k^*(n_1,n_2) = n_2.$$

Substituting this into (15)–(16), we obtain

$$u_2^*(n_1) = \mathbb{E}_{N_2}\big[R_2(n_1+N_2)\big], \qquad \mathsf{REV}_2^*(n_1) = \mathbb{E}_{N_2}\big[N_2 R_2(n_1+N_2)\big],$$

and therefore

$$\begin{aligned} B(n_1) &= n_1 u_2^*(n_1) + \mathsf{REV}_2^*(n_1) \\ &= \mathbb{E}_{N_2}\big[(n_1+N_2)R_2(n_1+N_2)\big] = \mathbb{E}_{N_2}\big[f_2(n_1+N_2)\big]. \end{aligned}$$

For each fixed n_2, the mapping $n_1 \mapsto f_2(n_1+n_2)$ is concave and hence discretely concave on integers; therefore their expectation $B(\cdot)$ is discretely concave on Ω_1^* (summation of concave functions preserve concavity). Similarly, $n_1 \mapsto f_1(n_1)$ is discretely concave on Ω_1^* because f_1 is concave on $[0,\infty)$.

By the decomposition (G.58), $F(\cdot;\gamma)$ is a convex combination of two discretely concave sequences and thus is itself discretely concave on Ω_1^*. By the argument above, discrete concavity implies unimodality. Thus, under condition (i), $F(\cdot;\gamma)$ is unimodal on Ω_1^*, and the unimodality condition (18) holds.

Step 2: Case (ii) – supply-adequate second stage

Condition (D.8) assumes that there exists $K \geq 0$ such that

$$\tau^*(n_1) = K, \qquad \forall n_1 \in \Omega_1^*,$$

and (D.9) requires that

$$\underline{n}_2 \geq K,$$

where $\underline{n}_2 = \min \Omega_2$. Fix any $n_1 \in \Omega_1^*$ and any $n_2 \in \Omega_2$. Then $n_2 \geq \underline{n}_2 \geq K$, so K is feasible in the constrained problem

$$\max_{k \in \{0,1,\ldots,n_2\}} g_{n_1}(k).$$

Because $g_{n_1}(k)$ is concave and unimodal in k with maximizer $K = \tau^*(n_1)$, the constrained maximizer is $k = K$. Thus, under condition (ii),

$$k^*(n_1, n_2) = K, \qquad \forall n_1 \in \Omega_1^*, \ \forall n_2 \in \Omega_2.$$

Substituting this into (15)–(16), we get, for each $n_1 \in \Omega_1^*$,

$$u_2^*(n_1) = R_2(n_1 + K), \qquad \mathsf{REV}_2^*(n_1) = K R_2(n_1 + K),$$

and therefore

$$B(n_1) = n_1 u_2^*(n_1) + \mathsf{REV}_2^*(n_1) = (n_1 + K) R_2(n_1 + K) = f_2(n_1 + K).$$

Because f_2 is strictly concave on $[0, \infty)$, $n_1 \mapsto f_2(n_1 + K)$ is strictly concave and hence discretely concave on integers. As before, f_1 is discretely concave as well, and by (G.58) the mapping $n_1 \mapsto F(n_1; \gamma)$ is a convex combination of two discretely concave sequences. Therefore $F(\cdot; \gamma)$ is discretely concave on Ω_1^* and thus unimodal. Hence the unimodality condition (18) holds under condition (ii).

Step 3: Case (iii) – high first-stage success probability

Finally, we consider the case where γ is close to 1. Strict concavity of f_1 implies that the sequence $n_1 \mapsto f_1(n_1)$ is strictly concave and hence strictly unimodal on Ω_1^*. That is, there exists $n_1^\dagger \in \Omega_1^*$ such that

$$f_1(0) < f_1(1) < \cdots < f_1(n_1^\dagger) > f_1(n_1^\dagger + 1) > \cdots > f_1(\bar{n}_0),$$

where we view $\Omega_1^* \subseteq \{0, 1, \ldots, \bar{n}_0\}$. Equivalently, the first differences

$$\Delta f_1(n_1) := f_1(n_1 + 1) - f_1(n_1)$$

satisfy

$$\Delta f_1(n_1) > 0 \quad \text{for } n_1 < n_1^\dagger, \qquad \Delta f_1(n_1) < 0 \quad \text{for } n_1 \geq n_1^\dagger,$$

and $\Delta f_1(n_1) \neq 0$ for all $n_1 \in \Omega_1^*$.

Using (G.58), the first difference of F in n_1 is

$$\Delta F(n_1;\gamma) := F(n_1+1;\gamma) - F(n_1;\gamma) = \gamma\,\Delta f_1(n_1) + (1-\gamma)\,\Delta B(n_1),$$

where $\Delta B(n_1) := B(n_1+1) - B(n_1)$. For each fixed n_1, $\Delta F(n_1;\gamma)$ is an affine (and hence continuous) function of γ on $[0,1]$.

At $\gamma = 1$, we have $\Delta F(n_1;1) = \Delta f_1(n_1)$, so

$$\Delta F(n_1;1) \begin{cases} > 0, & n_1 < n_1^\dagger, \\ < 0, & n_1 \geq n_1^\dagger. \end{cases}$$

Because $\Delta f_1(n_1) \neq 0$ for all n_1, continuity of $\Delta F(n_1;\gamma)$ implies that, for each n_1, there exists $\delta_{n_1} > 0$ such that the sign of $\Delta F(n_1;\gamma)$ coincides with that of $\Delta f_1(n_1)$ for all $\gamma \in [1-\delta_{n_1}, 1]$.

Let $\delta := \min_{n_1 \in \Omega_1^*} \delta_{n_1} > 0$ and define $\bar{\gamma} := 1 - \delta \in [0,1)$. Then, for every $\gamma \in [\bar{\gamma}, 1]$ and every $n_1 \in \Omega_1^*$,

$$\Delta F(n_1;\gamma) \begin{cases} > 0, & n_1 < n_1^\dagger, \\ < 0, & n_1 \geq n_1^\dagger. \end{cases}$$

In particular, for all $\gamma \in [\bar{\gamma}, 1]$ the sequence $n_1 \mapsto F(n_1;\gamma)$ is strictly increasing up to $n_1^\dagger$ and strictly decreasing afterwards, and hence strictly unimodal on Ω_1^*. Thus the unimodality condition (18) holds under condition (iii) whenever $\gamma \in [\bar{\gamma}, 1]$. □

G.8 Proofs for Appendix E

Proof of Lemma E.5. We derive three key *expected* utilities after receiving signals. After having these, the remaining proof is similar to Lemma 3.

(i) The providers' posterior belief, after receiving a signal $s_i = 1$, about the total number n and the recommended number of purchasing leads one k_1 is

$$\begin{aligned} p(n, k_1 \mid s_i = 1) &= \frac{p(s_i = 1 \mid n, k_1) \cdot p(n, k_1)}{p(s_i = 1)} \\ &= \frac{\frac{k_1}{n} \cdot (\mu_{\mathsf{p}}(n) \cdot p(k_1 \mid n))}{p(s_i = 1)} \\ &= \frac{\frac{k_1}{n} \cdot \left(\mu_{\mathsf{p}}(n) \cdot \sum_{k_2=0}^{n-k_1} \sigma(k_1, k_2 \mid n)\right)}{p(s_i = 1)} \\ &= \frac{\mu_0(n)}{p(s_i = 1) \cdot \mathbb{E}[N]} \cdot k_1 \cdot \left(\sum_{k_2=0}^{n-k_1} \sigma(k_1, k_2 \mid n)\right). \end{aligned}$$

(relation between μ_{p} and μ_0 from Lemma 1)

The expected utility when a provider receives a signal to purchase the lead one ($s_i = 1$), and he follows this signal is

$$\begin{aligned} &\sum_{n \in \Omega} \sum_{k_1=0}^{n} p(n, k_1 \mid s_i = 1) \cdot [R_1(k_1) - p_1] \\ &= \frac{1}{p(s_i = 1) \cdot \mathbb{E}[N]} \cdot \sum_{n \in \Omega} \mu_0(n) \sum_{k_1=0}^{n} k_1 \cdot \left(\sum_{k_2=0}^{n-k_1} \sigma(k_1, k_2 \mid n)\right) \cdot [R_1(k_1) - p_1] \end{aligned}$$

$$= \frac{1}{p(s_i=1)\cdot\mathbb{E}[N]}\cdot\sum_{n\in\Omega}\mu_0(n)\sum_{k_1+k_2\le n}\sigma(k_1,k_2\mid n)\cdot k_1\cdot[R_1(k_1)-p_1]\,.$$

The expected utility when a provider receives a signal to purchase the lead two ($s_i=2$), and he follows this signal is similar.

(ii) The providers' posterior belief, after receiving a signal $s_i=1$, about the total number n and the recommended number of purchasing leads one k_2 is

$$\begin{aligned}
p(n,k_2\mid s_i=1) &= \frac{p(s_i=1\mid n,k_2)\cdot p(n,k_2)}{p(s_i=1)}\\
&= \frac{\left(\sum_{k_1=0}^{n-k_2} p(s_i=1\mid n,k_2,k_1)\cdot p(k_1\mid n,k_2)\right)\cdot(\mu_{\mathsf{p}}(n)\cdot p(k_2\mid n))}{p(s_i=1)} \qquad \text{(law of total prob)}\\
&= \frac{\left(\sum_{k_1=0}^{n-k_2}\frac{k_1}{n}\cdot\frac{p(k_1,k_2\mid n)}{p(k_2\mid n)}\right)\cdot(\mu_{\mathsf{p}}(n)\cdot p(k_2\mid n))}{p(s_i=1)}\\
&= \frac{\left(\sum_{k_1=0}^{n-k_2}\frac{k_1}{n}\cdot\sigma(k_1,k_2\mid n)\right)\cdot\mu_{\mathsf{p}}(n)}{p(s_i=1)}\\
&= \frac{\mu_0(n)}{\mathbb{E}[N]\cdot p(s_i=1)}\cdot\sum_{k_1=0}^{n-k_2}k_1\cdot\sigma(k_1,k_2\mid n). \qquad \text{(relation between } \mu_{\mathsf{p}} \text{ and } \mu_0)
\end{aligned}$$

The expected utility when a provider receives a signal to purchase lead one ($s_i=1$), but he deviates to purchase lead two is

$$\begin{aligned}
&\sum_{n\in\Omega}\sum_{k_2=0}^{n}p(n,k_2\mid s_i=1)\cdot[R_2(k_2+1)-p_2]\\
&= \frac{1}{p(s_i=1)\cdot\mathbb{E}[N]}\cdot\sum_{n\in\Omega}\mu_0(n)\sum_{k_2=0}^{n}\left(\sum_{k_1=0}^{n-k_2}\sigma(k_1,k_2\mid n)\cdot k_1\right)\cdot[R_2(k_2+1)-p_2]\\
&= \frac{1}{p(s_i=1)\cdot\mathbb{E}[N]}\cdot\sum_{n\in\Omega}\mu_0(n)\sum_{k_1+k_2\le n}\sigma(k_1,k_2\mid n)\cdot k_1\cdot[R_2(k_2+1)-p_2]\,.
\end{aligned}$$

The expected utility when a provider receives a signal to purchase lead two ($s_i=2$), but he deviates to purchase lead one is similar.

(iii) The providers' posterior belief, after receiving a signal $s_i=0$, about the total number n and the recommended number of purchasing leads one k_1 is

$$\begin{aligned}
p(n,k_1\mid s_i=0) &= \frac{p(s_i=0\mid n,k_1)\cdot p(n,k_1)}{p(s_i=0)}\\
&= \frac{\left(\sum_{k_2=0}^{n-k_1} p(s_i=0\mid n,k_1,k_2)\cdot p(k_2\mid n,k_1)\right)\cdot(\mu_{\mathsf{p}}(n)\cdot p(k_1\mid n))}{p(s_i=0)}\\
&= \frac{\left(\sum_{k_2=0}^{n-k_1}\frac{n-k_1-k_2}{n}\cdot\frac{p(k_1,k_2\mid n)}{p(k_1\mid n)}\right)\cdot(\mu_{\mathsf{p}}(n)\cdot p(k_1\mid n))}{p(s_i=0)}\\
&= \frac{\left(\sum_{k_2=0}^{n-k_1}\frac{n-k_1-k_2}{n}\cdot\sigma(k_1,k_2\mid n)\right)\cdot\mu_{\mathsf{p}}(n)}{p(s_i=0)}
\end{aligned}$$

$$= \frac{\mu_0(n)}{p(s_i=0)\cdot\mathbb{E}[N]} \cdot \sum_{k_2=0}^{n-k_1} \sigma(k_1,k_2 \mid n)\cdot(n-k_1-k_2).$$

The expected utility when the provider receives a signal to quit ($s_i=0$), but he deviates to purchase lead one is

$$\begin{aligned}
&\sum_{n\in\Omega}\sum_{k_1=0}^{n} p(n,k_1 \mid s_1=0)\cdot[R_1(k_1+1)-p_1] \\
&= \frac{1}{p(s_i=0)\cdot\mathbb{E}[N]} \sum_{n\in\Omega}\mu_0(n) \sum_{k_1=0}^{n} \cdot \left(\sum_{k_2=0}^{n-k_1} \sigma(k_1,k_2 \mid n)\cdot(n-k_1-k_2)\right)\cdot[R_1(k_1+1)-p_1] \\
&= \frac{1}{p(s_i=0)\cdot\mathbb{E}[N]} \sum_{n\in\Omega}\mu_0(n) \sum_{k_1+k_2\le n} \sigma(k_1,k_2 \mid n)\cdot(n-k_1-k_2)\cdot[R_1(k_1+1)-p_1].
\end{aligned}$$

The expected utility when the provider receives a signal to quit ($s_i=0$), but he deviates to purchase lead two is similar. □

Proof of Proposition E.6 Based on the discussions before the proposition, it is equivalent to solve problem (E.10), and then use (E.11) to pin down the prices if $f_i(n)\neq 0$. Notice that the objective in (E.10) is a strictly concave quadratic in (f_1,f_2), and the feasible set $\{(f_1,f_2): 0\le f_i\le 1,\ f_1+f_2\le 1\}$ is a convex polytope. Hence the KKT conditions are necessary and sufficient.

For convenience, define

$$T := \frac{1}{2}\Big(\frac{r_1}{c_1}+\frac{r_2}{c_2}\Big).$$

Step 1 (KKT conditions and interior formulas). Introduce multipliers $\mu\ge 0$ for the sum constraint $f_1+f_2\le 1$, $\nu_i\ge 0$ for the lower bounds $f_i\ge 0$, and $\tau_i\ge 0$ for the upper bounds $f_i\le 1$. The Lagrangian for (E.10) is

$$\mathcal{L} = n\sum_{i=1}^{2}\left[(r_i-c_i)f_i - c_i(n-1)f_i^2\right] + \mu(1-f_1-f_2) + \sum_{i=1}^{2}\nu_i f_i + \sum_{i=1}^{2}\tau_i(1-f_i).$$

Stationarity with respect to f_i yields

$$n\big((r_i-c_i)-2c_i(n-1)f_i\big) - \mu + \nu_i - \tau_i = 0, \qquad i=1,2.$$

Whenever $0<f_i<1$, the complementary-slackness conditions imply $\nu_i=\tau_i=0$, so in that case

$$f_i = \frac{r_i-c_i-\lambda}{2c_i(n-1)}, \qquad \text{where} \quad \lambda := \frac{\mu}{n}\ge 0. \tag{G.59}$$

Complementary slackness for the sum constraint gives $\mu(1-f_1-f_2)=0$, so either $\mu=0$ and the sum constraint is slack, or $f_1+f_2=1$ and $\mu\ge 0$.

Step 2 (Case 1: $n>T$). First ignore the sum constraint $f_1+f_2\le 1$. Because the objective decouples across leads, the unconstrained maximizer for lead i is the single-lead optimum

$$f_i^0 = \frac{r_i-c_i}{2c_i(n-1)}.$$

Since $R_i(1) = r_i - c_i > 0$, we have $f_i^0 > 0$. Moreover, $f_i^0 \le 1$ if and only if

$$\frac{r_i - c_i}{2c_i(n-1)} \le 1 \quad \Longleftrightarrow \quad n \ge \frac{\frac{r_i}{c_i} + 1}{2}.$$

Because $r_i > c_i$, we have $\frac{r_i}{c_i} > 1$ and therefore

$$T = \frac{1}{2}\Big(\frac{r_1}{c_1} + \frac{r_2}{c_2}\Big) \ge \frac{1}{2}\Big(\frac{r_i}{c_i} + 1\Big)$$

for each i. Thus, whenever $n > T$, we have $n > \frac{1}{2}(\frac{r_i}{c_i} + 1)$ and hence $0 < f_i^0 < 1$ for both leads.

Furthermore,

$$f_1^0 + f_2^0 = \frac{\Big(\frac{r_1}{c_1} - 1\Big) + \Big(\frac{r_2}{c_2} - 1\Big)}{2(n-1)} \le 1 \quad \Longleftrightarrow \quad n \ge T.$$

At the boundary $n = T$ this holds with equality, while for $n > T$ we have $f_1^0 + f_2^0 < 1$, i.e., the sum constraint is slack. Hence, for $n > T$ the unconstrained pair (f_1^0, f_2^0) is feasible and, by strict concavity, is the unique maximizer of (E.10). In particular, at this optimum the sum constraint is slack, so $\mu = 0$ and hence $\lambda = 0$.

Thus, for $n > T$,

$$f_i^*(n) = f_i^0 = \frac{r_i - c_i}{2c_i(n-1)}, \qquad p_i^*(n) = \frac{r_i - c_i}{2},$$

and the equilibrium joining probabilities satisfy $f_1^*(n) + f_2^*(n) < 1$. Evaluating the objective gives

$$\mathsf{REV}_{\mathsf{cp}}^*(n) = n \sum_{i=1}^{2} f_i^*(n)\, p_i^*(n) = \frac{n}{4(n-1)} \sum_{i=1}^{2} \frac{(r_i - c_i)^2}{c_i},$$

which is exactly the expression stated in part 1 of the proposition.

Step 3 (Case 2: both leads active and $f_1 + f_2 = 1$). Now suppose $n \le T$ and consider an optimum where both leads are active: $0 < f_i^*(n) < 1$ and $f_1^*(n) + f_2^*(n) = 1$. Then the sum constraint binds, so complementary slackness implies $\mu \ge 0$, and the interior formulas (G.59) apply with a common $\lambda = \mu/n \ge 0$.

Summing (G.59) over i and using $f_1^* + f_2^* = 1$ yields

$$1 = \sum_{i=1}^{2} \frac{r_i - c_i - \lambda}{2c_i(n-1)} \quad \Longrightarrow \quad \lambda = \frac{\Big(\frac{r_1}{c_1} - 1\Big) + \Big(\frac{r_2}{c_2} - 1\Big) - 2(n-1)}{\frac{1}{c_1} + \frac{1}{c_2}} = \frac{\frac{r_1}{c_1} + \frac{r_2}{c_2} - 2n}{\frac{1}{c_1} + \frac{1}{c_2}} = \lambda(n).$$

Feasibility $0 < f_i^*(n) < 1$ then requires

$$r_i - c_i - \lambda(n) > 0, \quad \text{i.e.} \quad \lambda(n) < r_i - c_i, \qquad i = 1, 2.$$

This is precisely the condition $\lambda(n) < \min\{r_1 - c_1, r_2 - c_2\}$ in part 2 of the proposition.

Moreover, observe that $\lambda(n)$ is strictly decreasing in n. Hence for any integer $n \ge 2$ we have $\lambda(n) \le \lambda(2)$, so it suffices to verify that

$$\lambda(2) \le \max\{r_1 - c_1, r_2 - c_2\}.$$

Using the closed form

$$\lambda(2) = \frac{r_1 c_2 + r_2 c_1 - 4c_1 c_2}{c_1 + c_2},$$

consider first the case $r_1 - c_1 \geq r_2 - c_2$. Then, multiplying both sides of the desired inequality $\lambda(2) \leq r_1 - c_1$ by $c_1 + c_2 > 0$ and rearranging,

$$\begin{aligned}(r_1 - c_1)(c_1 + c_2) - \big(r_1 c_2 + r_2 c_1 - 4c_1 c_2\big) \\ = c_1\big(r_1 - c_1 - r_2 + 3c_2\big) = c_1\big((r_1 - c_1) - (r_2 - c_2) + 2c_2\big) \;\geq\; 2c_1 c_2 \;>\; 0,\end{aligned}$$

where we used $r_1 - c_1 \geq r_2 - c_2$ and $c_1, c_2 > 0$. Thus $\lambda(2) \leq r_1 - c_1$ in this case. If instead $r_2 - c_2 \geq r_1 - c_1$, the same argument with indices 1 and 2 swapped shows $\lambda(2) \leq r_2 - c_2$. In either case,

$$\lambda(2) \;\leq\; \max\{r_1 - c_1, r_2 - c_2\},$$

and therefore $\lambda(n) \leq \lambda(2) \leq \max\{r_1 - c_1, r_2 - c_2\}$ for all integers $n \geq 2$.

Substituting $\lambda = \lambda(n)$ into (G.59) we obtain

$$f_i^*(n) = \frac{r_i - c_i - \lambda(n)}{2c_i(n-1)} \in (0, 1).$$

The implementing prices again come from indifference:

$$p_i^*(n) = (r_i - c_i) - c_i(n-1) f_i^*(n) = \frac{r_i - c_i + \lambda(n)}{2}.$$

Finally,

$$\begin{aligned}\mathsf{REV}_{\mathsf{cp}}^*(n) &= n \sum_{i=1}^{2} f_i^*(n)\, p_i^*(n) \\ &= n \sum_{i=1}^{2} \frac{(r_i - c_i - \lambda(n))(r_i - c_i + \lambda(n))}{4c_i(n-1)} \\ &= \frac{n}{4(n-1)} \left(\sum_{i=1}^{2} \frac{(r_i - c_i)^2}{c_i} \;-\; \lambda(n)^2 \Big(\frac{1}{c_1} + \frac{1}{c_2} \Big) \right),\end{aligned}$$

which is exactly the revenue expression stated in part 2.

Note that at the boundary $n = T$ we have $\lambda(T) = 0$ and $f_i^*(T) = (r_i - c_i)/(2c_i(T-1))$, with $f_1^*(T) + f_2^*(T) = 1$, so the solution at $n = T$ indeed belongs to the present case rather than to part 1.

Step 4 (Case 3: corner with one lead excluded). Finally, suppose

$$n \;<\; T \quad \text{and} \quad \lambda(n) \geq \min\{r_1 - c_1, r_2 - c_2\},$$

as in part 3 of the proposition.

By definition,

$$\lambda(n) = \frac{\frac{r_1}{c_1} + \frac{r_2}{c_2} - 2n}{\frac{1}{c_1} + \frac{1}{c_2}},$$

so the denominator is strictly positive and the numerator is affine in n with slope -2. Hence $\lambda(n)$ is strictly decreasing in n, and for any integer $n \geq 2$,

$$\lambda(n) \;\leq\; \lambda(2).$$

We next show that $\lambda(2) < \max\{r_1 - c_1, r_2 - c_2\}$. Using the equivalent expression

$$\lambda(2) = \frac{r_1 c_2 + r_2 c_1 - 4c_1 c_2}{c_1 + c_2},$$

consider two cases.

Case 1: $r_1 - c_1 \geq r_2 - c_2$. Then $\max\{r_1 - c_1, r_2 - c_2\} = r_1 - c_1$ and

$$\begin{aligned} \lambda(2) < r_1 - c_1 &\iff \frac{r_1 c_2 + r_2 c_1 - 4c_1 c_2}{c_1 + c_2} < r_1 - c_1 \\ &\iff 0 < c_1\big((r_1 - c_1) - (r_2 - c_2) + 2c_2\big). \end{aligned}$$

Since $r_1 - c_1 \geq r_2 - c_2$ and $c_1, c_2 > 0$, the last term is strictly positive, so indeed $\lambda(2) < r_1 - c_1 = \max\{r_1 - c_1, r_2 - c_2\}$.

Case 2: $r_2 - c_2 \geq r_1 - c_1$. By symmetry (swapping indices 1 and 2), the same argument yields $\lambda(2) < r_2 - c_2 = \max\{r_1 - c_1, r_2 - c_2\}$.

Thus, in all cases,

$$\lambda(2) \;<\; \max\{r_1 - c_1, r_2 - c_2\},$$

and since $\lambda(n) \leq \lambda(2)$ for all integers $n \geq 2$, we obtain

$$\lambda(n) \;<\; \max\{r_1 - c_1, r_2 - c_2\}, \qquad \forall\, n \in \mathbb{Z}_{\geq 2}.$$

Combining the Case 3 assumption $\lambda(n) \geq \min\{r_1 - c_1, r_2 - c_2\}$ with the strict upper bound just derived, we have

$$\min\{r_1 - c_1, r_2 - c_2\} \;\leq\; \lambda(n) \;<\; \max\{r_1 - c_1, r_2 - c_2\}.$$

Therefore there exists at least one index $j \in \{1, 2\}$ with $\lambda(n) \geq r_j - c_j$, and at most one such index (because $\lambda(n)$ is strictly below the larger of the two numbers $r_1 - c_1, r_2 - c_2$). Thus there is a *unique* j such that $\lambda(n) \geq r_j - c_j$; for the other index $i \neq j$ we have $\lambda(n) < r_i - c_i$.

If lead j were active and interior, $0 < f_j^*(n) < 1$, then the interior formula (G.59) would give

$$f_j^*(n) = \frac{r_j - c_j - \lambda(n)}{2c_j(n-1)} \leq 0,$$

contradicting interiority. Hence lead j cannot be interior and must be pushed to the boundary $f_j^*(n) = 0$. The KKT conditions then imply a strictly positive multiplier $\nu_j > 0$ for the constraint $f_j \geq 0$, so lead j is indeed excluded at the optimum.

With $f_j^*(n) = 0$, the per-state program reduces to a single-lead problem for the remaining lead $i \neq j$:

$$\max_{0 \le f_i \le 1} \; n\Big[(r_i - c_i) f_i - c_i(n-1) f_i^2\Big].$$

This strictly concave quadratic has derivative zero at

$$f_i^0 = \frac{r_i - c_i}{2c_i(n-1)}.$$

As in the single-lead analysis, the maximizer is

$$f_i^*(n) = \begin{cases} 1, & \text{if } n \le \dfrac{\frac{r_i}{c_i} + 1}{2}, \\ f_i^0 = \dfrac{r_i - c_i}{2c_i(n-1)}, & \text{if } n > \dfrac{\frac{r_i}{c_i} + 1}{2}. \end{cases}$$

We now show that, under the Case 3 assumptions, only the first branch can occur, so that $f_i^*(n) = 1$.

Without loss of generality, suppose $r_1 - c_1 \le r_2 - c_2$, so that

$$\min\{r_1 - c_1, r_2 - c_2\} = r_1 - c_1 \quad \text{and} \quad \lambda(n) \ge r_1 - c_1.$$

By uniqueness, lead $j = 1$ is the excluded lead and $i = 2$ is the remaining lead. From the definition of $\lambda(n)$,

$$\begin{aligned} \lambda(n) \ge r_1 - c_1 &\iff \frac{\frac{r_1}{c_1} + \frac{r_2}{c_2} - 2n}{\frac{1}{c_1} + \frac{1}{c_2}} \;\ge\; r_1 - c_1 \\ &\iff n \;\le\; \frac{r_2 - r_1 + c_1 + c_2}{2c_2}. \end{aligned}$$

Compare this upper bound with the single-lead threshold for lead 2,

$$\frac{\frac{r_2}{c_2} + 1}{2} = \frac{r_2 + c_2}{2c_2}.$$

We have

$$\begin{aligned} \frac{r_2 - r_1 + c_1 + c_2}{2c_2} \le \frac{r_2 + c_2}{2c_2} &\iff r_2 - r_1 + c_1 + c_2 \le r_2 + c_2 \\ &\iff c_1 \le r_1, \end{aligned}$$

which holds strictly since $R_1(1) = r_1 - c_1 > 0$ implies $r_1 > c_1$. Therefore

$$n \;\le\; \frac{r_2 - r_1 + c_1 + c_2}{2c_2} \;<\; \frac{r_2 + c_2}{2c_2} = \frac{\frac{r_2}{c_2} + 1}{2},$$

so we are necessarily in the first regime of the single-lead solution, and hence

$$f_2^*(n) = 1.$$

If instead $r_2 - c_2 \le r_1 - c_1$, the same argument with indices 1 and 2 swapped shows that in Case 3 we must have $f_1^*(n) = 1$ and $f_2^*(n) = 0$.

For the active lead i, with $f_i^*(n) = 1$, the indifference condition gives

$$p_i^*(n) = R_i(n) = r_i - c_i n,$$

since exactly n suppliers purchase lead i. Any price $p_j^*(n) \geq R_j(1) = r_j - c_j$ implements $f_j^*(n) = 0$ for the excluded lead j. The per-state revenue is

$$\mathsf{REV}_{\mathsf{cp}}^*(n) = n\, p_i^*(n) = n\, R_i(n),$$

as claimed. □

G.9 Proofs for Appendix F

G.9.1 Proofs for Appendix F.1

Proof of Lemma F.7.

$$\begin{aligned}
\mathsf{W}(\sigma) &\overset{(a)}{=} \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot kR(k) \\
&\overset{(b)}{=} \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot k \left(\frac{r}{k} - c\right) \\
&= \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot (r - ck) \\
&= r \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) - c \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot k \\
&\overset{(c)}{=} r - c \cdot \mathsf{Sp}(\sigma),
\end{aligned}$$

where (a) and (c) are derived from the definition of welfare and supply under $p_0 = 0$ and signaling σ respectively, and (b) is derived from the definition of the reward function. □

Proof of Lemma F.6. Let $\mathsf{W}(\sigma) := \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot kR(k)$ be the welfare under the obedient mechanism σ. First, we show that given $p_0 = 0$, the welfare is strictly positive at optimality. We prove this by providing a *feasible* signaling mechanism σ to ($\mathcal{P}_{\mathsf{welfare}}$) that yields a strictly positive welfare. Define k_n as

$$k_n = \begin{cases} \min\{\lfloor R^{-1}(0) \rfloor, n\}, & \text{if } R^{-1}(0) \notin \mathbb{Z}, \\ \min\{R^{-1}(0) - 1, n\}, & \text{if } R^{-1}(0) \in \mathbb{Z}. \end{cases}$$

Consider a feasible signaling mechanism σ defined as

$$\sigma(k \mid n) = \begin{cases} 1, & k = k_n, \\ 0, & \text{otherwise.} \end{cases}$$

Note that by the definition of k_n and Assumption 1, we have $k_n R(k_n) > 0$ and $(n - k_n)R(k_n + 1) \leq 0$, $\forall n \in \Omega$. This implies that

$$\sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} k_n R(k) \sigma(k \mid n) > 0, \quad \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} (n - k) R(k+1) \sigma(k \mid n) \leq 0.$$

Therefore, σ satisfies the obedience constraints in (join) and (not-join) and is a feasible solution to problem ($\mathcal{P}_{\mathsf{welfare}}$). The welfare under σ is

$$\mathsf{W}(\sigma) = \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} kR(k) \sigma(k \mid n) = \sum_{n\in\Omega} \mu_0(n) k_n R(k_n) > 0.$$

This implies that the optimal welfare, which is no less than the welfare under σ, is strictly positive. Let σ^* be the welfare maximizing mechanism. We have

$$\mathsf{W}(\sigma^*) > 0 \implies \sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n} kR(k)\sigma^*(k\mid n) > 0 \implies \text{constraint (join) is non-binding.}$$

Since we have shown that this constraint is non-binding at optimality, we can remove it from the LP.

Next, we prove by contradiction that at optimality, constraint (not-join) is binding. Assume σ_1 is optimal and that the constraint (not-join) is not binding under σ_1. That is to say

$$\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}\sigma_1(k\mid n)\cdot(n-k)\cdot R(k+1) = -\epsilon < 0.$$

Thus, there exist $\hat{n}\in\Omega$ and $\hat{i}\in[\hat{n}]$ such that

$$\sigma_1(\hat{i}\mid\hat{n})\cdot(\hat{n}-\hat{i})\cdot R(\hat{i}+1) < 0,$$

which further implies that $\sigma_1(\hat{i}\mid\hat{n}) > 0$ and $R(\hat{i}+1) < 0$.

Pick $\hat{j}\le\lfloor R^{-1}(0)-1\rfloor$ so that $R(\hat{j}+1) > 0$. We then construct σ_2 as follows. For $n\neq\hat{n}$, $\sigma_2(k\mid n) = \sigma_1(k\mid n), \forall k\in[n]$, and for $n=\hat{n}$,

$$\sigma_2(k\mid n) = \begin{cases} \sigma_1(\hat{i}\mid\hat{n})-\delta, & k=\hat{i},\\ \sigma_1(\hat{j}\mid\hat{n})+\delta, & k=\hat{j},\\ \sigma_1(k\mid\hat{n}), & k\in[\hat{n}], k\neq\hat{i}, k\neq\hat{j}, \end{cases}$$

where

$$\delta = \min\left\{\sigma_1(\hat{i}\mid\hat{n}), \frac{\epsilon}{\mu_0(\hat{n})\left[(\hat{n}-\hat{j})R(\hat{j}+1)-(\hat{n}-\hat{i})R(\hat{i}+1)\right]}\right\}.$$

In other words, comparing σ_2 with σ_1, we transfer δ from the probability of recommending $\hat{i}$ of $\hat{n}$ providers to the probability of recommending $\hat{j}$ of $\hat{n}$ providers. Note from $R(\hat{j}+1)\ge 0$ and $R(\hat{i}+1) < 0$ that δ is positive. σ_2 is clearly a feasible probability distribution. To see that σ_2 satisfies (not-join), notice that

$$\sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}(n-k)R(k+1)\left(\sigma_2(k\mid n)-\sigma_1(k\mid n)\right) = \delta\cdot\mu_0\cdot(\hat{n})\cdot\left[(\hat{n}-\hat{j})R(\hat{j}+1)-(\hat{n}-\hat{i})R(\hat{i}+1)\right] \overset{(a)}{\le} \epsilon,$$

where inequality (a) follows from the definition of δ. Hence, the left-hand side of (not-join) under σ_2 increases at most by ϵ compared to σ_1 and remains feasible. Moreover, σ_2 results in a higher welfare as

$$\mathsf{W}(\sigma_2)-\mathsf{W}(\sigma_1) = \sum_{n\in\Omega}\mu_0(n)\sum_{k=0}^{n}kR(k)\left(\sigma_2(k\mid n)-\sigma_1(k\mid n)\right) = \delta\cdot\mu_0(\hat{n})\cdot\underbrace{\left(\hat{j}R(\hat{j})-\hat{i}R(\hat{i})\right)}_{\text{positive}} \ge 0.$$

Note that $\hat{i}+1 > R^{-1}(0)$ and $kR(k)$ is a concave function (by Assumption 1 (ii)) that is decreasing at $R^{-1}(0)$. As a result, for $\hat{j} < \hat{i}$, we have $\hat{j}R(\hat{j}) > \hat{i}R(\hat{i})$. Hence, $\mathsf{W}(\sigma_2)$ is higher than $\mathsf{W}(\sigma_1)$ which contradicts the optimality of σ_1. We conclude that (not-join) should be binding at optimality. $\square$

Proof of Theorem F.1. By Lemma F.6, (join) is non-binding at optimality and can be dropped without affecting the optimal objective value, while (not-join) is binding at optimality. As a result, welfare maximization problem ($\mathcal{P}_{\mathsf{welfare}}$) is equivalent to the following problem:

$$\max_{\sigma} \quad \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} k \cdot R(k) \cdot \sigma(k \mid n) \tag{G.60a}$$

$$\text{s.t.} \quad \sum_{n\in\Omega} \mu_0(n) \sum_{k=0}^{n} \sigma(k \mid n) \cdot (n-k) \cdot R(k+1) = 0 \tag{G.60b}$$

$$\sum_{k=0}^{n} \sigma(k \mid n) = 1, \quad \forall n \in \Omega \tag{G.60c}$$

$$\sigma(k \mid n) \geq 0, \quad \forall n \in \Omega, k \in [n]^+. \tag{G.60d}$$

This LP has $|\Omega|+1$ constraints, and therefore, there are $|\Omega|+1$ basic variables. Note that for any $n \in \Omega$, constraint (G.60c) implies that $\sigma(k \mid n)$ must take a non-zero value (basic variable) for at least one $k \in [n]^+$; this results in at least $|\Omega|$ non-zero basic variables. Recall that there are $|\Omega|+1$ basic variables in total. This means we have exactly one state in set Ω that has two associated basic variables, one of which can be zero. This leads to Theorem F.1 (i). For the remaining states, there is one (non-zero) basic variable. This leads to Theorem F.1 (ii). $\square$

Proof of Proposition F.7. We will use the approach discussed in the Appendix G.1.1 and work with beliefs directly instead of signals. Assume $R(x) = r - \alpha x$. Let μ be some posterior belief under a public signaling mechanism for the platform. Therefore, μ belongs to probability simplex $\Delta^{|\Omega|-1}$. For the ease of notation, we refer to $\mathbb{E}_{N\sim\mu}[N]$ and $\mathbb{E}_{N\sim\mu}[N^2]$ by $\mathbb{E}_{\mu}[N]$ and $\mathbb{E}_{\mu}[N^2]$ respectively in this proof. Note that in Appendix F.1, we assumed that $p_0 = 0$; hence, by Equation (G.61) and Lemma G.1

$$f(\mu) = \min\left\{ \frac{(R^{-1}(0)-1)\,\mathbb{E}_{\mu}[N]}{\mathbb{E}_{\mu}[N^2] - \mathbb{E}_{\mu}[N]},\ 1 \right\}. \tag{G.61}$$

Now, we characterize the interim welfare function $V(\mu)$.

$$\mathsf{W}(\mu) = \sum_{n\in\Omega} \mu(n)\mathbb{E}[XR(X)] \overset{(a)}{=} \sum_{n\in\Omega} \mu(n)\left(r\mathbb{E}(X) - \alpha\mathbb{E}[X^2]\right), \tag{G.62}$$

where X follows a binomial distribution $\mathsf{Bin}(n, f(\mu))$ and (a) holds by linearity of R. Since X is a binomial distribution, we have

$$\mathbb{E}[X] = nf(\mu), \quad \mathbb{E}[X^2] = nf(\mu)(1-f(\mu)) + n^2 f^2(\mu).$$

After re-arrangement, it can be shown that

$$\mathsf{W}(\mu) = \max\left\{0,\ \underbrace{\sum_{n\in\Omega} \mu(n) n R(n)}_{:=B(\mu)}\right\} := h(B(\mu)),$$

where $h(x) = \max\{0, x\}$ is convex and non-decreasing. Since the composition of a convex nondecreasing function with an affine map (note that $B(\mu)$ is linear in μ) is convex, we conclude that $V(\mu)$ is convex. Thus, full information is optimal. □

G.9.2 Proofs for Appendix F.2

Proof of Lemma F.8. The formal proof can be obtained by following an approach similar to Lemma 1 in McAfee and McMillan (1987), where the underlying states are rigorously defined as the *set* of providers. Without introducing the complex notation, we focus on a special case to illustrate the result. Suppose we have a pool of $\bar{N}_\ell$ type-ℓ providers for $\ell \in [L]$.

Let $\mu_0(n)$ be the probability that some $n = (n_1, \ldots, n_L)$ providers from the pool $\bar{N} = (\bar{N}_1, \ldots, \bar{N}_L)$ are interested in the lead. Within each type, providers are interested in the lead in a uniformly random manner. That is, given there are n_ℓ type-ℓ providers selected, a focal type-ℓ provider is present interested in the lead with probability $n_\ell/\bar{N}_\ell$. Let B_ℓ be the event that a focal type-ℓ provider is present. Then we have

$$\mathbb{P}\{B_\ell \mid N = n\} = \frac{n_\ell}{\bar{N}_\ell},$$

independent of $n_{-\ell}$. Using this relation, we have, for all $n \in \Omega$, $\ell \in [L]$,

$$\begin{aligned}
\mu_{\mathsf{p}}^{\ell}(n) = \mathbb{P}\{N = n \mid B_\ell\} &= \frac{\mathbb{P}\{N = n\} \cdot \mathbb{P}\{B_\ell \mid N = n\}}{\mathbb{P}\{B_\ell\}} \\
&= \frac{\mu_0(n) \cdot n_\ell/\bar{N}_\ell}{\sum_{\hat{n}\in\Omega} \mathbb{P}\{B_\ell \mid N = \hat{n}\} \cdot \mathbb{P}\{N = \hat{n}\}} \\
&= \frac{\mu_0(n) \cdot n_\ell/\bar{N}_\ell}{\sum_{\hat{n}\in\Omega} \mu_0(\hat{n}) \cdot \hat{n}_\ell/\bar{N}_\ell} \\
&= \frac{n_\ell \mu_0(n)}{\mathbb{E}[N_\ell]}.
\end{aligned}$$

This completes the proof. □

Proof of Proposition F.8. In what follows, we will first derive the optimal solution of problem ($\mathcal{P}_{\mathsf{hetero}}$) without constraints (not-join-type-ℓ). We then show that this optimal solution indeed satisfies constraints (not-join-type-ℓ).

For (join-type-ℓ) with type-ℓ, we have

$$\sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \cdot k_\ell \cdot R_\ell(k) \geq p_\ell \cdot \sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \cdot k_\ell, \tag{G.63}$$

where the right-hand side is the revenue contributed by type-ℓ providers in the objective. Summing over all $\ell \in [L]$ on both sides of (G.63), we get

$$\sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \sum_{\ell\in[L]} k_\ell \cdot R_\ell(k) \geq \sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \sum_{\ell\in[L]} k_\ell \cdot p_\ell,$$

where the right-hand side is exactly the platform's expected revenue as in the objective of ($\mathcal{P}_{\text{hetero}}$). Thus, at optimality, constraints (join-type-ℓ) must be binding for all $\ell \in [L]$. This leads to the following problem:

$$\max_{\sigma \text{ satisfies (distribution)}} \sum_{n\in\Omega} \mu_0(n) \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \sum_{\ell\in[L]} k_\ell \cdot R_\ell(k).$$

Since there are no constraints across states, we can do state-wise maximization. Specifically, in state $n \in \Omega$, it is equivalent to solving the following problem:

$$\max_{\sigma(\cdot|n)\in\Delta(\mathcal{N}(n))} \sum_{k\in\mathcal{N}(n)} \sigma(k \mid n) \sum_{\ell\in[L]} k_\ell \cdot R_\ell(k).$$

The objective function is a convex combination of $\sum_{\ell\in[L]} k_\ell \cdot R_\ell(k)$ for different $k \in \mathcal{N}(n)$ with weights $\sigma(\cdot \mid n)$. This leads to the optimal $\sigma^*(\cdot \mid n)$ in Proposition F.8. The corresponding optimal price p^*_ℓ can be found by solving (G.63) under equality.

We now turn back to show that the above solution is indeed feasible. Since p^* is derived by making constraints (join-type-ℓ) binding, obviously, (σ^*, p^*) satisfies constraints (join-type-ℓ). We are left to show constraints (not-join-type-ℓ) are also satisfied. To see this, for any type $\ell \in L$ in state $n \in \Omega$,

$$\begin{aligned}
&\sum_{k\in\mathcal{N}(n)} \sigma^*(k \mid n) \cdot (n_\ell - k_\ell) \cdot [R_\ell(1+k_\ell, k_{-\ell}) - p^*_\ell] = (n_\ell - k^*_\ell(n)) \cdot \left[R_\ell(1+k^*_\ell(n), k^*_{-\ell}(n)) - p^*_\ell\right] \\
&= (n_\ell - k^*_\ell(n)) \cdot \left[R_\ell(1+k^*_\ell(n), k^*_{-\ell}(n)) - \frac{\sum_{\tilde{n}\in\Omega} \mu_0(\tilde{n}) \cdot k^*_\ell(\tilde{n}) \cdot R_\ell(k^*(\tilde{n}))}{\sum_{\tilde{n}\in\Omega} \mu_0(\tilde{n}) \cdot k^*_\ell(\tilde{n})}\right] \\
&= \frac{n_\ell - k^*_\ell(n)}{\sum_{\tilde{n}\in\Omega} \mu_0(\tilde{n}) \cdot k^*_\ell(\tilde{n})} \cdot \sum_{\tilde{n}\in\Omega} \mu_0(\tilde{n}) \cdot k^*_\ell(\tilde{n}) \cdot \left[R_\ell(1+k^*_\ell(n), k^*_{-\ell}(n)) - R_\ell(k^*(\tilde{n}))\right].
\end{aligned}$$

To show the above term is non-positive, it is sufficient to show that for all $\ell \in [L]$,

$$\underbrace{[n_\ell - k^*_\ell(n)]}_{:=A} \cdot \left[\underbrace{R_\ell(1+k^*_\ell(n), k^*_{-\ell}(n)) - R_\ell(k^*(\tilde{n}))}_{:=B}\right] \le 0, \ \forall \tilde{n} \in \Omega. \tag{G.64}$$

By Assumption F.1, we have, for any $n \in \Omega$,

$$k^*(n) = \begin{cases} n & \text{if } n \le \bar{\tau} \\ \bar{\tau} & \text{if } n > \bar{\tau} \end{cases},$$

where the inequalities are element-wise. This leads to, for any $\ell \in [L]$,

$$k^*_\ell(n) = \begin{cases} n_\ell & \text{if } n \le \bar{\tau} \\ \bar{\tau}_\ell & \text{if } n > \bar{\tau} \end{cases}. \tag{G.65}$$

With the above property, we now analyze (G.64):

(i) if $n \le \bar{\tau}$, by (G.65), we have $k^*_\ell(n) = n_\ell$ for any $\ell \in [L]$. Thus, the left-hand side of (G.64) is zero for any $\tilde{n} \in \Omega$.

(ii) if $n > \bar{\tau}$, by (G.65), we have $k^*_\ell(n) = \bar{\tau}_\ell \le n_\ell$ for any $\ell \in [L]$. Thus, $A \ge 0$ in this case. For any $\tilde{n} \in \Omega$, either $\tilde{n} \ge \bar{\tau}$ or $\tilde{n} < \bar{\tau}$ by Assumption F.1 (ii), Then, by (G.65), we have

$$k^*_\ell(\tilde{n}) \le \bar{\tau}_\ell, \ \forall \ell \in [L]. \tag{G.66}$$

Regarding B, for any $\tilde{n} \in \Omega$ and any $\ell \in [L]$,

$$\begin{aligned} B = R_\ell(1 + k_\ell^*(n), k_{-\ell}^*(n)) - R_\ell(k^*(\tilde{n})) &= R_\ell(1 + \bar{\tau}_\ell, \bar{\tau}_{-\ell}) - R_\ell(k^*(\tilde{n})) \\ &\overset{(a)}{\leq} R_\ell(\bar{\tau}_\ell, \bar{\tau}_{-\ell}) - R_\ell(k^*(\tilde{n})) \\ &\overset{(b)}{\leq} R_\ell(\bar{\tau}_\ell, \bar{\tau}_{-\ell}) - R_\ell(\bar{\tau}) = 0, \end{aligned}$$

where inequality (a) holds by the decreasing property of $R_\ell(\cdot)$, and inequality (b) holds by the decreasing property of $R_\ell(\cdot)$ and by (G.66).

This completes the proof. □

Proof of Proposition F.9. Based on Proposition F.8, this result follows a similar proof to Theorem 4 under the homogeneous case. □